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Fundamental Concepts of Enamel and Dentin Adhesion

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The classic concepts of operative dentistry were challenged in the 1980s and 1990s by the introduction of new adhesive techniques, first for enamel and then for dentin. This chapter discusses the differences in enamel and dentin adhesion and developments in adhesive systems over time. Current adhesive systems applied using several adhesion strategies with their indications and limitations, based on in vitro and clinical evidence, are discussed.

Basic Concepts of Adhesion

The American Society for Testing and Materials (specification D907) defines adhesion as "the state in which two surfaces are held together by interfacial forces which may consist of valence forces or interlocking forces or both."[1] The word *adhesion* comes from the Latin *adhaerere* ("to stick to"). An adhesive is a material, frequently a viscous fluid, that joins two substrates together by solidifying and transferring a load from one surface to the other. Adhesion or adhesive strength is the measure of the load-bearing capacity of an adhesive joint.[2] Four different mechanisms of adhesion have been described, as follows[3]:

1. ***Mechanical adhesion***—interlocking of the adhesive with irregularities in the surface of the substrate, or adherend
2. ***Adsorption adhesion***—chemical bonding between the adhesive and the adherend; the forces involved may be primary (ionic and covalent) or secondary (hydrogen bonds, dipole interaction, or van der Waals) valence forces
3. ***Diffusion adhesion***—interlocking between mobile molecules, such as the adhesion of two polymers through diffusion of polymer chain ends across an interface
4. ***Electrostatic adhesion***—an electrical double layer at the interface of a metal with a polymer that is part of the total bonding mechanism

In dentistry, bonding of resin-based materials to tooth structure is a result of four possible mechanisms, as follows[4]:

1. ***Mechanical***—penetration of resin and formation of resin tags within the tooth surface
2. ***Adsorption***—chemical bonding to the inorganic component (hydroxyapatite) or organic components (mainly type I collagen) of tooth structure
3. ***Diffusion***—precipitation of substances on the tooth surfaces to which resin monomers can bond mechanically or chemically
4. A ***combination*** of the previous three mechanisms

For good adhesion, close contact must exist between the adhesive and the substrate (enamel or dentin). The surface tension of the adhesive must be lower than the surface energy of the substrate. Failures of adhesive joints occur in three locations, which are generally combined when an actual failure occurs: (1) cohesive failure in the substrate; (2) cohesive failure within the adhesive; and (3) adhesive failure, or failure at the interface of substrate and adhesive.

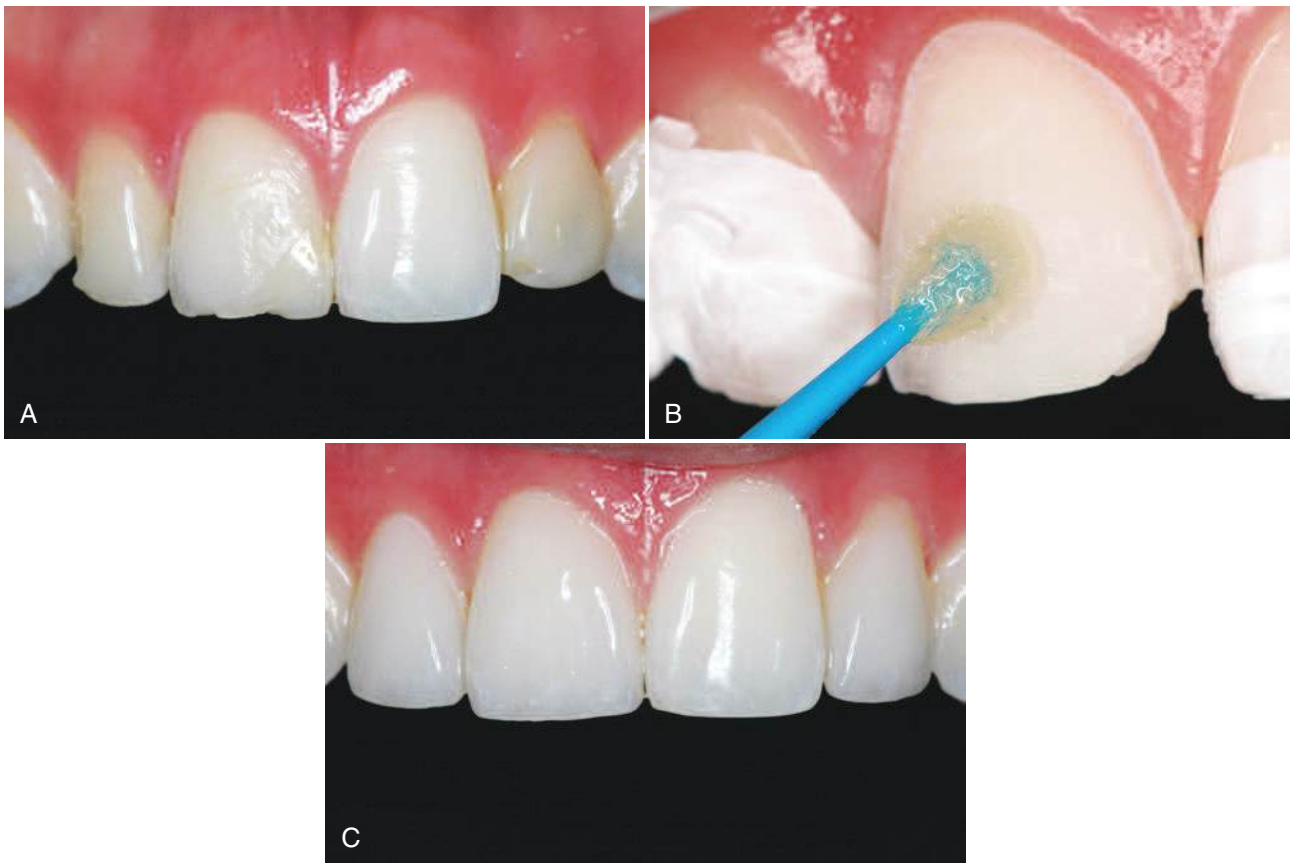
A major problem in bonding resins to tooth structure is that all methacrylate-based dental resins shrink during free-radical addition polymerization.[5] Dental adhesives must provide a strong initial bond to resist the stresses of resin shrinkage.

Trends in Restorative Dentistry

The introduction of enamel bonding, the increasing demand for restorative and nonrestorative esthetic treatments, and the ubiquity of fluoride have combined to transform the practice of operative dentistry.[6,7] The classic concepts of tooth preparation were advocated in the early 1900s,[8] but these have changed drastically. This transformation in philosophy has resulted in a more conservative approach to tooth preparation, with regard to not only the basic concepts of retention form but also the resistance form of the remaining tooth structure. Bonding techniques allow more conservative tooth preparations, less reliance on macromechanical retention, and less removal of unsupported enamel.

The availability of new scientific information on the etiology, diagnosis, and treatment of carious lesions and the introduction of reliable adhesive restorative materials have substantially reduced the need for extensive tooth preparations. With improvements in materials, indications for resin-based materials have progressively shifted from the anterior segment only to posterior teeth as well. Adhesive restorative techniques currently are used to accomplish the following:

1. Restore Class I, II, III, IV, V, and VI carious lesions or traumatic defects.
2. Change the shape and the color of anterior teeth (e.g., with full or partial resin veneers).[9,10]
3. Improve retention for porcelain-fused-to-metal (ceramometal) or metallic crowns.[11,12]
4. Bond all-ceramic restorations (Fig. 5.1).[13,14]
5. Seal pits and fissures.[15]
6. Bond orthodontic brackets.[16]



• **Fig. 5.1** A, Preoperative view of anterior teeth in a 24-year-old patient with defective composite restorations. The treatment plan included bonded porcelain veneers on teeth #7, #8, and #10 to match the natural aspect of tooth #9. B, Porcelain veneers were bonded with a two-step etch-and-rinse adhesive and a light-activated resin cement. C, Final aspect 1 week after the bonding procedure.

7. Bond splints for tooth luxations and periodontally involved anterior teeth and conservative tooth-replacement prostheses.[17-20]
8. Repair existing restorations (composite, amalgam, ceramic, or ceramometal).[21-24]
9. Provide foundations for crowns.[25,26]
10. Desensitize noncarious cervical lesions (NCCLs) and exposed root surfaces.[27-30]
11. Impregnate enamel and dentin making them less susceptible to caries.[31,32]
12. Bond fractured fragments of anterior teeth (Fig. 5.2).[33]
13. Bond prefabricated fiber, metal, and cast posts.[34]
14. Reinforce fragile endodontically treated roots internally.[35,36]
15. Seal root canals during endodontic therapy.[37,38]
16. Seal surgically resected root apices.[39,40]

Enamel Adhesion

Inspired by the industrial use of 85% phosphoric acid to facilitate adhesion of paints and resins to metallic surfaces, Buonocore envisioned the use of acids to etch enamel for sealing pits and fissures.[6,41] Since Buonocore's introduction of the acid-etch technique, many dental researchers have attempted to achieve methods for reliable and durable adhesion between resins and tooth structures.

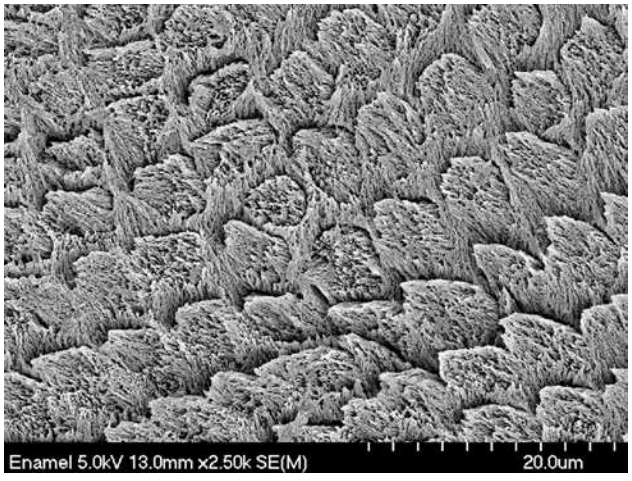
Various concentrations of phosphoric acid have been used to etch enamel. Most current phosphoric acid gels have concentrations of 30% to 40%, with 37% being the most common, although some studies using lower concentrations have reported similar adhesion values.[42-45] An etching time of 60 seconds originally was recommended for permanent enamel using 30% to 40% phosphoric acid. Although one study concluded that shorter etch times resulted in lower bond strengths, other studies using scanning electron microscopy (SEM) showed that a 15-second etch resulted in a surface roughness similar to that provided by a 60-second etch.[46-49] Other in vitro studies have shown similar bond strengths and leakage for etching times of 15 and 60 seconds.[50-54]

As measured in the laboratory, shear bond strengths of composite to phosphoric acid-etched enamel usually exceed 20 megapascals (MPa) and can range up to 53 MPa, depending on the test method used.[55-58] Such bond strengths provide adequate retention for a broad variety of procedures and prevent leakage around enamel margins of restorations.[53]

Acid etching transforms the smooth enamel into an irregular surface (Fig. 5.3) and increases its surface-free energy. When a fluid resin-based material is applied to the irregular etched surface, the resin penetrates into the surface, aided by capillary action. Monomers in the material polymerize, and the material becomes interlocked with the enamel surface (Fig. 5.4).[41] The formation of resin microtags within the enamel surface is the fundamental mechanism of resin-enamel adhesion.[41,46,59] Fig. 5.5 shows a replica of an etched enamel surface visualized through the extensions of resin that penetrated the irregular enamel surface. The acid-etch technique has revolutionized the practice of restorative dentistry.



• **Fig. 5.2** A, Intraoral frontal view of a 20-year-old female presenting with complicated crown fracture on tooth #9 after endodontic treatment. The fracture extends subgingivally on the mesial aspect of the lingual surface. B, A two-step etch-and-rinse ethanol-based adhesive applied to the crown fragment and tooth. C, Extraoral view 6 months after rebonding. D, Extraoral view 3 years after rebonding. (From Macedo GV, Ritter AV: Essentials of rebonding tooth fragments for the best functional and esthetic outcomes, *Pediatr Dent* 31(2):110–116, 2009.)



• **Fig. 5.3** Scanning electron micrograph of enamel etched with 35% phosphoric acid (3M Oral Care) for 15 seconds.

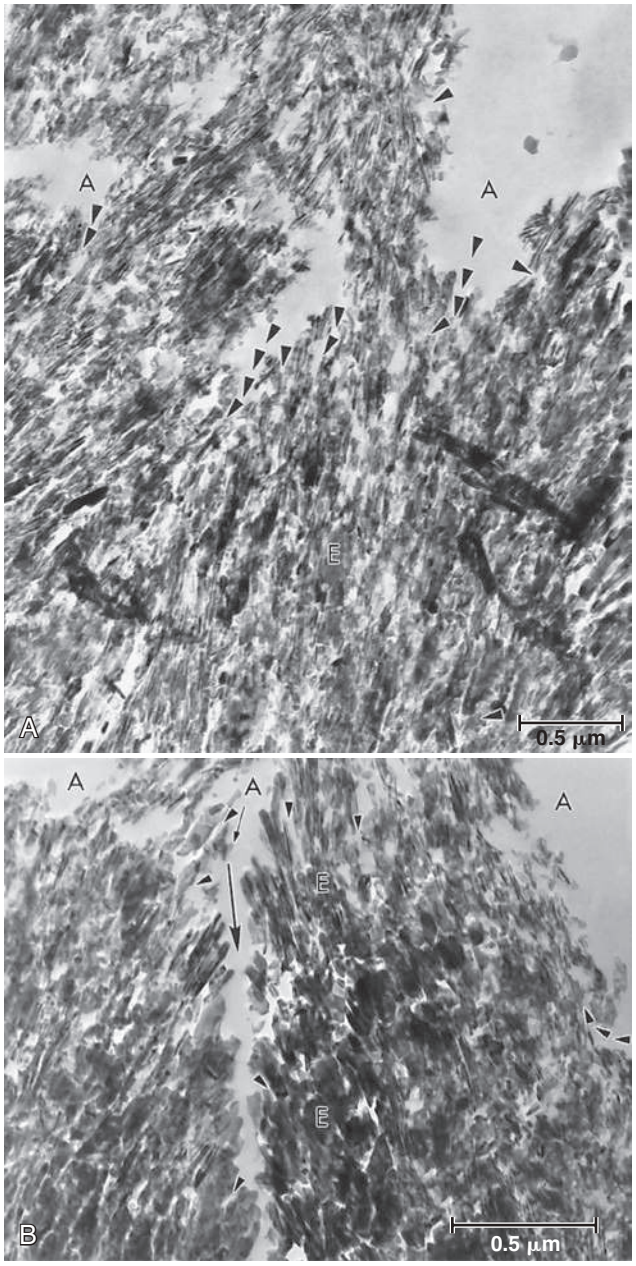
Dentin Adhesion

The classic concepts of operative dentistry were challenged in the 1980s and 1990s by the introduction of new adhesive techniques, first for enamel and then for dentin. Nevertheless, adhesion to dentin remains difficult. Adhesive materials can interact with dentin in different ways—mechanically, chemically, or both.[7,60-66] The importance of micromechanical bonding, similar to what occurs in enamel bonding, has become accepted.[60,67] Dentin adhesion relies primarily on the penetration of adhesive monomers into the network of collagen fibrils left exposed by acid etching (Fig. 5.6).[67,68] However, for adhesive materials that do not require etching, such as glass ionomer cements and some phosphate-based self-etch adhesives, chemical bonding between polycarboxylic or phosphate monomers and hydroxyapatite has been shown to be an important part of the bonding mechanism.[62-66] Contemporary strategies for bonding to dentin are summarized in Table 5.1.

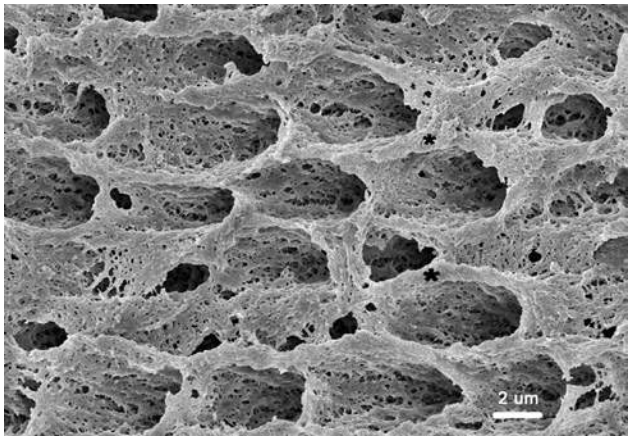
Challenges in Dentin Bonding

Substrate

Bonding to enamel is a relatively simple process, without major technical requirements or difficulties. Bonding to dentin presents a much greater challenge. Several factors account for this difference between enamel and dentin bonding. Enamel is a highly mineralized tissue composed of more than 90% (by volume) hydroxyapatite, whereas dentin contains a substantial proportion of water and organic material, primarily type I collagen (Fig. 5.7).[69] Dentin also contains a dense network of tubules that connect the pulp with the dentinoenamel junction (DEJ) (Fig. 5.8). A cuff of hypermineralized dentin called *peritubular dentin* lines the tubules. The less mineralized intertubular dentin contains collagen fibrils with the characteristic collagen banding (Fig. 5.9). Intertubular dentin is penetrated by submicron channels, which allow the passage of



• **Fig. 5.4** A and B, Transmission electron micrographs of the enamel-adhesive interface after application of Adper Single Bond (3M Oral Care) as per manufacturer's instructions. Acid etching with 35% phosphoric acid opened spaces between enamel prisms *(arrows),* allowing the permeation of resin monomers between the crystallites *(arrowheads). A*, Adhesive; *E*, enamel.



• **Fig. 5.5** Replica of enamel etched with 35% phosphoric acid. Enamel was dissolved completely in 6N hydrochloric acid for 24 hours. The resin extensions correspond to the interprismatic spaces *(asterisks).*

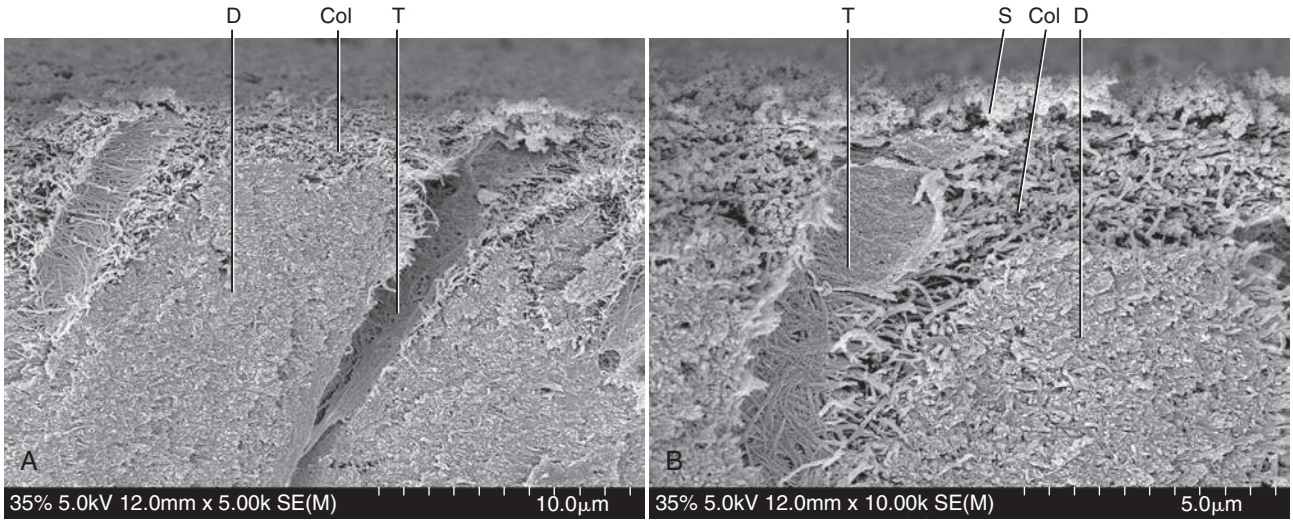
tubular liquid and fibrils between neighboring tubules, forming intertubular anastomoses.

Dentin is an intrinsically hydrated tissue, penetrated by a maze of fluid-filled tubules. Movement of fluid from the pulp to the DEJ is a result of a slight but constant pulpal pressure.[70] Pulpal pressure has a magnitude of 25 to 30 mm Hg or 34 to 40 cm H_2O.[71,72] Dentinal tubules enclose cellular extensions from the odontoblasts and are in direct communication with the pulp (Fig. 5.10). Inside the tubule lumen, other fibrous organic structures such as the lamina limitans are present, which substantially decreases the functional radius of the tubule. The relative area occupied by dentin tubules decreases with increasing distance from the pulp. The number of tubules decreases from about 45,000/mm^2 close to the pulp to about 20,000/mm^2 near the DEJ.[73] The tubules occupy an area of only 1% of the total surface near the DEJ, whereas they occupy 22% of the surface close to the pulp.[74] The average tubule diameter ranges from 0.63 µm at the periphery to 2.37 µm near the pulp.[75]

Adhesion can be affected by the remaining dentin thickness after tooth preparation. Bond strengths are generally less in deep dentin than in superficial dentin.[76-78] Some dentin adhesives, including one-step self-etch adhesives, do not seem to be affected by dentin depth.[79] When the concentration of cross-linked carboxyterminal telopeptide of type I collagen was evaluated using an immunoassay to detect the degree of collagen degradation, significantly increased degradation was measured for a two-step etch-and-rinse adhesive compared to a one-step self-etch adhesive.[80] In the same study, collagen in deep dentin underwent significantly more degradation than collagen in superficial dentin after thermal fatigue.[80] Degradation of collagen by dentin proteinases is known to negatively affect the bonded interface.

Whenever tooth structure is prepared with a bur or other instrument, residual organic and inorganic components form a "smear layer" of debris on the surface.[81,82] The smear layer fills the orifices of dentin tubules, forming "smear plugs" (Fig. 5.11), and decreases dentin permeability by nearly 90% in vitro.[83] The composition of the smear layer is basically hydroxyapatite and altered denatured collagen. This altered collagen can acquire a gelatinized consistency because of the friction and heat created by the preparation procedure.[84] Submicron porosity of the smear layer still allows for diffusion of dentinal fluid.[85] Removal of the smear layer and smear plugs with acidic solutions results in an increase of the fluid flow onto the exposed dentin surface. This fluid can interfere with adhesion because hydrophobic resins do not adhere to hydrophilic substrates, even if resin tags are formed in the dentin tubules.[7,86]

Several additional factors affect dentin permeability. Besides the use of vasoconstrictors in local anesthetics, which decrease pulpal pressure and fluid flow in the tubules, factors such as the radius and length of the tubules, the viscosity of dentin fluid, the pressure gradient, the molecular size of the substances dissolved in the tubular fluid, and the rate of removal of substances by the blood vessels in the pulp affect permeability. All of these variables

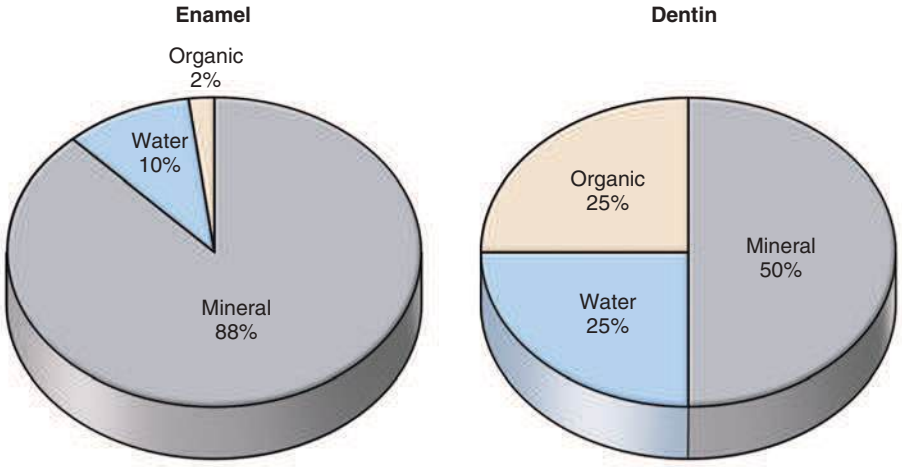


• **Fig. 5.6** A, Dentin etched with 35% phosphoric acid. B, Higher magnification view of etched dentin. *Col*, Collagen exposed by the acid; *D*, normal dentin; *T*, dentinal tubule; *S*, residual silica particles used as acid gel thickener.

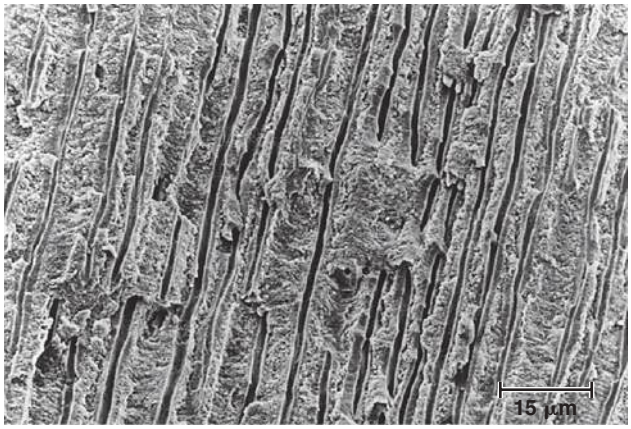
TABLE 5.1 Current Strategies for Adhesion of Resins to Dentin

	Etchant (E)	Primer (P)	Bonding Agent (B)
Three-step etch-and-rinse[a] E + P + B	Removes the smear layer Exposes intertubular and peritubular collagen Opens tubules in a funnel configuration Decreases surface-free energy	Includes bifunctional molecules (simultaneously hydrophilic and hydrophobic) Envelops the external surface of collagen fibrils Reestablishes surface-free energy to levels compatible with a more hydrophobic restorative material	Includes monomers that are mostly hydrophobic such as Bis-GMA; however, can contain a small percentage of hydrophilic monomers such as HEMA Copolymerizes with the primer molecules Penetrates and polymerizes into the interfibrillar spaces to serve as a structural backbone to the hybrid layer
Two-step etch-and-rinse E + [PB]	Removes the smear layer Exposes intertubular and peritubular collagen Opens tubules in a funnel configuration Decreases surface-free energy	Penetrates into the dentin tubules to form resin tags The first coat applied on etched dentin works as a primer—it increases the surface-free energy of dentin The second coat (and third, fourth, and so on) acts as the bonding agent used in three-step systems—it fills the spaces between the dense network of collagen fibrils	
Two-step self-etch [EP] + B	Enamel etch is typically shallow The self-etching primer (SEP) does not remove the smear layer, but fixes it and exposes about 0.5–1 μm of intertubular collagen because of its acidity (pH = 1.2–2) The smear plug is impregnated with acidic monomers, but it is not removed; some SEP monomers bond chemically to dentin When it impregnates the smear plug, the SEP prepares the pathway for the penetration of the subsequently placed fluid resin into the microchannels that permeate the smear plug	Uses the same type of bonding agent included in the three-step etch-and-rinse systems The resin tags form on resin penetration into the microchannels of the primer-impregnated smear plug	
One-step self-etch [EPB]	Etches enamel, but etch pattern is typically shallow Incorporates the smear layer into interface Being an aqueous solution of a phosphonated monomer, it demineralizes and penetrates dentin simultaneously, leaving a precipitate on the hybrid layer Forms a thin layer of adhesive, leading to low bond strengths; a multicoat approach is recommended; an extra layer of a hydrophobic bonding resin improves bond strengths and clinical performance Incompatible with self-cure composite resins unless coated with an hydrophobic bonding resin		

[a]Although the meaning of the two terms is the same, the term "etch-and-rinse" is preferred over "total-etch."



• **Fig. 5.7** Composition of enamel and dentin by volume percent.

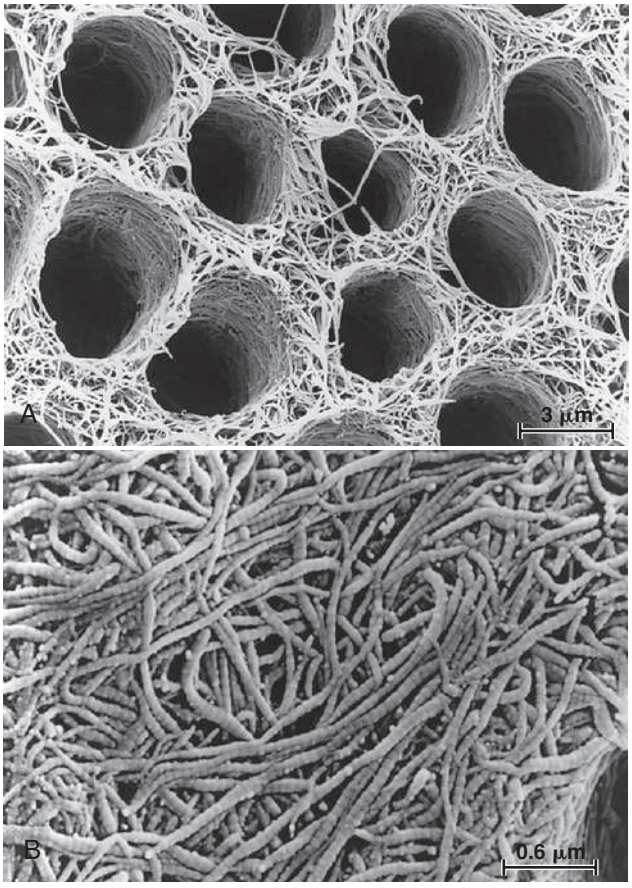


• **Fig. 5.8** Scanning electron micrograph of dentin that was fractured longitudinally to show dentinal tubules.

make dentin a dynamic substrate and consequently a difficult substrate for bonding.[74,87]

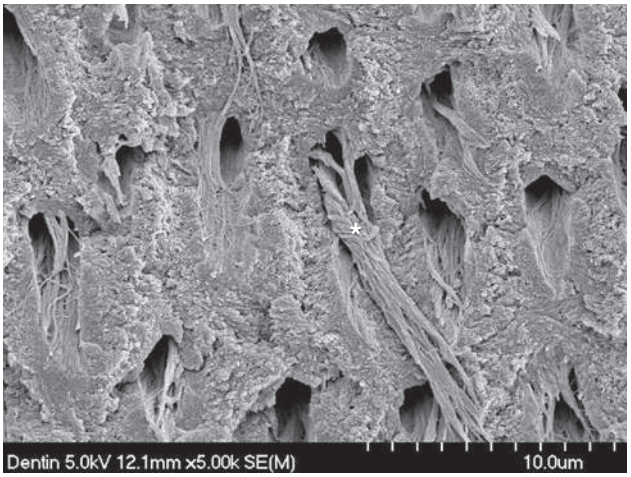
Stresses at the Resin–Dentin Interface

Composites shrink as they polymerize, creating considerable stresses within the composite mass, depending on the configuration of the preparation.[88-91] When the composite is bonded to one surface only (e.g., for a direct facial veneer), stresses within the composite are relieved by flow from the unbonded surface. Stress relief within a three-dimensional bonded restoration is limited however by its configuration factor (C-factor).[92] In an occlusal preparation, composite is bonded to five tooth surfaces—mesial, distal, buccal, lingual, and pulpal. The occlusal surface of the composite is the only "free" or unrestrained surface. In such a situation, the ratio between the number of bonded surfaces and the number of unbonded surfaces is 5:1, giving the restoration a configuration factor equal to 5. Stress relief is limited because flow can occur only from the single free surface.[92,93] Unrelieved stresses in the composite contribute to internal bond disruption and marginal gaps around restorations that may increase microleakage and potential postoperative sensitivity.[94] The C-factor might be partially responsible for the decrease in bond strengths observed when deep dentin is bonded as part of a three-dimensional preparation.[95]

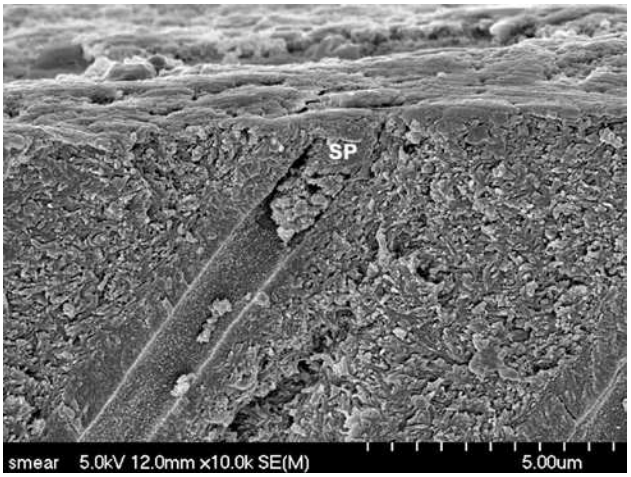


• **Fig. 5.9** A, Scanning electron micrograph of etched dentin showing exposed collagen fibrils. B, Higher magnification shows the characteristic collagen banding in intertubular collagen. Superficial collagen was dissolved by collagenase to remove the most superficial collagen fibrils that were damaged by tooth preparation.

In addition to the C-factor, the magnitude of the polymerization shrinkage stress depends on several other variables, including the rate of the polymerization reaction, the degree of conversion, the composite stiffness and its rate of acquisition of stiffness during polymerization, the nature and relative volume of the inorganic filler, the type of monomers in the composite, the composite



• **Fig. 5.10** Scanning electron micrograph of deep dentin displaying an odontoblastic process in a dentinal tubule *(asterisk)*.



• **Fig. 5.11** Scanning electron micrograph of a smear plug blocking the entrance of a dentinal tubule. *SP*, Smear plug.

insertion technique, and the opacity of the composite.[96-100] It has been reported that immediate bond strengths of approximately 17 MPa are necessary to resist the contraction stresses that develop in the composite during polymerization, to prevent marginal debonding.[90,101] However, this magnitude of bond strengths may not be enough to counteract the polymerization contraction stresses in some composite restorations, as stiffer composite materials, in general, result in increased stresses than those generated by more elastic composite materials.[102]

The water absorption in composites results in partial relief of the contraction residual stresses from polymerization.[103] However, the magnitude of hygroscopic expansion from water absorption depends on the volume of the restoration and the specific monomers used in the composition of composites,[104,105] as more hydrophilic materials may actually result in a net expansion.[103] Composite materials that contain bisphenol A ethoxylated dimethacrylate (Bis-EMA) undergo less water sorption than composites with traditional monomers such as bisphenol A-glycidyl methacrylate (Bis-GMA) and urethane dimethacrylate (UDMA).[104] Silorane-based composites have the lowest hygroscopic expansion.[105]

Several clinical strategies have been suggested to reduce the effects of polymerization shrinkage stress, including control of curing light irradiance,[106,107] incremental insertion techniques,[102,108] and application of intermediate low-modulus liners.[109] None of these methods has been effective in completely eliminating the deleterious effects of polymerization shrinkage stress, which include cuspal deflection, tooth cracking, marginal leakage and gap formation, reduction of adhesive bond strengths, and decreased physical properties of the resin restorative material.[110] Nevertheless, polymerization shrinkage stresses of composites remain a concern, and stress relief in a restoration is important. Polymerization is initiated on the surface of the restoration, close to the light source, eliminating this surface as a potential stress relief pathway.[111] Several methods have been advocated to improve the flow capacity of composites used in Class II tooth preparations. One of those methods is the use of a flowable composite between the composite and the tooth wall. Conceptually, this flowable low-modulus composite would serve as a shock absorber and simultaneously protect the interface against fatigue stresses.[112,113] The use of flowable composites as an intermediate layer in NCCLs or as the gingival increment of Class II preparations however has not been proved effective clinically.[114-118] Likewise, the application of a glass ionomer cement base underneath posterior composite restorations does not improve the restoration survival rate at 18 years.[119] Also, modifications of the current resin-based composites have been proposed as a means to reduce stress without compromising mechanical properties.[120] Studies have shown that some commercial composites specifically designed to have reduced polymerization contraction and/or stress produce less deleterious outcomes on marginal adaptation and cuspal deflection.[121-123] However, further studies demonstrated that, in spite of the lower total shrinkage obtained with the recent low-shrinkage composite, shrinkage stress is not necessarily reduced.

Each time a restoration is exposed to wide temperature variations in the oral environment (e.g., drinking coffee and eating ice cream), the restoration undergoes volumetric changes of different magnitude compared with those of the tooth structure. This occurs because the linear coefficient of thermal expansion (CTE) of hybrid and microfilled composites is about two to three times and four times greater than that of dentin, respectively.[124-126] Microleakage around dentin margins is potentiated by this discrepancy in linear coefficient of thermal expansion between the restoration and the substrate.[127,128] Enamel bond strengths usually are sufficient to prevent the formation of marginal gaps by polymerization contraction stresses. These stresses might however be powerful enough to cause enamel defects at the margins.[107] Although the extension of the enamel cavosurface bevel helps improve the enamel peripheral seal in vitro and reduce the stresses,[88,129,130] recent systematic reviews have questioned the need for enamel beveling prior to acid etching.[131,132]

Loading and unloading of restored teeth can result in transitional or permanent interfacial gaps.[133] Additionally, the tooth substrate itself might be weakened by cyclic loading.[134] A study found that 71% of Class V composite restorations in third molars with antagonists had significantly more leakage than restorations placed in teeth without opposing contact.[135] Another study found that cyclic loading and preparation configuration significantly reduced the bond strengths of self-etch and etch-and-rinse adhesives.[136,137]

Development

Beginning

During the 1950s, it was reported that a resin containing glycerophosphoric acid dimethacrylate (GPDM) could bond to a hydrochloric acid–etched dentin surface.[138] (*Note*: A complete listing of the chemical names mentioned in this chapter is provided in

TABLE 5.2 Abbreviations Commonly Used in Dentin/Enamel Adhesion Literature and in This Chapter

Abbreviation	Chemical Name
Bis-EMA	Bisphenol A ethoxylated dimethacrylate
Bis-GMA	Bisphenol A-glycidyl methacrylate
EDTA	Ethylenediamine tetra-acetic acid
GPDM	Glycerophosphoric acid dimethacrylate
HEMA	Hydroxyethyl methacrylate
10-MDP	10-Methacryloyloxy decyl dihydrogen phosphate
4-META	4-Methacryloxyethyl trimellitate anhydride
NPG-GMA	N-phenylglycine glycidyl methacrylate
PENTA	Dipentaerythritol penta-acrylate monophosphate
Phenyl-P	2-(Methacryloxy) ethyl phenyl hydrogen phosphate
UDMA	Urethane dimethacrylate

Table 5.2.) The bond strengths of this primitive adhesion technique were severely reduced by immersion in water. A few years before that report, another researcher had used the same monomer chemically activated with sulfinic acid, and that combination would later be known commercially as Sevriton Cavity Seal (Amalgamated Dental Company, London, England).[139,140]

First Generation

The development of the surface-active comonomer NPG-GMA was the basis for Cervident (S.S. White Burs, Inc., Lakewood, NJ), which is considered the first-generation dentin bonding system.[141,142] Theoretically, this comonomer could chelate with calcium on the tooth surface to generate water-resistant chemical bonds of resin to dentinal calcium.[143,144] The in vitro dentin bond strengths of this material were, however, in the range of only 2 to 3 MPa.[145] Likewise, the in vivo results were discouraging as Cervident had poor clinical results when used to restore NCCLs without mechanical retention.[146]

Second Generation

In 1978, the Clearfil Bond System F[c] was introduced in Japan (Kuraray Co., Ltd., Osaka, Japan). Generally recognized as the first product of the second generation of dentin adhesives, it was a phosphate-ester material (phenyl-P and hydroxyethyl methacrylate [HEMA] in ethanol). Its mechanism of action was based on the polar interaction between negatively charged phosphate groups in the resin and positively charged calcium ions in the smear layer.[145] The smear layer was the weakest link in the system because of its relatively loose attachment to the dentin surface. Examination of both sides of failed bonds revealed the presence of smear layer debris.[147]

Several other phosphate-ester dentin bonding systems were introduced in the early 1980s, including Scotchbond (3M Dental Products, St. Paul, MN), Bondlite (Kerr Corporation, Orange, CA), and Prisma Universal Bond (DENTSPLY Caulk, Milford, DE). These second-generation dentin bonding systems typically had in vitro bond strengths of only 1 to 5 MPa, which was considerably below the 10-MPa value estimated as the threshold value for acceptable in vivo retention.[84,148] In addition to the problems caused by the loosely attached smear layer, these resins were relatively devoid of hydrophilic groups and had large contact angles on intrinsically moist surfaces.[149] They did not wet dentin well, did not penetrate the entire depth of the smear layer, and therefore could not reach the superficial dentin to establish ionic bonding.[84] Whatever bonding did occur was due to interaction with calcium ions in the smear layer.[150]

The in vitro performance of second-generation adhesives after 6 months was unacceptable.[151] The bonding material tended to peel from the dentin surface after water storage, indicating that the interface between dentin and some types of chlorophosphate ester–based materials (e.g., second-generation dentin bonding systems) was unstable.[151,152] The in vivo performance of these materials was found to be clinically unacceptable 2 years after placement in cervical tooth preparations without additional retention, such as beveling and acid etching.[153,154]

Third Generation

The concept of phosphoric acid etching of dentin before application of a phosphate ester–type bonding agent was introduced by Fusayama et al. in 1979.[155] Because of the hydrophobic nature of the bonding resin however, acid etching did not produce a significant improvement in dentin bond strengths despite the flow of the resin into the open dentinal tubules.[86,156] Pulpal inflammatory responses were thought to be triggered by the application of acid on dentin surfaces, providing another reason to avoid etching.[157,158] Nevertheless, continuing the etched dentin philosophy, Kuraray introduced Clearfil New Bond in 1984. This phosphate-based material contained HEMA and a 10-carbon molecule known as *10-MDP* (10-methacryloyloxydecyl dihydrogen phosphate), which includes a long hydrophobic component and a short hydrophilic component.[143] Treatment of the smear layer with acidic primers was proposed using an aqueous solution of 2.5% maleic acid, 55% HEMA, and a trace of methacrylic acid (Scotchbond 2, 3M Dental Products).[95] Scotchbond 2 was the first dentin bonding system to receive "provisional" and "full acceptance" from the American Dental Association (ADA).[159] With this type of smear layer treatment, manufacturers effectively combined the dentin etching philosophy advocated in Japan with the more cautious approach advocated in Europe and the United States. The result was preservation of a modified smear layer with slight demineralization of the underlying intertubular dentin surface. Clinical results were mixed, with some reports of good performance and some reports of poor performance.[153,154]

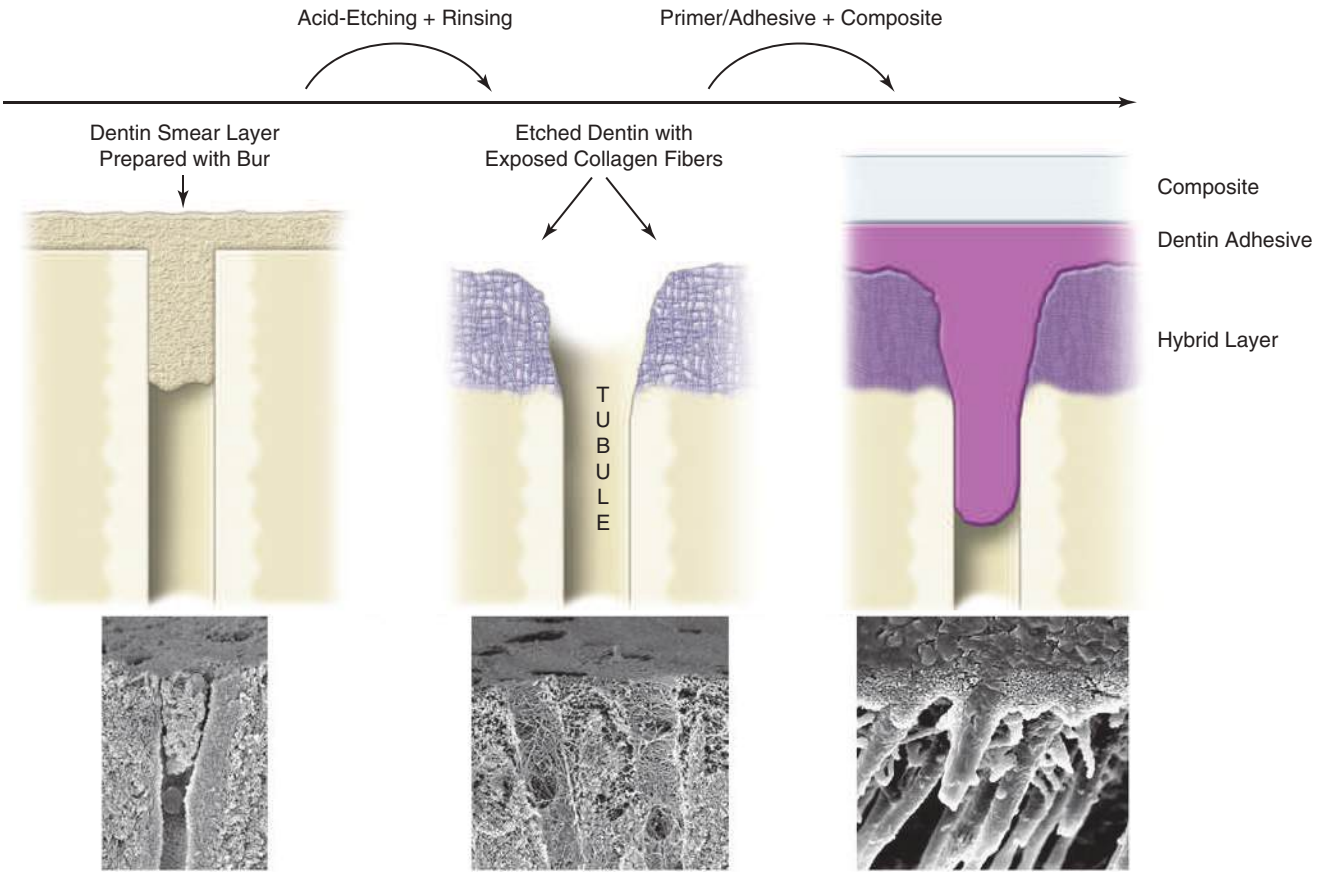
The removal of the smear layer using chelating agents such as EDTA was recommended in the original Gluma system (Bayer Dental, Leverkusen, Germany) before the application of a primer solution of 5% glutaraldehyde and 35% HEMA in water. The effectiveness of this system might have been impaired however by the manufacturer's questionable recommendation of placing the composite over uncured unfilled resin.[154]

Most other third-generation materials were designed not to remove the entire smear layer, but to modify it and allow penetration of acidic monomers such as phenyl-P or PENTA (see Table 5.2). Despite promising laboratory results, some of the bonding mechanisms never resulted in satisfactory clinical results.[154]

Current Options for Resin–Dentin Bonding

Three-Step Etch-and-Rinse Adhesives

Although the smear layer acts as a "diffusion barrier" that reduces the permeability of dentin, it may be considered an obstacle that

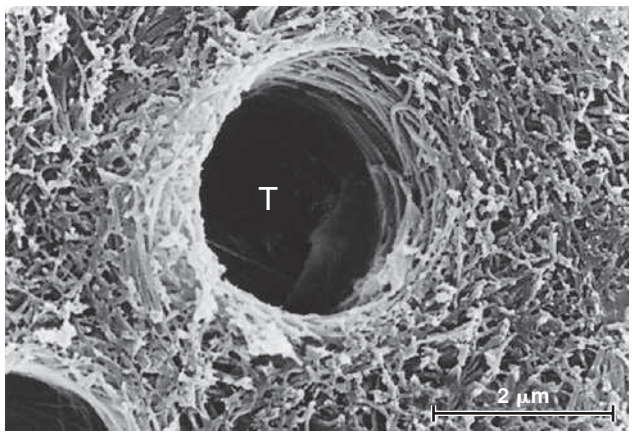


• **Fig. 5.12** Bonding of resin to dentin using an etch-and-rinse technique.

must be removed or chemically modified to permit resin bonding to the underlying dentin substrate.[65,83,160,161] Based on that consideration, a new generation of dentin adhesives was introduced for use on acid-etched dentin.[162] Removal of the smear layer via acid etching led to improvements in the in vitro bond strengths of resins to dentin.[163-166] Because the clinical technique involves simultaneous application of an acid to enamel and dentin, this method was originally known as the "total-etch" technique. Now more commonly called *etch-and-rinse technique*, it was the most popular strategy for dentin bonding during the 1990s and remains somewhat popular today (Fig. 5.12).

Application of acid to dentin results in partial or total removal of the smear layer and demineralization of the underlying dentin.[167] Acids demineralize intertubular and peritubular dentin, open the dentin tubules, and expose a dense filigree of collagen fibrils (see Fig. 5.6), increasing the microporosity of the intertubular dentin (Fig. 5.13).[67,167] Dentin is demineralized by up to approximately 7.5 µm depending on the type of acid, application time, and concentration.[67,167]

Despite the obvious penetration of early adhesives into the dentinal tubules, etching did not result in a significant improvement in bond strengths, possibly as a result of the hydrophobic nature of the phosphonated resin.[156] On the basis of concerns about the potential for inflammatory pulpal responses, acids were believed to be contraindicated for direct application on dentin, and the total-etch technique was not readily accepted in Europe or the United States. Adhesive systems based on the total-etch philosophy have proved successful however in vitro and in vivo.[154,168-170]



• **Fig. 5.13** Scanning electron micrograph of dentin that was kept moist after rinsing off the etchant. The abundant intertubular porosity serves as a pathway for the penetration of the dentin adhesive. *T*, Dentinal tubule.

Adhesive systems such as All-Bond 2 and All-Bond 3 (Bisco, Inc., Schaumburg, IL), OptiBond FL (Kerr Corporation), and Scotchbond Multi-Purpose (3M Oral Care, St. Paul, MN) (Table 5.3) are described by some authors as fourth-generation adhesives. However, because they include three essential components that are applied sequentially, they are more accurately described as three-step etch-and-rinse systems. The three essential components are (1) a phosphoric acid–etching gel that is rinsed off; (2) a primer

TABLE 5.3 Adhesive Systems Categorized by Adhesion Strategy

Type	Brand Names (Manufacturer)	COMPONENTS 1	2	3
Three-step etch-and-rinse	Adper Scotchbond Multi-Purpose (3M Oral Care) All-Bond 2 (Bisco, Inc.) All-Bond 3 (Bisco, Inc.) OptiBond FL (Kerr Corporation) Solobond Plus (Voco) Syntac (Ivoclar Vivadent)	Phosphoric acid	Hydrophilic primer	Bonding resin
Two-step etch-and-rinse	Admira Bond (Voco) Adper Single Bond Plus (3M Oral Care) ExciTE F (Ivoclar Vivadent) One-Step Plus (Bisco, Inc.) OptiBond SOLO Plus (Kerr Corporation) Prime & Bond NT (DENTSPLY Sirona) Prime & Bond XP (DENTSPLY Sirona)	Phosphoric acid	Combined (hydrophilic primer + bonding resin)	
Two-step self-etch	AdheSE (Ivoclar Vivadent) All-Bond SE (Bisco) Clearfil SE Bond (Kuraray Noritake Dental Inc) Clearfil SE Protect (Kuraray Noritake Dental Inc) OptiBond XTR (Kerr Corporation)		Self-etching primer	Bonding resin
One-step self-etch	Clearfil S^3 Bond Plus (Kuraray Noritake Dental Inc) Futurabond M (Voco) iBond (Heraeus Kulzer) OptiBond All-in-One (Kerr Corporation) Xeno IV (DENTSPLY Sirona) Xeno V$^+$ (DENTSPLY Sirona)		All-in-one Self-etch hydrophilic primer/bonding resin	
Two-step etch-and-rinse or One-step self-etch	Adhese Universal (Ivoclar Vivadent) All-Bond Universal (Bisco, Inc.) Clearfil Universal Bond (Kuraray Noritake Dental Inc) Futurabond U (Voco) One Coat 7 Universal (Coltene) Prime & Bond Elect (DENTSPLY Sirona) Scotchbond Universal Adhesive (3M Oral Care) Prime & Bond Active (DENTSPLY Sirona)	Phosphoric acid	Hydrophilic bonding resin Self-etch hydrophilic primer/bonding resin	

containing reactive hydrophilic monomers in ethanol, acetone, or water; and (3) a nonsolvated unfilled or filled resin bonding agent. Some authors refer to this third step as *adhesive*. It contains hydrophobic monomers such as Bis-GMA, frequently combined with hydrophilic molecules such as HEMA.

The acid-etching step not only alters the mineral content of the dentin substrate but also changes its surface-free energy.[77,162] The latter is an undesirable effect because for good interfacial contact, any adhesive must have a low surface tension, and the substrate must have a high surface-free energy.[84,152,171] Phosphoric acid etching of dentin removes hydroxyapatite, which has high surface energy, exposing the low surface energy collagen. A correlation exists between the ability of an adhesive to spread on the dentin surface and the concentration of calcium on that same surface.[172] The primer in a three-step system is designed to increase the critical surface tension of dentin, and a direct correlation between surface energy of dentin and shear bond strengths has been shown.[77]

When primer and bonding resin are applied to etched dentin, they penetrate the intertubular dentin, forming a resin–dentin interdiffusion zone or hybrid layer. They also penetrate and polymerize in the open dentinal tubules, forming resin tags. For most etch-and-rinse adhesives, the ultramorphologic characterization of the transition between the hybrid layer and the unaffected dentin suggests that an abrupt shift from hybrid tissue to mineralized tissue occurs, without any empty space or pathway that could result in leakage (Figs. 5.14 and 5.15). The demarcation line seems to consist of hydroxyapatite crystals embedded in the resin from the hybrid layer (see Fig. 5.15B). For self-etch adhesive systems, the transition is more gradual, with a superficial zone of resin-impregnated smear residues and a deeper zone, close to the unaffected dentin, rich in hydroxyapatite crystals (Fig. 5.16).

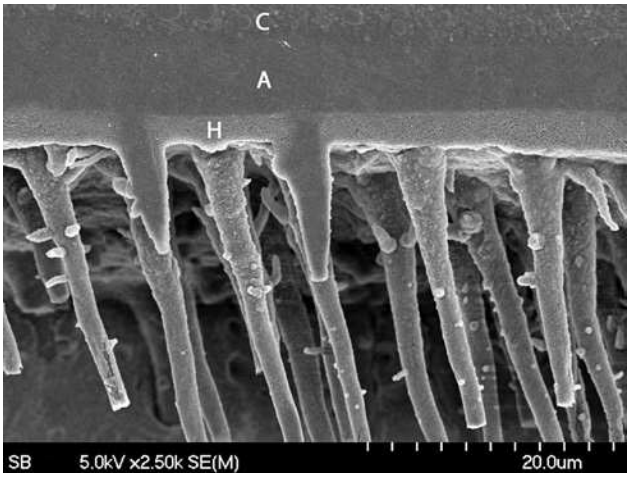
Two-Step Etch-and-Rinse Adhesives

Much of the research and development (R&D) has focused on the simplification of the bonding procedure. A number of dental materials manufacturers market a simplified, two-step etch-and-rinse adhesive system. Some authors refer to these as fifth-generation adhesives. A separate etching step still is required. In vitro dentin bond strengths obtained with two-step etch-and-rinse adhesives have improved so much that they approach the level of enamel bonding.[56] However, three-step etch-and-rinse adhesives result in better laboratory and clinical performance than two-step etch-and-rinse adhesives.[170,173-175]

Numerous two-step etch-and-rinse adhesive systems are available (see Table 5.3), including One-Step Plus (Bisco, Inc.), Prime & Bond NT (DENTSPLY Sirona, York, PA, USA), Adper Single

Bond Plus (3M Oral Care), OptiBond SOLO Plus (Kerr Corporation), PQ1 (Ultradent Products, South Jordan, UT), ExciTE F (Ivoclar Vivadent, Schaan, Liechtenstein), One Coat Bond (Coltene, Altstätten, Switzerland), and Prime & Bond XP (DENTSPLY Sirona).

Different from three-step etch-and-rinse adhesives, these simplified etch-and-rinse materials do not contain a hydrophobic bonding agent as the last bonding step. The primer/adhesive single solution in two-step etch-and-rinse adhesives contains solvents and hydrophilic components from the primer that are mixed with the hydrophobic monomers from the bonding agent. For this reason, hydrolytic degradation, which directly relates to the hydrophilicity of the adhesive,[176] is more evident in two-step etch-and-rinse adhesives when compared to their nonsimplified predecessors.[174,176] Active rubbing application and the use of multiple coats of the adhesive have been shown to somewhat compensate for the lack of a hydrophobic resin bonding agent layer as the last step in two-step etch-and-rinse adhesives.[177,178]

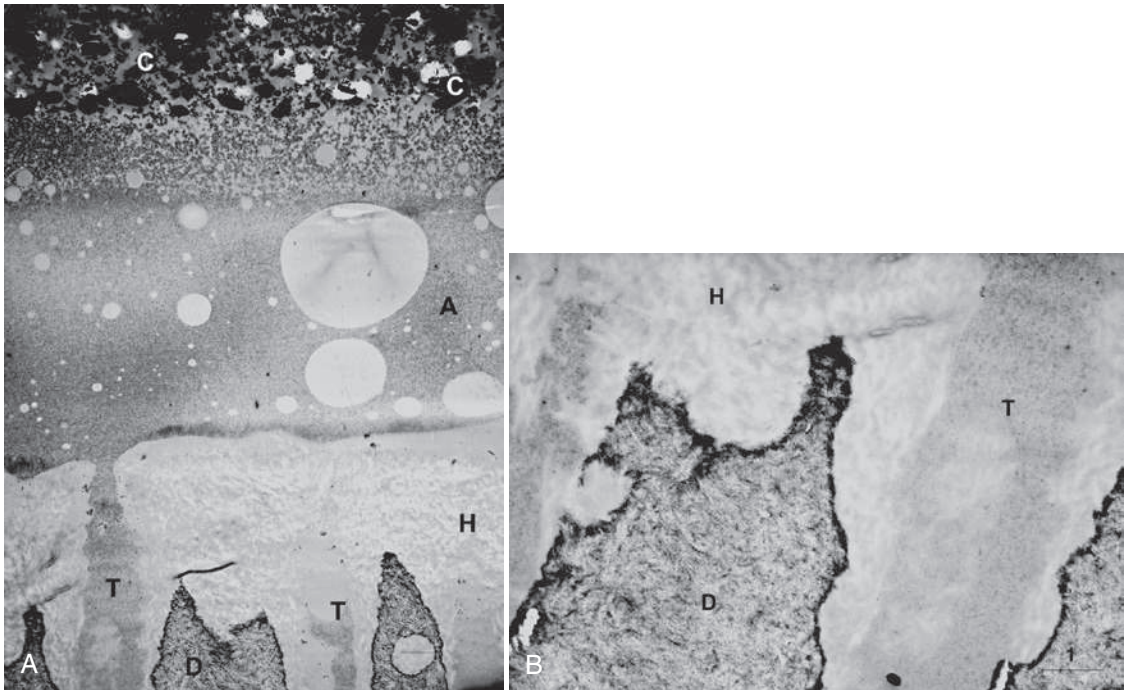


• **Fig. 5.14** Scanning electron micrograph of the transition between composite resin *(C)*–adhesive *(A)*, adhesive–hybrid layer *(H)*, and hybrid layer–dentin.

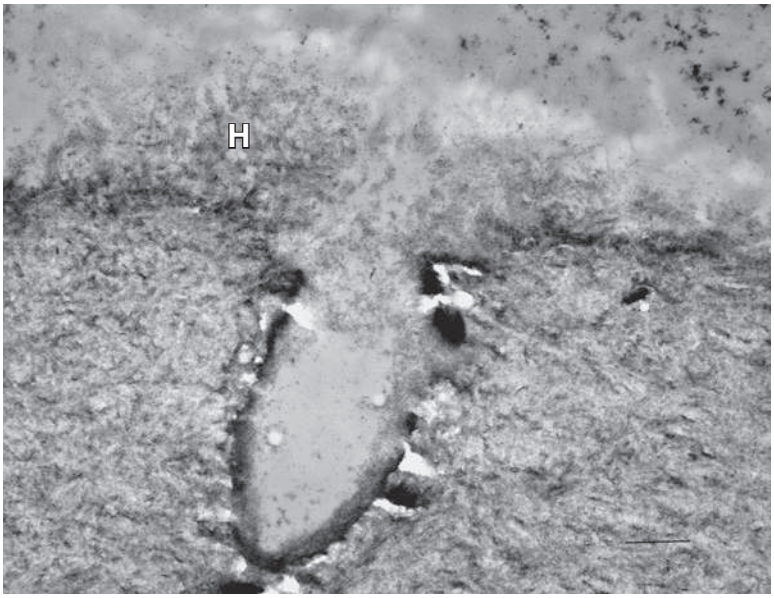
Two-Step Self-Etch Adhesives

An alternative bonding strategy is the self-etch approach (Fig. 5.17; see Fig. 5.16).

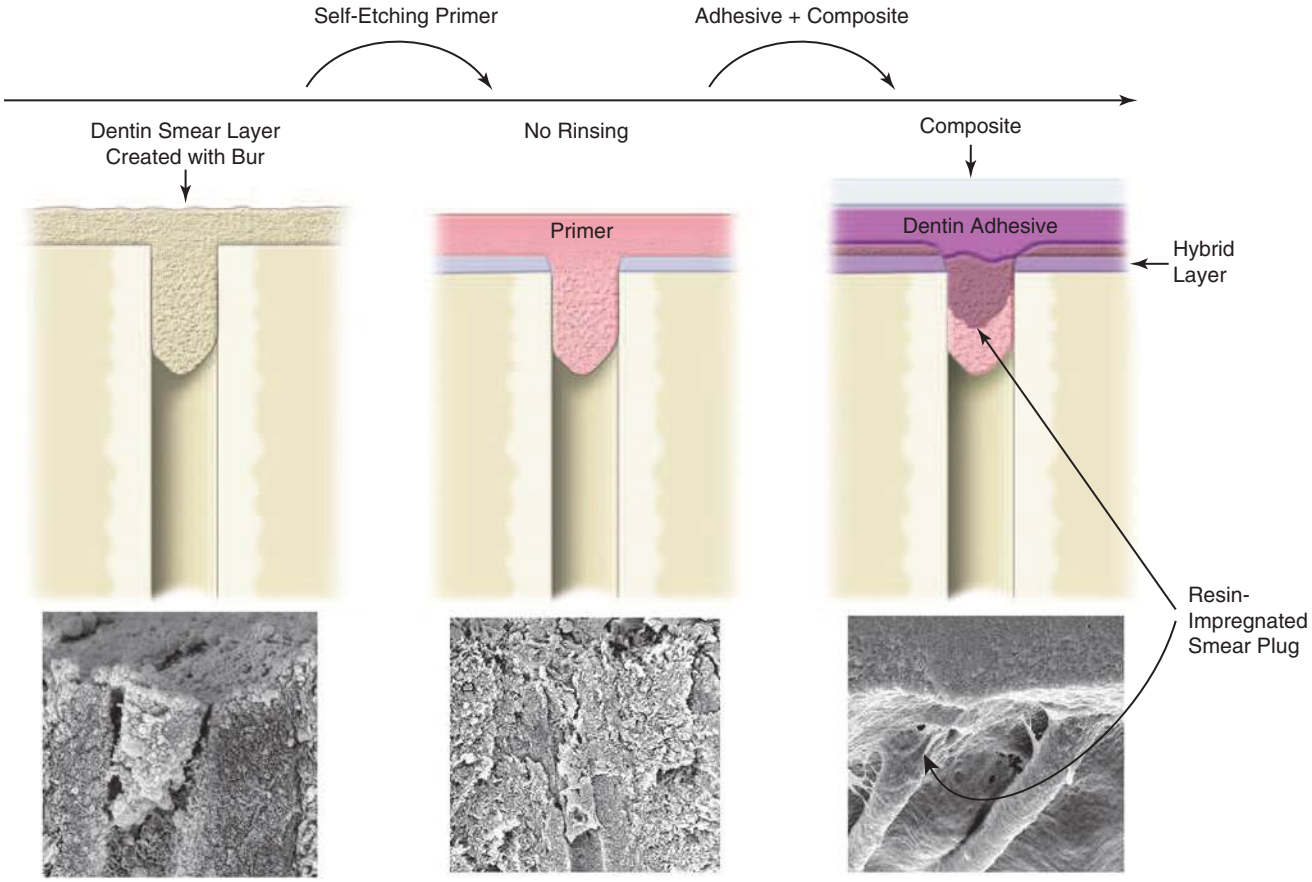
Introduced in Japan, two-step self-etch adhesives (SEAs) contain an acidic monomer that functions as a self-etching primer and a hydrophobic nonsolvated bonding resin. The acidic primers include a phosphonated and/or carboxylated resin molecule that performs two functions simultaneously—etching and priming of dentin and enamel. In contrast to conventional etchants, the acidic primers are not rinsed off. The bonding mechanism of SEAs is based on the simultaneous etching and priming of enamel and dentin, forming a continuum in the substrate and incorporating smear plugs into the resin tags (Fig. 5.18).[179,180] In addition to simplifying the bonding technique, the elimination of rinsing and drying steps reduces the possibility of overwetting or overdrying dentin, either of which can affect adhesion adversely.[164,165] Also, water is always a component of the acidic primer because it is needed for the monomers to ionize and trigger demineralization of hard dental tissues; this makes SEAs less susceptible to variations in the degree of substrate moisture but more susceptible to chemical instability due to hydrolytic degradation.[176,181,182]



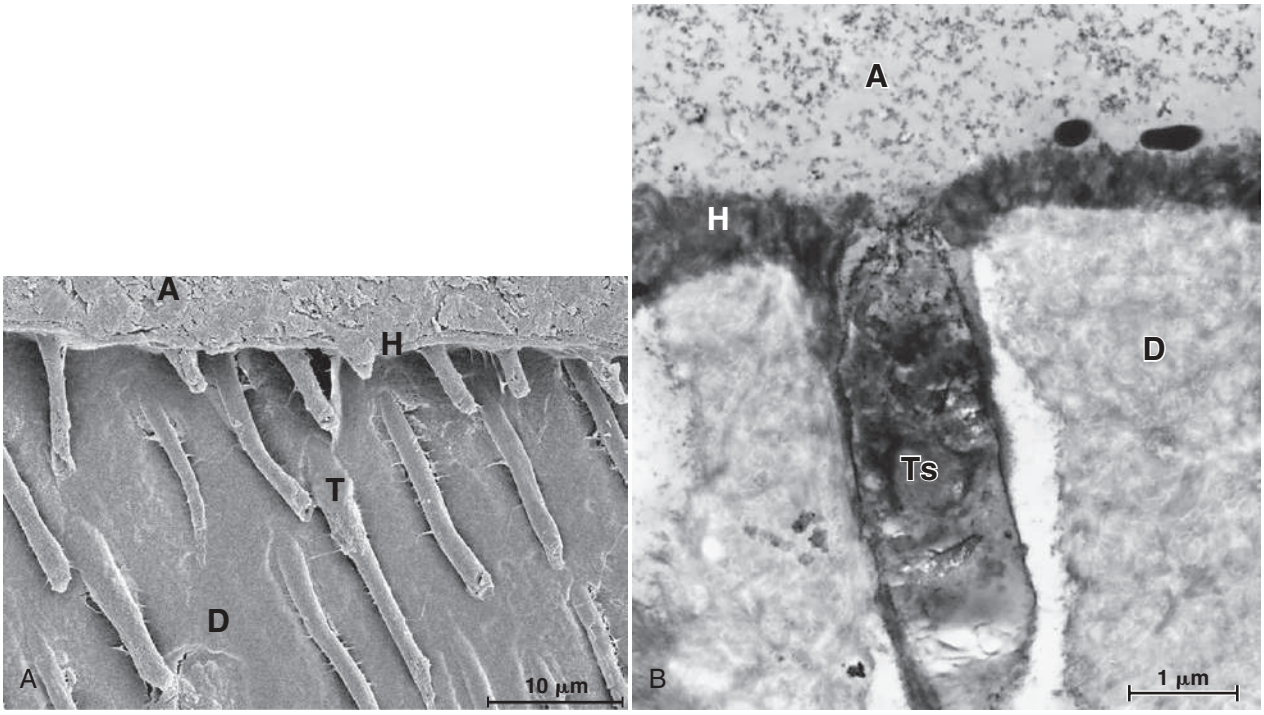
• **Fig. 5.15** Transmission electron micrograph of a resin–dentin interface formed by the etch-and-rinse adhesive Adper Single Bond Plus (3M Oral Care). This specimen was not decalcified or stained; the unaltered dentin appears darker and the hybrid layer appears lighter. A, General view showing the composite *(C)*, the adhesive *(A)*, the hybrid layer *(H)*, a filled resin tag *(T)*, and the unaffected dentin *(D)*. B, Higher magnification of the transition between the hybrid layer and unaffected dentin. Note the filler in the resin tag as small dark dots (nanofiller).



• **Fig. 5.16** Transmission electron micrograph of a resin–dentin interface formed with the two-step self-etch adhesive Clearfil SE Bond (Kuraray Noritake Dental Inc.). Residual hydroxyapatite crystals and residual components of the smear layer are embedded in the resin within the hybrid layer *(H)*.



• **Fig. 5.17** Bonding to dentin using a self-etch primer.



• **Fig. 5.18** A, Scanning electron micrograph of a resin–dentin interface formed with Clearfil SE Bond (Kuraray Noritake Dental Inc.) on chemical dissolution of the superficial dentin. B, Transmission electron micrograph of a resin–dentin interface formed with Clearfil SE Bond on EDTA decalcification and staining with uranyl acetate and lead citrate. *A*, Adhesive; *D*, residual dentin (appears gray in *B* because it was decalcified with EDTA); *H*, hybrid layer (appears dark in *B* because of decalcification followed by staining); *T*, resin tag; *Ts*, resin tag that incorporates the smear plug.

Because they are user friendly and do not require the etching and rinsing step, SEAs have become very popular.[183] One disadvantage is that some SEAs do not etch enamel as well as phosphoric acid, particularly if the enamel has not been instrumented.[184,185] However, some of the in vitro enamel bond strengths obtained with SEAs may not reflect their actual bonding effectiveness as the use of silicon carbide (Si-C) paper in in vitro experiments creates smear layers that are not clinically relevant.[160] If enamel is not etched adequately, the seal of enamel margins in vivo might be compromised.[186,187] When enamel bonds are stressed in the laboratory by thermal cycling, some of the first SEAs were more likely than etch-and-rinse systems to undergo deterioration.[188] This decrease in bond strengths with thermal fatigue might be a sign that a potential exists for enamel microleakage when SEAs are employed to bond to enamel. In a 10-year recall of an older generation SEA, 39 of 44 restorations had marginal discoloration.[189] The enamel bond strengths of some newer SEAs approach the enamel bond strengths of phosphoric acid–based adhesives however, suggesting that SEAs are gradually being developed to replace etch-and-rinse adhesive systems.

Several two-step SEAs are available, including AdheSE (Ivoclar Vivadent), All-Bond SE (Bisco, Inc.), Clearfil SE Bond or Clearfil SE Bond 2 (Kuraray Noritake Dental Inc., Tokyo, Japan), Clearfil SE Protect (Kuraray Noritake Dental Inc.), and OptiBond XTR (Kerr Corporation) (see Table 5.3). Clearfil SE Bond contains an aqueous mixture of a phosphoric acid ester monomer (10-MDP), with a much higher pH than that of phosphoric acid etchants.[190] Although the pH of a 34% to 37% phosphoric acid gel is much lower than 1.0, the pH of Clearfil SE Primer (Kuraray Noritake Dental Inc.) is 1.9 to 2.0.[167,191] SEAs have been classified according to their pH in three categories: *mild*, *moderate*, and *aggressive*, with Clearfil SE Bond being a mild SEA.[191] Mild SEAs tend to provide excellent dentin bond strengths and poorer enamel bonds, whereas more aggressive self-etch systems provide the reverse. Clearfil SE Bond resulted in 93% retention rate in Class V composite restorations at 13 years with selective enamel etching of the margins versus 86% without enamel etching, but the difference was not statistically significant.[192] Selective enamel etching improved marginal adaptation and marginal discoloration compared to the self-etch strategy.[192] In posterior restorations, Clearfil SE Bond resulted in 100% retention rate at 2 years with a tendency for deterioration of the composite margins compared with the etch-and-rinse control Adper Single Bond.[168] The clinical success of Clearfil SE Bond might be a result of its chemical composition, specifically the monomer 10-MDP, which has been shown to bond chemically to calcium in hydroxyapatite through a mechanism known as nanolayering.[65,193]

SEAs are less technique sensitive than are etch-and-rinse adhesives. Additionally, SEAs are less likely to result in a discrepancy between the depth of demineralization and the depth of resin infiltration because SEAs demineralize and infiltrate dentin simultaneously.[190] Recently, however, that concept has been challenged by showing that Clearfil SE Bond may also present partially demineralized dentin beneath the hybrid layer.[194] SEAs do not remove the smear layer from dentin completely (see Figs. 5.16 and 5.17), and in some cases smear layer remnants may interfere with in vitro bond strengths.[161] For some self-etch adhesives, collagen fibrils in the hybrid layer are not totally enveloped by the adhesive, leaving a partially demineralized dentin layer.[195] Strong one-step SEAs continue the demineralization of the adjacent dentin structure

in the tubules, which may also result in exposed collagen fibers.[196] The presence of a hybrid layer that is not completely infiltrated by the adhesive may increase the stresses in resin–dentin interfaces.[194]

Research has been conducted on the potential for remineralization of the partially infiltrated demineralized collagen inside the hybrid layer. One of the methods used to remineralize exposed dentin collagen fibrils is the application of casein phosphopeptide-amorphous calcium phosphate (CPP-ACP), which has been shown to have the potential to induce biomimetic mineralization of dentin collagen fibrils.[197] Another method with promising results is to obtain intrafibrillar and interfibrillar remineralization of a 5-µm-thick layer of demineralized dentin with the inclusion of carboxylic acid–containing and phosphonic acid–containing polyelectrolytes as biomimetic analogues in a phosphate-containing fluid.[198] More recently it has been demonstrated that intermediate precursors of calcium phosphate biomineralization may be prefabricated for loading and release with the goal of achieving collagen intrafibrillar mineralization with released fluidic intermediate precursors.[199] These authors reported that collagen fibrils can be mineralized using polyacid-stabilized amorphous calcium phosphate loaded with amine-functionalized mesoporous silica nanoparticles. These nanoparticles may be included in the composition of dentin adhesives as controlled release devices for the delivery of the amorphous calcium phosphate prenucleation clusters to remineralize deficiently infiltrated hybrid layers.[200]

Despite the prevailing opinion that SEAs cause less postoperative sensitivity compared with etch-and-rinse systems, recent clinical studies have shown no relationship between the type of adhesive and the occurrence of postoperative sensitivity.[201-206]

One-Step Self-Etch Adhesives

Continuing the trend toward simplification, no-rinse self-etch materials that incorporate the fundamental steps of etching, priming, and bonding into one solution have become increasingly popular. In contrast to conventional adhesive systems that contain an intermediate light-cured, low-viscosity bonding resin to join the composite restorative material to the primed dentin–enamel substrate, these one-step SEAs contain uncured ionic monomers that contact the composite restorative material directly.[207,208] Their acidic unreacted monomers are responsible in part for the incompatibility between these simplified adhesives and self-cured composites (discussed later).[209] Additionally, one-step SEAs tend to behave as semipermeable membranes, facilitating hydrolytic degradation of the resin–dentin interface.[181] Because these adhesives must be acidic enough to be able to demineralize enamel and penetrate dentin smear layers, the hydrophilicity of their resin monomers, usually organophosphates and carboxylates, also is high. Some of these resin monomers are too hydrophilic, which makes them liable to water degradation.[176,210]

One-step SEAs, which have etching, priming, and bonding functions delivered in a single solution, include AdheSE One F (Ivoclar Vivadent), All-Bond SE (Bisco Inc.), Bond Force (Tokuyama Dental Corporation Inc., Tokyo, Japan), Clearfil S^3 Bond Plus (Kuraray Noritake Dental Inc.), iBOND Self-Etch (Heraeus Kulzer GmbH, Hanau, Germany), OptiBond All-in-One (Kerr Corporation), and Xeno IV (DENTSPLY Sirona) (see Table 5.3). The pH of one-step SEAs affects their clinical properties. Also, application of multiple coats significantly increases dentin bond strengths and decreases leakage, suggesting that some of the one-step SEAs might not coat the dentin surface uniformly.[211]

The in vitro and clinical behavior of one-step SEAs improves with the addition of an extra coat of a hydrophobic bonding layer.[212-214] In a clinical study in Class V lesions, the one-step SEA iBond resulted in 40% retention rate at 18 months.[212] For the group to which an extra layer of a thick bonding resin was added (Scotchbond Multi-Purpose Adhesive), the retention rate increased to 83.3% at 18 months. This behavior of one-bottle SEAs may be related to their behavior as semipermeable membranes in vitro and in vivo.[181,215] Simplified SEAs do not provide a hermetic seal for vital deep dentin as demonstrated by transudation of dentinal fluid across the polymerized adhesives to form fluid droplets on the surface of the adhesive.[176]

Universal Adhesives

Dentists have used dentin adhesives following the adhesion strategy recommended by the respective manufacturer. With the advent of universal adhesives dentists are now using the same dentin adhesive for different adhesion strategies (i.e., self-etch, etch-and-rinse, or selective enamel etch adhesive), according to each specific clinical situation. Because of their multistrategy approach, this new generation of one-bottle dental adhesives has become very popular in dentistry.

These adhesives are basically one-step self-etch adhesives that can be used under different adhesion strategies (see Table 5.3).[216-218] As one-step SEAs, universal adhesives also benefit from the application of an extra hydrophobic bonding resin.[219] The major difference between traditional one-step SEAs and universal adhesives is that most universal adhesives contain 10-MDP (and/or other monomers), which is capable of bonding ionically to hydroxyapatite through nanolayering.[65,220] The 10-MDP molecule forms stable calcium-phosphate salts without causing strong decalcification. The chemical bonding formed by 10-MDP is more stable in water than that of other monomers used in the composition of SEAs, such as 4-META and phenyl-P.[221] Because the nature of the chemical bonding provided by 10-MDP depends on the concentration of this molecule, the extent of chemical bonding in universal adhesives is very weak compared to that observed for the two-step SEA Clearfil SE Bond.[220] Additionally, the presence of HEMA may hamper the chemical bonding ability of 10-MDP–containing universal adhesives,[222] because it reduces the formation of MDP-Ca salts.

The active application (rubbing) of 10-MDP–containing adhesives results in more intense nanolayering than passive application.[65] It has also been shown that active application results in higher bond strengths to intact enamel for most universal adhesives compared to passive application.[223] These improvements from rubbing the adhesive on the bonding substrate may be caused by higher concentration of 10-MPD molecules in intimate contact with the hydroxyapatite crystals in addition to higher rate of solvent evaporation from the enamel and dentin surfaces.[65]

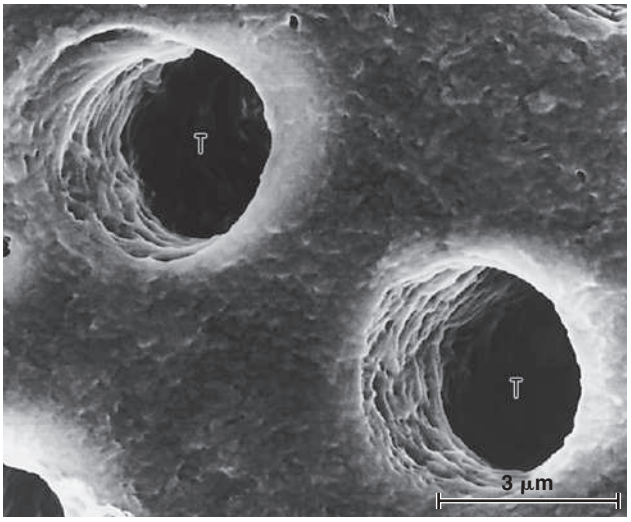
Differences in the hydroxyapatite structure in dentin and enamel dictate the interaction pattern of 10-MDP with these substrates. The lesser amount and smaller hydroxyapatite crystals in dentin and their criss-cross orientation compared to a more parallel orientation in enamel makes dentin more receptive to the chemical interaction between 10-MDP and hydroxyapatite. This interaction discloses the formation of a nanolayered structure, which is reduced in enamel compared to dentin.[65]

Universal adhesives show signs of interfacial bonding degradation after 12-month water storage when applied either as self-etch or etch-and-rinse adhesives, although the self-etch strategy caused less nanoleakage.[224] In spite of the simplification of the bonding protocol, universal adhesives will likely

undergo the same degradation pattern observed with older one-step SEAs.[218]

Moist Versus Dry Dentin Surfaces With Etch-and-Rinse Adhesives

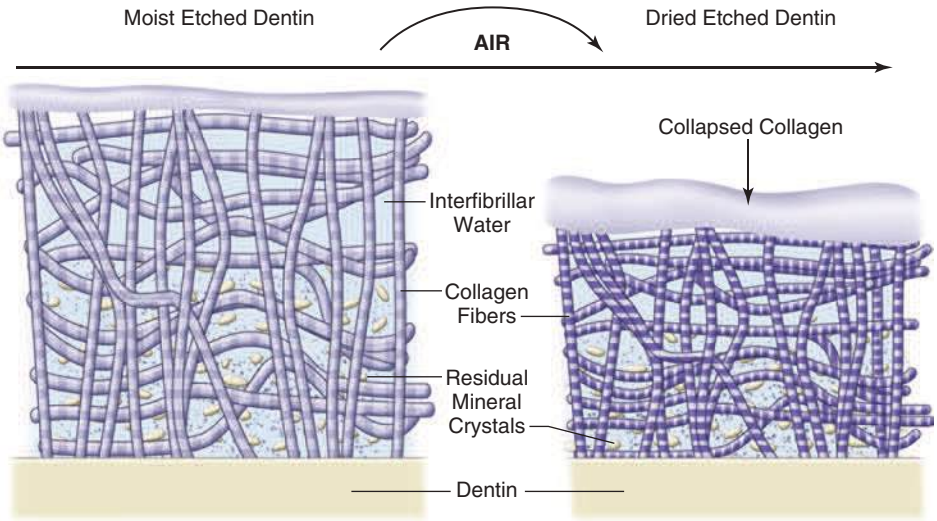
Because vital dentin is inherently wet, complete drying of dentin is difficult to achieve clinically.[165,225] Water has been considered an obstacle for attaining an effective adhesion of resins to dentin, so research has shifted toward the development of dentin adhesives that are compatible with humid environments. The "moist bonding" technique used with etch-and-rinse adhesives prevents the spatial alterations (i.e., collagen collapse) that occur on drying demineralized dentin (Fig. 5.19; compare with Fig. 5.13).[165] Such alterations might prevent the monomers from penetrating the labyrinth of nanochannels formed by dissolution of hydroxyapatite crystals between collagen fibrils.[226,227]



• **Fig. 5.19** Scanning electron micrograph of dentin collagen after acid etching with 35% phosphoric acid. Dentin was air-dried. The intertubular porosity disappeared as a consequence of the collapse of the collagen secondary to the evaporation of water that served as a backbone to keep collagen fibrils raised.

The use of etch-and-rinse adhesive systems on moist dentin is made possible by incorporation of the organic solvents acetone or ethanol in the primers or adhesives. Because the solvent can displace water from the dentin surface and the moist collagen network, it promotes the infiltration of resin monomers throughout the nanospaces of the dense collagen web. The moist bonding technique has been shown repeatedly to enhance bond strengths of etch-and-rinse adhesives in vitro because water preserves the porosity of the collagen network available for monomer interdiffusion.[165,225,228] Studies showed that the excess water after rinsing the etching gel can be removed with a damp cotton pellet, high-volume suction, disposable brush, or laboratory tissue paper without adversely affecting bond strengths.[229,230] If the dentin surface is dried with air in vitro, the collagen mesh may undergo immediate collapse and prevent resin monomers from penetrating (Fig. 5.20).[231,232]

When etched dentin is dried using an air syringe, in vitro bond strengths decrease substantially, especially for acetone-based and (to a lesser extent) ethanol-based dentin adhesive systems.[164,232,233] When water is removed, the elastic characteristics of collagen may be lost. While in a wet state, wide gaps separate the collagen molecules from each other.[234] In a dry state, the molecules are arranged more compactly. This is because extrafibrillar spaces in hydrated type I collagen are filled with water, whereas dried collagen has fewer extrafibrillar spaces open for the penetration of the monomers included in the adhesive systems.[235] During air-drying, water that occupies the interfibrillar spaces previously filled with hydroxyapatite crystals is lost by evaporation, resulting in a decrease of the volume of the collagen network to about one third of its original volume.[231] Under the scanning electron microscope, the adhesive does not seem to penetrate etched intertubular dentin that has been dried.[232] Under the transmission electron microscope (TEM), collagen fibrils coalesce into a structure without individualized interfibrillar spaces.[232] When air-dried demineralized dentin is rewetted with water, the collagen matrix may reexpand and recover its primary dimensions to the levels of the original hydrated state.[227,231,236] This spatial reexpansion is a result of the spaces between fibrils being refilled with water, but also reexpansion occurs because type I collagen itself is capable of undergoing expansion on rehydration.[237] The stiffness of decalcified dentin increases when the tissue is dehydrated chemically in water-miscible solvents or physically in air.[227] The increase in stiffness is reversed when specimens are



• **Fig. 5.20** Collapse of etched dentin by air-drying.



• **Fig. 5.21** Clinical aspect of moist dentin—a glistening appearance without accumulation of water. (From Rubinstein S, Nidetz A: The art and science of the direct posterior restoration: Recreating form, color, and translucency, *Alpha Omegan* 100(1):30–35, 2007.)

rehydrated in water. Rewetting dentin after air-drying to check for the enamel frosty aspect is an acceptable clinical procedure.[68,238] In addition, pooled moisture should not remain on the tooth because excess water can dilute the primer and render it less effective.[233,239] A glistening hydrated surface is preferred (Fig. 5.21).[240]

Although having adequate dentin surface moisture is important, agitation of the hydrophilic primer/adhesive during application of two-step etch-and-rinse adhesives may be critical for optimal adhesive infiltration into the demineralized collagen network. That procedure also may aid in the evaporation of residual water in the adhesive and hybrid layers preventing nanoleakage.[177,178,215] It is speculated that active rubbing of the adhesive compresses the demineralized collagen network, which is filled with the adhesive upon pressure release and expansion of the collagen. A recent clinical trial compared the performance of Prime & Bond NT using no rubbing action, slight rubbing action, and vigorous rubbing action in the restoration of NCCLs. While retention rates of 82.5% were found for the no rubbing action and slight rubbing action groups, 92.5% of the restorations in the vigorous rubbing action group were retained after 24 months of clinical service.[177]

In case universal adhesives are applied using an etch-and-rinse strategy, it may not be necessary to leave dentin moist for two reasons:

(1) Universal adhesives contain 10% to 20% water, which may be able to rewet dried dentin. Their bond strengths to dentin that has been air-dried for 10 seconds is similar to their bond strengths to moist dentin.[217]

(2) Manufacturers recommend the evaporation of the solvent with air for 5 seconds after the application of the universal adhesive (10 seconds for All-Bond Universal); however, 5 seconds is not enough to evaporate the water added to the composition of the adhesive.[241] If dentists leave dentin moist prior to applying universal adhesives, the amount of residual water left into the dentin substrate after air-drying will reduce bond strengths and substantially increase hydrolytic degradation of the bonded interface.[242]

Recently, some in vitro research has evaluated the possibility of replacing water with ethanol in the etched dentin collagen network, a technique known as "ethanol wet-bonding."[243,244] When acid-etched dentin is saturated with 100% ethanol instead of water, the bond strengths of both hydrophilic and hydrophobic resins increase significantly.[243,244] Although ethanol wet-bonding appears promising, it involves an extra step of replacing rinse water with 100% ethanol, and no clinical studies are available. Additionally, the time needed to replace water with ethanol in the dentin collagen network would make the technique difficult to implement in a clinical setting.

Role of Water in Self-Etch Adhesives

Water plays different roles in the bonding mechanisms of self-etch adhesives and etch-and-rinse adhesives. Unlike etch-and-rinse adhesives, self-etch systems do not include separate acid-etching and rinsing steps. The functions of etching and priming are simultaneously performed by the acidic monomers. Water (10–30 weight percent [wt%]) is added to the hydrophilic formulations to ionize the acidic methacrylate monomers (usually phosphate or carboxylic) and to solubilize calcium and phosphate ions that form from the interaction of the monomers with dentin and enamel.[245,246] When SEAs are formulated, a compromise must be made to provide sufficient water for adequate ionization of the acidic monomers without lowering the monomer concentration to levels that would jeopardize the bonding efficacy. Increasing the water concentration from 0 to 60 volume percent (vol%) in the acidic primer resulted in improved acidic monomer ionization and increased depth of dentin demineralization created by the acidic monomers.[246] However, increasing the water concentration dilutes the concentration of the acidic monomer, thereby lowering the bonding efficacy of the respective adhesive system.

The mechanical properties of one-step SEAs might be significantly compromised in the presence of water, which is less likely to occur with two-step SEAs.[247] One-step SEAs have higher water absorption or solubility than two-step self-etch adhesives.[175,248]

Role of Proteins in Dentin Bonding

Despite efforts to generate new dental materials that can enhance dentin bonding, the stability of the hybrid layer and thus long-term acceptable bond strength are still a challenge. Stability of the hybrid layer due to hydrolytic degradation of dentin collagen fibrils that have been exposed by acid etching of dentin and not completely embedded by adhesive during the bonding procedure continues to be extensively studied. The degradation is enzyme driven and compromises the hybrid layer, which becomes the restoration's weakest link.[249,250] This observation is important because acid etching demineralizes dentin and may leave a layer of exposed collagen at the bottom of the hybrid layer.[251,252] However, it has been reported that when demineralized dentin is restored with an adhesive system, the demineralized layer might undergo remineralization within 4 months.[253]

Some evidence also suggests that phosphoric acid causes denaturation of collagen fibrils in dentin.[254,255] Concerns about potential effects of phosphoric acid on dentin collagen arose from its known susceptibility to other acids such as bacterial lactic acid in caries processes.[256] However, it has been demonstrated that treatment of collagen with phosphoric acid does not significantly alter the biochemistry of dentin collagen as per analyses of collagen intramolecular and intermolecular cross-links.[257] Although evidence suggests that longer etching times might denature the collagen fibril, the normal 15-second etch does not change the spatial configuration of the collagen molecule. Etching for 15 seconds does not compromise the bonding substrate.[258]

Collagen type I is the main organic component of dentin.[69] The mineralized dentin structure carries a well-compressive load, as mineral crystals minimize movement and deformation of the collagen. In physiologic conditions (dentin formation), particles

are deposited within (intrafibrillar) and on the surface (interfibrillar) of collagen fibrils.[259,260] Structures such as bone and dentin have different mechanical properties from nonmineralized tissues. Mineralized structures are much stiffer, and although they tolerate compression well, they cannot extend much in tension.[261] When phosphoric acid treatment removes the mineral around and within collagen fibrils, the tension and compressive strength of the collagen change. Then, the cross-linking of collagen fibrils is extremely important in providing the mechanical and adaptive response.[262,263] In this regard, hydration also plays a major role. In hydrated and nonmineralized collagen, hydrogen bonds form between collagen and water, which allows for slipping and movement. However, as the collagen is dried, bonds form directly between collagen molecules and within fibrils, preventing sliding and stiffening the structure. With dehydration, the stress point lowers and collagen fibrils break more easily.[264] In dentin bonding, ideally collagen should be demineralized as minimally as possible, keeping the demineralized collagen hydrated and infiltrated with a hydrophilic adhesive to ensure proper penetration within the exposed collagen fibrils.

Ninety percent of the protein content of dentin is type I collagen and the remaining are noncollagenous proteins. The noncollagenous proteins fall into several categories: proteoglycans (PGs), SIBLINGs, growth factors, matrix metalloproteinases (MMPs), serum proteins, among others. A few of these noncollagenous proteins act as nucleators for mineral growth and dentin mineralization. The most characterized dentin mineral nucleator is dentin phosphoprotein (DPP). This protein is able to bind calcium and present it to collagen at the mineralization front during the formation of dentin.[265] It contains a large number of aspartic acid residues and thus is acidic. Biomimetic analogs of such proteins that are as acidic as DPP are able to create amorphous calcium phosphate precursors and to bind to dentin collagen.[266-269] Polyacrylic, polycarboxylic, and polyvinylphosphonic acids have been used as analogs of dentin noncollagenous phosphoproteins and have been able to induce intrafibrillar mineralization within hybrid layers when exposed to a remineralization media containing a calcium hydroxide–releasing material.[270-272]

Another noncollagenous protein category of importance in dentin bonding is the MMP family. Matrix metalloproteinases are zinc-dependent and calcium-dependent endopeptidases capable of degrading all extracellular matrix components.[273-275] In 1999, one study suggested that the direct inhibition of the MMP activity by chlorhexidine might explain the beneficial effects of chlorhexidine in the treatment of periodontitis.[276] Chlorhexidine was first used in dentin bonding as a dentin disinfectant prior to the application of the dentin adhesive. SEM revealed that chlorhexidine debris remained on the dentin surface and within the tubules of etched dentin after rinsing, but chlorhexidine had no significant effect on the dentin shear bond strengths.[277]

More recently, research has shifted toward the preservation of the hybrid layer through the inhibition of specific dentin proteases capable of degrading collagen, using chlorhexidine as a protease inhibitor.[278] Collagen fibrils that are not encapsulated by resin might be vulnerable to degradation by endogenous MMPs after acid etching.[273] Collagenolytic and gelatinolytic activities found in partially demineralized dentin imply the existence of MMP in human dentin.[273] Dentin contains gelatinases (MMP-2 and MMP-9), collagenase (MMP-8), and enamelysin (MMP-20).[273-275] These enzymes are trapped within the mineralized dentin matrix during odontogenesis.[273,275]

Dentin collagenolytic and gelatinolytic activities can be overcome by protease inhibitors, indicating that MMP inhibition might preserve the integrity of the hybrid layer and reduce the rate of resin–dentin bond degradation within the first few months after restoration.[278,279] When chlorhexidine is used, the integrity of the hybrid layer and the magnitude of bond strengths are preserved in aged resin–dentin interfaces.[280-282] When phosphoric acid is applied without the subsequent application of chlorhexidine, it does not inhibit the collagenolytic activity of mineralized dentin. In contrast, the use of chlorhexidine after acid etching—even in very low concentrations—strongly inhibits that activity.

However, the role of MMPs in dentin bonding is not completely clear for several reasons: (1) The immunoreactivity of MMP-2 is localized preferably in predentin and around the DEJ in teeth from subjects age 12 to 30 years; (2) MMP-2 and MMP-9 are both gelatinases and are unable to degrade the collagen fibrils directly, so the initial degradation step has to be performed by another mechanism; (3) MMPs do not inhibit the degradation of bonded interfaces created by self-etch adhesives; and (4) preservation of the hybrid layer can occur even in the absence of MMP inhibitors.[283-285]

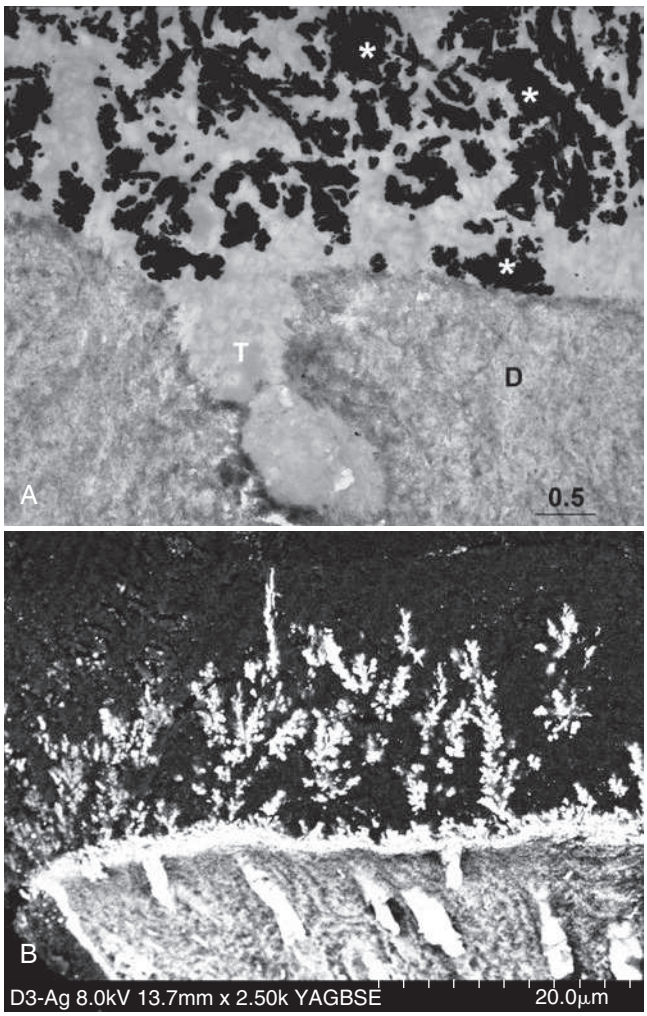
Microleakage and Nanoleakage

"Microleakage" is defined as the passage of bacteria and their toxins between restoration margins and tooth preparation walls. Clinically, microleakage becomes important when one considers that pulpal irritation is more likely caused by bacteria than by chemical toxicity of restorative materials.[286-288] An adhesive restoration might not bond sufficiently to etched dentin to prevent gap formation at margins.[289] Several studies have shown that the pulpal response to restorative materials is related to the degree of marginal leakage.[291-294] Bacteria are able to survive and proliferate within the fluid-filled marginal gaps under composite restorations. If the restoration is hermetically sealed, bacteria may not be able to survive.[286,293]

Despite the common presence of marginal gaps, their occurrence at the resin–dentin interface does not necessarily cause debonding of the restoration. Despite having shown excellent marginal seal in vitro, OptiBond (Kerr Corporation) does not completely seal the interface in vivo.[94] Other reports showed excellent clinical retention of OptiBond and OptiBond FL in Class V lesions at 12 and 13 years, respectively.[94,169,170] If a dentin adhesive system does not adhere intimately to the dentin substrate, an interfacial gap eventually develops; bacteria may penetrate through this gap and cause demineralization of the substrate around the restoration margins.[295] Despite the probability of an incomplete dentin margin seal, Class V clinical studies using etch-and-rinse dentin adhesive systems reported no findings of pulpal inflammation or necrosis.[76,154]

Important to point out is that in vitro microleakage measurements for a specific adhesive material often do not correlate with clinical behavior of the same material.[296] Additionally, silver nitrate penetration may be a particularly demanding test of marginal seal because silver ions are smaller than the bacteria that colonize the oral cavity.[297]

When all margins of the restoration are in enamel, the quality and integrity of the bonds remain unchanged with time, at least in vitro.[298] Degradation of the bonds might result from hydrolysis, which occurs either in the adhesive resin or in the collagen fibril that are not fully enveloped by the adhesive in the hybrid layer, especially when margins are in dentin.[298-300] The degree of degradation of the bonded interface is more pronounced with simplified adhesives (i.e., one-step self-etch and two-step etch-and-rinse



• **Fig. 5.22** Nanoleakage under the electron microscope. A, Spotted pattern in the hybrid layer formed by a one-step self-etch adhesive under the transmission electron microscope. B, Reticular pattern and "water trees" in the adhesive layer formed by a one-step self-etch adhesive under the scanning electron microscope in backscattered mode.

adhesives). A nearly 50% reduction in bond strengths of the 24-hour control has been reported at 1 year with a one-step SEA.[299]

The term "nanoleakage" has been used to describe small porosities in the hybrid layer or at the transition between the hybrid layer and the mineralized dentin that allow the penetration of minuscule particles of a silver nitrate dye.[301] When ammoniacal silver nitrate is used, silver deposits penetrate the hybrid layer formed by either etch-and-rinse or self-etch adhesive materials.[302] Penetration of ammoniacal silver nitrate results in two distinct patterns of nanoleakage: (1) a spotted pattern in the hybrid layer of self-etch adhesives, which might be caused by incomplete resin infiltration (Fig. 5.22A), and (2) a reticular pattern that occurs in the adhesive layer, most likely caused by areas where water was not totally removed from the bonding area (see Fig. 5.22B).[303] The term "water trees" is associated with porosities in the polymerized adhesive layer.[303] Silver uptake in hybrid layers formed by one-step SEAs is associated with areas of increased permeability within the polymerized resin from which water was incompletely removed. The residual water prevents complete polymerization.[303] There has been, however, some discussion regarding the clinical relevance of nanoleakage and the methods used to assess it.[304]

Biocompatibility

Besides demineralizing the dentin surface, phosphoric acid removes the smear layer and opens the orifices of the tubules (see Fig. 5.12).[67,305] Despite past apprehension about potential acid penetration into the dentin tubules and pulp space, the interaction of etchants with dentin is limited to the superficial 1 to 7 µm.[67,167] It is unlikely that the acid is directly responsible for any injury to the pulp.[306,307] Acid penetration occurs primarily along the tubules, with penetration of intertubular dentin occurring at a lower rate.[307,308] The effects of etchants on dentin are limited by the buffering effect of hydroxyapatite and other dentin components, including collagen, which may act as a barrier that reduces the rate of demineralization.[309,310] Marshall et al. elucidated the importance of pH with regard to the effects of acids on dentin surfaces.[307] Etching rates increase dramatically with lower pH. Small differences in pH between acidic gels of similar phosphoric acid concentration may be responsible for distinct depths of dentin demineralization. Manufacturers add thickeners to facilitate handling and other modifiers (e.g., buffers, surfactants, and colorants) to their etching gels, which may contribute to that phenomenon.

Several early studies suggested that acidic components included in restorative materials such as silicate cements would trigger adverse pulp reactions.[157,311] For several decades, the development of adhesive systems was limited by the belief that acids applied to dentin during restorative procedures caused pulpal inflammation. The use of bases and liners was considered essential to protect the pulp from the toxicity of restorative materials. This concept has, however, changed over the years.[286,287,312,313]

Dentin adhesive systems are well tolerated by the pulp-dentin complex in the absence of bacterial infection.[313] To prevent bacterial infection, restorations must be hermetically sealed. The pulp response to dentin adhesives, when teeth are restored in an ideal clinical environment, has been studied using histologic assessment of animal pulps or in human premolars extracted for orthodontic reasons and in third molars extracted for surgical reasons.[288,289,312-319] Some clinical studies also have reported normal pulp responses after the application of adhesive on the dentin-pulp complex when the pulp is macroscopically exposed, although reports involved only one tooth.[320,321] Another study showed that the newest dentin adhesive systems are not harmful when applied to exposed pulps.[322] Several reports have shown however that etching the pulp and applying a dentin adhesive directly on the exposed pulp tissue results in severe inflammation and eventual formation of pulpal abscesses.[323-326] The solution for this disparity would be long-term follow-up of patients in whom the pulp was treated with acid and adhesive. Ethical concerns do not allow the routine use of pulpal etching in patients. It is known however that the thicker the remaining dentin left between the pulpal aspect of the preparation and the pulp, the better the prognosis for that specific pulp.[324] The concept of pulp capping remains a controversial topic.

Adverse pulpal reactions after a restorative procedure are not caused by the material used in that procedure but by bacteria remaining in, or penetrating, the preparation. In some cases, adverse reactions are caused by a combination of factors, as follows:

1. Bacterial invasion of the pulp, either from the tooth preparation or from an existing carious lesion[286,327]
2. Bacterial penetration into the pulp caused by a faulty restoration[286]
3. Pressure gradient caused by excessive desiccation or by excessive pressure during cementation[328,329]
4. Traumatic injuries[330,331]

5. Iatrogenic tooth preparation—excessive pressure, heat, or friction[328,332]
6. Pulpal temperature rise induced during the polymerization of composites, especially flowable composites[333,334]
7. Stress derived from polymerization contraction of composites and adhesives[335]
8. Unpolymerized resin monomers[336]

With regard to the biocompatibility issue, tooth preparations with enamel peripheries are important. When all margins are in enamel, polymerization shrinkage stresses at the interface are counteracted by strong enamel adhesion. Marginal gaps are less likely to form, and the restoration is sealed against bacteria.

Relevance of In Vitro Studies

One of the major concerns with laboratory bond strength testing is the wide range of results obtained for the same material in different testing sites. It is not an uncommon occurrence for the same dentin adhesive system to have average shear bond strengths of 20 MPa in one laboratory and bond strengths less than 10 MPa in another.[84,142] Also, dentin bond strengths for a specific dentin adhesive vary depending on the specific test used—microtensile, shear, or microshear.[337] Some perplexity exists that no correlation can be established between bond strength and degree of resin penetration into the hybrid layer.[338,339] To illustrate this discrepancy, some reports have suggested that dentin adhesives do not penetrate the whole depth of the demineralized dentin layer but still result in bond strengths greater than 20 MPa.[340,341] Intuitively, one would expect an inverse relationship between bond strength and microleakage, but that relationship has not been confirmed.[342]

The microtensile bond strength testing methodology has become popular in recent decades.[343] This method allows for the assessment of bond strengths using bonded surfaces with a cross-sectional area in the range of 1 to 1.5 mm^2 or even less (Fig. 5.23). Microtensile testing has several advantages over conventional shear and tensile bond strength methods for the following reasons:

1. It permits the use of only one tooth to fabricate several bonded dentin-resin rods.
2. It allows for testing substrates of clinical significance, such as carious dentin, cervical sclerotic dentin, and enamel.[344]
3. It results in fewer defects occurring in the small-area specimens, as reflected in higher bond strengths.[345]
4. It allows for the testing of regional differences in bond strengths within the same tooth.[346]

However, the results of the microtensile bond strength methodology vary in function of several testing parameters, including storage time, the cross-sectional shape of the microspecimen, remaining dentin thickness, and jig design.[347,348]

Although in vitro bond strength studies are only rough categorizing tools for evaluating the relative efficacy of bonding materials, they are excellent tools for screening new materials and for comparing the same parameter among different adhesive systems.[349] The results of in vitro bond strength tests have been validated with clinical results because improvements seen in the laboratory environment from the earlier generations to contemporary adhesive systems have been confirmed in clinical trials.[154]

A systematic analysis of the correlation between in vitro marginal adaptation and the outcome of clinical trials of Class V restorations revealed that the correlation is weak and only present for studies that used the same composite for the in vitro and in vivo evaluation.[350] Another systematic review found a correlation between bond strength data and clinical retention rates of Class V restorations, specifically when the bond strength specimens were aged prior to testing.[349] The clinical parameter in Class V restorations that is more directly related to bond strength data is marginal adaptation.[338] Clinical studies with dentin adhesive systems are expensive for manufacturers and take at least 18 months to 3 years. Cost is a major concern, in part because of the constant developments in the area of adhesion, making new materials quickly obsolete. No financial incentive exists for the manufacturer to invest in a clinical study of a material that may not be on the market by the time the study is concluded. Consequently, in vitro studies are still used predominantly by manufacturers to anticipate the clinical behavior of their materials.

Several factors contribute to the questionable use of in vitro tests to predict clinical behavior. Among others, variables including age and storage conditions of the teeth used, dentin depth, degree of sclerosis, tooth surface to be bonded, dentin roughness, and type of test used frequently are not controllable.[4,5,351,352] According to some authors, one of the major drawbacks of laboratory bond strength testing is the usual lack of simulated pulpal pressure to replicate the pulpal pressure that occurs in vivo. Other authors have reported, however, that the pulpal pressure does not interfere significantly with bond strength results.[353]



• **Fig. 5.23** Preparation of specimens for microtensile bond strength testing.

Clinical Performance

Several clinical factors may influence the success of an adhesive restoration. The mineral content of dentin increases in different situations, including aged dentin, dentin beneath a carious lesion, and dentin exposed to the oral cavity in NCCLs in which the tubules become obliterated with tricalcium phosphate crystals.[343,354,355] The dentin that undergoes these compositional changes is called sclerotic dentin and is much more resistant to acid etching than "normal" dentin.[163,356] Consequently, the penetration of a dentin adhesive is limited.[356-358] Irrespective of the use of a etch-and-rinse or a self-etch technique, bonding to sclerotic dentin in NCCLs has resulted in low bond strengths.[355,359] Additionally, the clinical effectiveness of dentin adhesives is less in sclerotic cervical lesions than in normal dentin.[360,361] Nevertheless, some specific dentin adhesives may perform better in sclerotic dentin than in normal dentin.[362]

Some evidence suggests that masticatory forces not only might cause NCCLs but also might contribute to the failure of Class V restorations.[359,363,364] Bruxism or any other eccentric movement may generate lateral forces that cause concentration of stresses around the cervical area of the teeth. Although this stress may be of very low magnitude, the fatigue caused by cyclic stresses may cause failure of bonds between resin and dentin. Considering the challenges to successfully bond to NCCLs, these have been used often as ultimate clinical test to assess adhesives.

The type of composite used might play an important role in clinical longevity of Class V, as well as Class II, restorations.[350,365] Composites shrink as they polymerize, but the amount of shrinkage

depends on the inorganic load of each specific composite. Microfilled composites have a low elastic (or Young) modulus, which means that they are better able to relieve stresses caused by polymerization or by tooth flexure.[366,367] Materials that have a higher Young modulus do not relieve stresses by flow; they are unable to compensate for the stresses accumulated during polymerization. These stresses subsequently might be transferred to the adhesive interface and cause debonding. As adhesives have improved however, restorative material stiffness might be less important. A 2-year clinical study of a three-step etch-and-rinse adhesive showed no difference in retention rates of Class V composite restorations based on stiffness of the restorative material.[368] Another clinical study reported that composite stiffness did not affect the clinical longevity of cervical composite restorations.[369]

A systematic review published in 2005[370] analyzed the literature published from January 1998 to May 2004 on the clinical effectiveness of adhesives to restore NCCLs to determine if simplified adhesives were as clinically effective as conventional three-step adhesives in terms of retention rate. Although GIC-based materials are usually outperformed by adhesively bonded restorations in in vitro studies, their clinical retention to tooth structure is more effective and durable than any other type of adhesive material. Three-step etch-and-rinse adhesives and two-step SEAs showed a clinically reliable clinical performance. The clinical effectiveness of two-step etch-and-rinse adhesives was less favorable, while the worst clinical performance was found for one-step SEAs.

A meta-analysis published in 2010, for which 50 clinical studies and 40 adhesive systems matched the inclusion criteria, reported that the clinical performance of dentin adhesives in NCCLs up to 3 years is significantly influenced by the type of adhesive system used and whether or not the dentin/enamel is roughened.[371] GIC-based materials, two-step SEAs, and three-step etch-and-rinse adhesives performed better than adhesives of the other categories. The use of rubber dam isolation did not increase the chance of survival of restorations in this meta-analysis.

Peumans et al.[174] conducted a systematic review in which they analyzed annual failure rates in NCCLs in publications, including AADR/IADR abstracts, up to 2013.[174] Dentin adhesive materials were divided into six main categories: three-step etch-and-rinse adhesives, two-step etch-and-rinse adhesives, two-step SEAs, one-step SEAs, GIC-based materials, and self-adhesive composites. Both classes of SEAs were further subdivided as "mild" and "intermediately strong" (pH ≥1.5) and "strong" (pH <1.5).[174] GIC-based materials had a 2% annual failure rate (AFR), which was the lowest for all adhesive classes analyzed. The AFR for two-step mild SEAs was 2.5% and for three-step etch-and-rinse adhesives was 3.1%. Higher AFRs were measured for one-step strong SEAs (5.4%), two-step etch-and rinse adhesives (5.8%), and two-step strong SEAs (7.9%). Selective enamel etching did not significantly influence the retention rate of SEAs.

Another recent meta-analysis analyzed prospective clinical trials on restorations of NCCLs with an observation period of at least 18 months.[132] Eighty-one studies involving 185 experiments for 47 adhesives matched the inclusion criteria. The respective authors concluded that 12.3% of the cervical restorations were lost, 27.9% exhibited marginal discoloration, and 34.6% exhibited marginal deterioration after 5 years of service. The clinical index[a] was 17.4% of failures after 5 years and 32.3% after 8 years; however, a great variability was measured for retention loss and marginal discoloration. This meta-analysis concluded that roughened dentin and rubber-dam use resulted in a statistically significantly higher retention rate than unprepared dentin or no rubber dam. Enamel beveling however had no influence on any of the examined variables. Overall, one-step SEAs had a significantly worse clinical index than two-step SEAs and three-step etch-and-rinse adhesives.

Because universal adhesives are fairly recent materials, systematic reviews of clinical studies are not yet available. Two clinical trials have reported acceptable clinical effectiveness for the first universal adhesive introduced to the market, Scotchbond Universal Adhesive (Single Bond Universal in some regions) (3M Oral Care).[372,373] A 3-year clinical study compared this adhesive applied in different adhesion strategies (self-etch, selective enamel etch, etch-and-rinse applied on moist dentin, etch-and-rinse applied on air-dried dentin) and concluded that while there was no statistical difference among bonding strategies, there were signs of marginal degradation when the universal adhesive was applied in full self-etch mode.[372]

In summary, results from recent clinical studies and systematic reviews suggest that chemical bonding to dentin calcium may play a crucial role in stable adhesion to NCCLs, as GIC-based materials and mild two-step SEAs result in excellent retention rate. Roughening dentin in NCCLs may increase retention rate, whereas enamel beveling does not seem to be a significant factor. Although there is conflicting data on the influence of rubber-dam isolation, its use may decrease the failure rate of restorations in NCCLs. GIC-based materials are still the standard to restore NCCLs in terms of retention rate.

Incompatibility Issues With Self-Cure and Dual-Cure Composites

Chemically activated and dual-activated composites still have significant use in restorative dentistry, especially in areas of preparations with limited access to light. Examples include crown foundations, bonded posts, and ceramic and composite inlays, onlays, and crowns.

Several studies have reported incompatibility between specific light-cured adhesives and chemically activated composites.[209,374,375] In one study, Prime & Bond NT, which contains PENTA, a monomer with an acidic phosphate group, did not bond to a self-cured composite unless the adhesive was mixed with a sulfinic acid activator.[375] In another study, the reductions in mean bond strength of adhesives decreased by 45% to 91% when self-cured composite was used instead of light-cured composite.[374] The most drastic reduction was associated with Prime & Bond NT. The inhibition of polymerization of the self-cured composites by adhesives with specific compositions seems to be related directly to the pH of the adhesive.[374] One-Step, which caused the least reduction in bond strengths between self-cured and light-cured composite in that specific study, was the adhesive with the highest pH. Prime & Bond NT had the lowest (more acidic) pH.

Similarly, an adverse chemical interaction occurs between catalytic components of chemically cured composites and acidic one-step self-etch adhesives.[208,209] In contrast, despite the acidity of their primers, some two-step self-etch adhesives might be compatible with self-cure and dual-cure composites, owing to the presence of a thick resin layer that is less permeable and more hydrophobic than the layer formed with all-in-one systems.[208,209]

The issue related to the acidity of one-step adhesives also applies to recently introduced universal adhesives. This issue is particularly relevant because the indication for universal adhesives has been expanded to indirect restorations, including those made of

[a]Clinical index = (4xR + 2xMD + 1xMI) / 7 (R = Retention; MD = marginal discoloration; MI = detectable margins).

glass-matrix ceramics and oxide-based ceramics, without the need for additional primers.[376-378] While some current universal adhesives contain different functional monomers, the compatibility of universal adhesives with chemically activated and dual-cured resin-based composite materials has not been thoroughly investigated, especially taking into consideration that the pH of universal adhesives varies from to 1.6 to 3.2.[379,380] Among others, Clearfil Universal Bond (pH = 2.3, Kuraray Noritake Dental Inc.), One Coat 7 Universal (pH = 2.8, Coltene), Prime & Bond Elect (pH = 2.5, DENTSPLY Sirona), and Scotchbond Universal Adhesive (pH = 2.7, 3M Oral Care) are examples of universal adhesives for which the respective manufacturer recommends mixing the universal adhesive with a chemical activator when used with composite materials that contain a chemical activator, such as dual-cured buildup composite materials and dual-cured resin cements. AdheSE Universal (pH = 2.5–3, Ivoclar Vivadent) and All-Bond Universal (pH = 3.1–3.2, Bisco Inc.) are examples of universal adhesives for which the respective manufacturer does not recommend adding a chemical activator due the relatively high pH of the adhesive solution. The manufacturer of the universal adhesive Xeno Select (DENTSPLY Sirona, also known as Prime & Bond One Select in some countries) does not recommend this adhesive for any indirect restorations due to its low pH (1.6).[380]

All current universal adhesives that are recommended for indirect restorations resulted in high bond strengths to sandblasted zirconia if used in light-cured mode (without adding a dual-cured activator) with a dual-cured amine-free resin cement (NX3, Kerr Corporation).[380] This may explain why the manufacturer of Scotchbond Universal Adhesive recommends combining the adhesive with its proprietary amine-free dual-cured resin cement without the need to mix the adhesive with the respective chemical activator. The same adhesive must be mixed with the chemical activator if it is used with a conventional dual-cured resin cement based on the peroxide-amine redox system.

Expanded Clinical Indications for Dentin Adhesives

Desensitization

Dentin hypersensitivity is a common clinical condition that is difficult to treat because the treatment outcome is not consistently successful. Most authorities agree that the hydrodynamic theory best explains dentin hypersensitivity.[357] The equivalency of various hydrodynamic stimuli has been evaluated from measurements of the fluid movement induced in vitro and relating this to the hydraulic conductance of the same dentin specimen.[381] There are other mechanisms responsible for tooth sensitivity including those triggered by dental whitening agents.[382]

Patients may complain of discomfort when teeth are subjected to temperature changes, osmotic gradients such as those caused by sweet or salty foods, or even tactile stimuli. The cervical area of teeth is the most common site of hypersensitivity. Cervical hypersensitivity may be caused not only by chemical erosion but also by mechanical abrasion or even occlusal stresses.[383,384]

Theories about the transmission of pain stimuli in dentin sensitivity suggest that pain is amplified when the dentinal tubules are open to the oral cavity.[385,386] Dentin hypersensitivity can be a major problem for periodontal patients who frequently have gingival recession and exposed root surfaces. The relationship between dentin hypersensitivity and the patency of dentin tubules in vivo has been established; occlusion of the tubules seems to decrease that sensitivity.[387]

The use of dentin adhesives to treat hypersensitive root surfaces has gained popularity.[27,28] Reductions in sensitivity can result from formation of resin tags and a hybrid layer[388] when a dentin adhesive is used. The precipitation of proteins from the dentinal fluid in the tubules may also account for the efficacy of desensitizing solutions.[389] Other factors may be involved in the action of dentin desensitizing solutions.[390] The primer of the multibottle adhesive system All-Bond 2 has a desensitizing effect, even without consistent resin tag formation.[391] In a clinical study using the primer of the original GLUMA adhesive system (an aqueous solution of 5% glutaraldehyde and 35% HEMA, currently marketed as GLUMA Desensitizer [Heraeus Kulzer GmbH]), the desensitizing solution was applied to crown preparations.[392] The authors concluded that GLUMA primer reduced dentin sensitivity through a protein denaturation process with concomitant changes in dentin permeability. Glutaraldehyde has long been used as a fixative that cross-links proteins.[393] This theory has been supported by studies using confocal microscopy, which found the formation of transversal septa occluding the dentinal tubules after application of GLUMA Desensitizer.[394] Another study evaluated dentin permeability in dogs up to 3 months. At the end of this period, GLUMA Desensitizer had the lowest permeability value, providing a longer lasting tubule-occluding effect.[395] Most desensitizing agents, however, are able to greatly decrease dentin permeability.[396]

The same glutaraldehyde-based desensitizing agent has been suggested as a rewetting agent on etched dentin to help prevent postoperative sensitivity under posterior composite restorations.[397,398] In spite of the favorable in vitro bond strengths, one clinical pilot study found that the operative technique might be more relevant to prevent postoperative sensitivity than the use of the glutaraldehyde-based desensitizer.[399] A randomized clinical trial of Class I restorations in molars and premolars included a glutaraldehyde-containing two-step etch-and-rinse adhesive, a glutaraldehyde-containing one-step SEA, and a glutaraldehyde-free two-step etch-and-rinse adhesive. There was no difference in postoperative sensitivity after 48 hours and after 7 days.[400]

The use of a dentin desensitizer before cementing full-coverage crowns is supported by studies that showed dentin-desensitizing solutions do not interfere with crown retention, regardless of the type of luting cement used.[401,402]

Adhesion to Root Canal Dentin

Adhesive strategies used in root dentin are similar to those used in coronal dentin. There are however differences in terms of bonding substrate and the restorative technique.

A. Substrate for Adhesion

Dentin located in the root walls of teeth that have been endodontically treated contains 9% less moisture than root dentin of vital teeth.[403] That is a positive feature for dentin adhesion. However, the area occupied by the more calcified transparent dentin in the root, also called sclerotic dentin,[404,405] widens with aging. Additionally the apical area of root dentin has fewer tubules than the occlusal part.[406]

The cutting instruments and cutting speed used for root dentin during endodontic therapy, as well as the irrigation methods (including chelating and deproteinizing solutions), are different from those used for coronal dentin, which results in different composition of the respective smear layers and dentin substrate.[407,408] Differently from the smear layer formed during a conventional coronal preparation, the morphology of the smear layer formed in the root canal walls includes two zones. The deepest part of endodontic smear

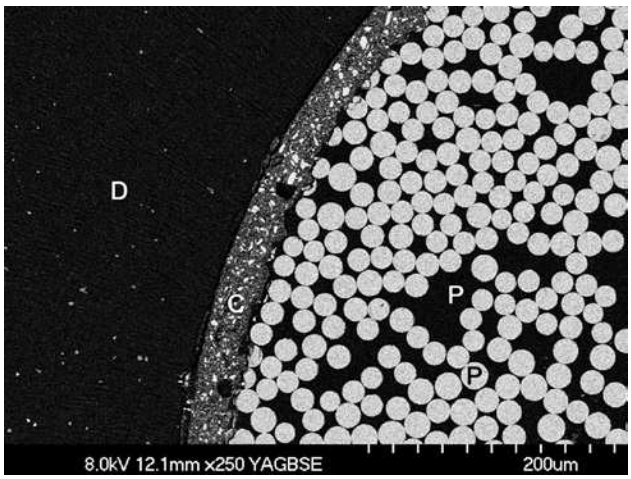
layer is 1 to 2 µm thick and similar to that formed in coronal dentin. The most superficial zone forms smear plugs into the dentinal tubules to a depth of at least 10 µm,[409] which are not as accessible to dentin adhesives. Furthermore, there is a great variability in the composition of this deepest smear layer[407,410] aside from the presence of more organic tissue remnants and residual gutta-percha and endodontic sealer in some occasions.[410]

B. Restorative Technique

As discussed, stress relief within a three-dimensional bonded restoration is limited by its C-factor.[92] While the magnitude of the C-factor typically varies from 1 to 5 in direct coronal restorations, the estimated C-factor in the root canal may exceed 200, which is extremely unfavorable for restorative procedures that involve composite-based materials.[411] The polymerization shrinkage stress also depends on the degree of compliance (or deformation) of the substrate, which provides stress relief for luted restorations.[412] As this compliance is minimal in the root canal space, all root canal luting procedures that involve radical polymerization resins are challenging.

The concept of anatomic or relined fiber post (a glass fiber post relined with composite to reduce the volume of resin luting cement to retain the glass fiber post in the root canal)[413-415] has been recently advocated as a technique to increase the pushout bond strengths and decrease the polymerization stress associated with increased volume of resin cement. This concept however does not seem to take into account the slow chemical polymerization of resin luting cements used in the root canal. The use of chemically cured composites, as well as the utilization of hand-mixed resin materials, may partially compensate for the polymerization stresses that develop in the interface formed by root dentin and the composite material.[412,416] The polymerization stresses decrease and the development of the curing stress decreases with increasing of the layer thickness of chemically cured composites up to a thickness of 2.7 mm.[412] There is no strong evidence to recommend this technique over the use a nonrelined glass fiber post.

Although both the etch-and-rinse and the self-etch strategies are still often used to bond to root canal dentin, more recently the use of self-adhesive resin cements (i.e., no separate dentin conditioner or adhesive required) has gained popularity for their user-friendliness and better sealing ability than conventional adhesive systems (Fig. 5.24).[417,418] A systematic review of in vitro studies also suggested that self-adhesive resin cements may improve the retention of glass fiber posts into the root canal,[419] while being less technique sensitive.[420]



• **Fig. 5.24** Scanning electron micrograph (backscattered mode) of a dentin/self-adhesive resin cement/fiber post interface. *D*, Root dentin; *C*, self-adhesive resin cement; *P*, fiber post.

There is some controversy regarding the use of eugenol-containing sealers and their influence on the retention of luted posts, as eugenol is a phenolic component that can scavenge free radicals and delay or inhibit the polymerization of resin-based materials.[421] Some authors have reported lower bond strengths to root canal dentin when eugenol-based sealers were used with the root canal treatment.[422-424] Other authors have reported no significant differences in bond strengths to root canal dentin when eugenol-based sealers were used.[425,426]

Unfilled controls revealed significantly higher retentive post strengths compared to root-filled groups indicating that remnants of sealer might hamper adhesion inside the root canal.[425,427] When the post is luted 1 week after the completion of the root canal treatment, bond strengths to root canal dentin are not compromised regardless of the presence of eugenol in the endodontic sealer.[428,429]

Indirect Adhesive Restorations

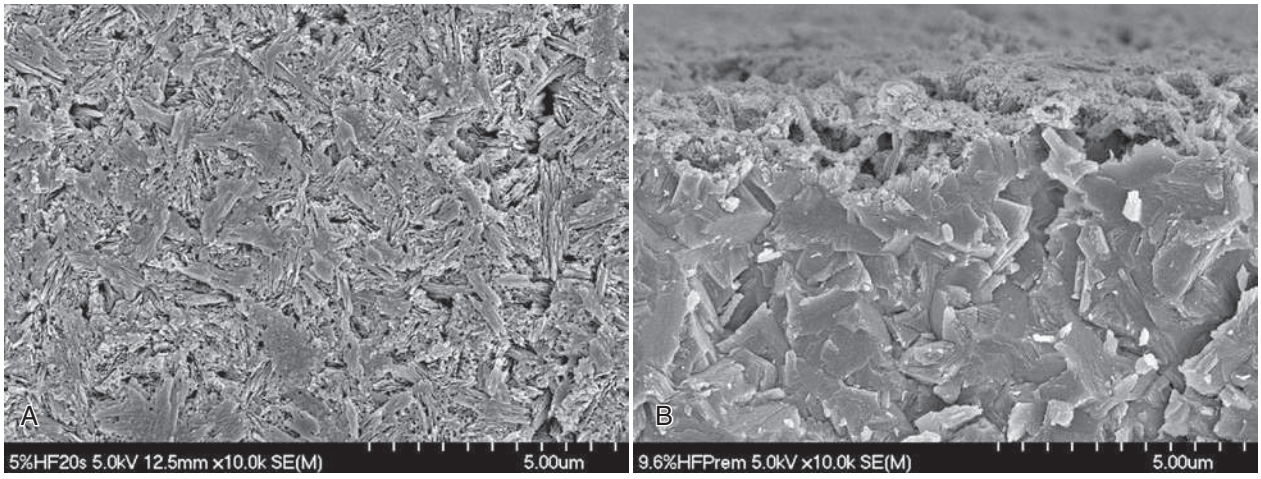
Some current dentin adhesive systems bond to various substrates besides dentin.[7,418,430-433] Developments in adhesion technology have led to new indications for bonding to tooth structure, such as indirect ceramic and resin-based restorations (crowns, inlays, onlays, and veneers). The use of a dentin/enamel adhesive system in conjunction with a resin cement provides durable bonding of indirect restorations to tooth structure.[13,431]

A. Glass-Matrix Ceramics

Silica-based or glass-matrix ceramics[434] are still extensively used for porcelain veneers, inlays/onlays, and single crowns. Examples of glass-matrix ceramics are feldspathic porcelain, leucite-reinforced, and lithium disilicate–reinforced ceramics. Glass-matrix ceramic restorations must be etched internally with ~5% to ~10% hydrofluoric acid (HF) for 20 to 180 seconds [435-437] to create retentive microporosities (Fig. 5.25) analogous to those created in enamel by phosphoric acid etching.[438] HF must be rinsed off thoroughly with running water.[439] The adhesion of a resin luting cement to glass-matrix ceramics is achieved through a combination of mechanical retention from HF etching and chemical adhesion provided by the silane coupling agent to achieve durable adhesion.[13,431] Etched porcelain is an inorganic substrate, which silane makes more receptive to organic materials, the adhesive system, and resin cement.

Silane coupling agents were introduced in 1952 to bond organic with inorganic substances.[440] In 1962, this technology was transferred to dentistry to couple inorganic filler particles with Bis-GMA resin to form a composite resin.[441-444] After rinsing off the HF and air-drying, a silane coupling agent is applied on the etched glass-matrix ceramics intaglio surface and air-dried. The silane acts as a primer because it modifies the surface characteristics of etched glass-matrix ceramics. The application of a silane solution on HF-etched surfaces increases bond strengths between resin-based composite materials and glass-matrix ceramics up to 50% compared to HF alone, including bond strengths obtained with feldspathic porcelain and lithium disilicate–reinforced ceramic blocks.[437,442]

With the introduction of universal one-bottle adhesives and new silane solutions containing 10-MDP, some manufacturers do not recommend the application of a separate silane coupling agent to HF-etched glass-matrix ceramic surface. Clearfil Universal Bond and Scotchbond Universal Adhesive are examples of universal adhesives that contain a silane in their composition; however, the combination of resin monomers with a silane in the same solution



• **Fig. 5.25** Scanning electron micrograph of lithium disilicate–reinforced ceramics (IPS e.max CAD, Ivoclar Vivadent). A, Top view, etched with 5% hydrofluoric acid for 20 seconds. B, Lateral view, etched with 9.6% hydrofluoric acid for 20 seconds.

remains controversial.[378,445] Bond strengths to lithium disilicate–reinforced and leucite-reinforced ceramics are lower when a silane-containing universal adhesive is used compared to the application of a silane solution followed by a conventional multistep adhesive or a universal adhesive.[378,446] Contact angle and bond strength measurements have demonstrated that a silane primer may not be compatible mixed with resin monomers in the same solution.[445]

Some clinicians use sandblasting with aluminum oxide particles (airborne-particle abrasion) in the internal surface of glass-matrix ceramics restorations to roughen the intaglio surface. Mean bond strengths decrease however when hydrofluoric acid etching is not used.[437] Additionally airborne-particle abrasion of glass-matrix ceramics with alumina particles results in immediate lower intrinsic strength[447] triggered by the reduction in flexural strength of feldspathic porcelains[448] as well as of lithium disilicate–reinforced ceramics.[431]

B. Polycrystalline Ceramics

The discovery of a stress-induced transformation from the metastable tetragonal form to the stable monoclinic form *(t→m)*[449-451] has led to a new class of strong, tough, dense, relatively flaw-tolerant oxide-based or polycrystalline ceramics known as zirconium oxide or zirconia.[449-451] This *t→m* phase transformation can occur in the vicinity of a propagating crack, causing an increase in volume, thereby closing the crack tip and preventing further crack propagation.[450,451] These high-strength ceramic materials have become extremely popular in dentistry, especially yttria-stabilized tetragonal zirconia polycrystal (Y-TZP) and ceria-stabilized tetragonal zirconia polycrystal/alumina (Ce-TZP/Al_2O_3).[452-454]

HF etching does not improve bond strengths to oxide-based or polycrystalline ceramics,[434,435,455-457] since these substrates do not contain a glass matrix. Accordingly, other surface conditioning methods have been used in the last two decades to improve bond strengths of resin luting cements to polycrystalline ceramics.[435,456,458,459] Some of the newest protocols include airborne-particle abrasion, tribochemical silica coating, and primers or silanes mixed with functional monomers such as the phosphate ester monomer 10-MDP.[430,459-461] Recent meta-analysis and systematic reviews of in vitro studies[430,458,459] concluded that the combination of mechanical and chemical pretreatment is recommended for bonding to zirconia. Although the luting material may not be a determining factor after aging, as long as resin-based cement is used, adhesion to zirconia is increased after mechanical and chemical surface pretreatment and the use of a 10-MDP–based resin cement. The recommended chemical treatment of the Y-TZP intaglio is the application of a 10-MDP–based primer or a 10-MDP–based universal adhesive solution.[376,377,430,460,462] Chemical bonding between 10-MDP solutions and Y-TZP substrates was confirmed with time-of-flight secondary ion mass spectroscopy.[460] When dual-cure resin cements were used on Y-TZP surfaces that had not been air-abraded, higher bond strengths were achieved when a specific 10-MDP–containing zirconia primer was used,[462] confirming its chemical bonding ability. Nevertheless, micromechanical retention plays a more prominent role in adhesion to Y-TZP than chemical bonding.[463]

As surface contamination and aging in water have negative effects on adhesion to zirconia, the residual alumina particles and other surface contamination must be removed prior to adhesive luting.[459] The protocol for removing surface contaminants from intaglio surfaces of Y-TZP restorations includes cleaning the restoration in distilled water or ethanol for at least 3 minutes in an ultrasonic device, followed by rinsing with copious amounts of water spray and drying the surface with a strong stream of oil-free air.[464]

The durability of Y-TZP–based biomaterials has been questioned. In 1981 a phenomenon currently known as low-temperature degradation of Y-TZP was first described for applications near 250°C.[465] It was revealed that Y-TZP could undergo a slow superficial *t→m* transformation in a humid atmosphere followed by microcracking and loss in strength. In 1999 the *t→m* transformation of a 3Y-TZP from 70°C to 130°C in water and in steam was reported, confirming the susceptibility of Y-TZP to degradation in humid environments.[466] Stress corrosion by water molecules was also described as responsible for slow crack propagation in Y-TZP under cyclic loading.[467] This slow aging under humid environment was associated with roughening and microcracking, which led to increased wear and the release of wear debris in the body from hip replacement zirconia prostheses.[468]

Regarding the oral environment, Y-TZP–based materials may undergo fatigue and subcritical crack growth under functional loading with subsequent reduction in flexural strength.[469,470] New

ceria-stabilized zirconia (Ce-TZP) materials do not undergo the same reduction in flexural strength compared to Y-TZP.[471] Significant decrease in bond strengths were noticed after aging through water storage at 37°C for 3 days or 150 days and 37,500 thermal cycles, regardless of the surface conditioning method (airborne-particle abrasion or tribochemical silica coating abrasion) and the luting system used.[376]

The clinical relevance of these structural changes in Y-TZP used in dentistry is not clear, as most in vitro studies have tested Y-TZP without a glaze coating.[472,473] The Y-TZP structure is not usually exposed directly to the oral environment except for zirconia abutments[474] and for glazed monolithic zirconia restorations that are adjusted prior to cementation.

C. Resin-Matrix Ceramics

Resin-based CAD/CAM materials for indirect restorations have become recently popular. Ambarino High-Glass (Creamed GmbH & Co. Produktions, Marburg, Germany), Brilliant Crios (Coltene), Ceramill COMP (Amann Girrbach AG, Herrschaftswiesen, Austria), Cerasmart (GC Corporation, Tokyo, Japan), and Lava Ultimate (3M Oral Care) are examples of CAD/CAM resin blocks currently available. These new resin-based CAD/CAM materials have a much lower modulus of elasticity compared to CAD/CAM ceramic materials.[446,447] It is recommended to treat the intaglio surface with airborne particle abrasion without any chemical etching.[475,476] Air abrasion with alumina particles improves fracture resistance of Lava Ultimate compared to polished specimens,[447] which may be a result of better mechanical interlocking of the bonding resins into the resin microporosities and better stress distribution at higher loads. Most universal adhesives also seem to bond effectively to air-abraded resin-based CAD/CAM materials.[380]

Hybrid resin-ceramic materials consisting of a polymer-infiltrated ceramic network (PICN) have been recently introduced with the goal to more closely mimic the physical properties of the natural tooth compared to other tooth-colored restorative materials.[477] Vita Enamic (VITA Zahnfabrik H. Rauter GmbH & Co. KG, Bad Säckingen, Germany) is a PICN containing an aluminum oxide–enriched feldspathic porcelain (86 wt%) into which 14 wt% urethane dimethacrylate and triethylene glycol dimethacrylate polymer material is injected.[478] The bond strengths obtained with HF-etched PICN materials are similar to those obtained with lithium disilicate–reinforced ceramics.[475]

Indirect tooth-colored restorations are luted with low-viscosity composite resin materials. Many of these resin cements are dual cured—that is, they polymerize both chemically and by light activation. Some materials marketed as "dual cure" may not polymerize efficiently in the absence of a curing light.[479-481] Self-adhesive cements, a recent category of resins, have become very popular to cement oxide–based ceramic restorations. Self-adhesive cements are dual-cured phosphate monomer-based resin cements (BisCem, Bisco Inc.; G-CEM LinkAce, GC Corporation; Maxcem Elite, Kerr Corporation; Panavia SA Cement Plus, Kuraray Noritake Dental Inc.; RelyX Unicem 2, 3M Oral Care; SmartCem2, DENTSPLY Sirona; Solocem, Coltene; SpeedCEM Plus, Ivoclar Vivadent) that do not require any pretreatment of the tooth substrate. The acidic phosphate groups react with the filler and simultaneously etch enamel or dentin in the same manner as do self-etch adhesives. A chemical interaction between self-adhesive cements and hydroxyapatite has been reported,[481,482] but this interaction depends of the activation mode and the specific self-adhesive resin cement tested.[482] The pH of some self-adhesive cements increases from 1 to 5 or 6 during their acid-base setting reaction.[483] The setting pH profiles of self-adhesive resin cements depend on the brand and mode of cure. In spite of their dual-curing ability, the physical properties improve significantly when light activated.[482,484] Self-adhesive resin cements result in lower bond strengths to etched glass-matrix ceramics than traditional resin cements.[437]

Summary

Reliable bonding of resins to enamel and dentin has revolutionized the practice of operative dentistry. Improvements in dentin bonding materials and techniques are likely to continue. Even as the materials themselves become better and easier to use however, proper attention to technique and a good understanding of the bonding process remain essential for clinical success, with emphasis in evidence-based dentistry.

References

1. Packham DE: Adhesion. In Packham DE, editor: *Handbook of adhesion*, Essex, UK, 1992, Longman Scientific & Technical, pp 18–20.
2. Akinmade AO, Nicholson JW: Glass-ionomer cements as adhesives: part I. Fundamental aspects and their clinical relevance. *J Mater Sci Mater Med* 4:95–101, 1993.
3. Allen KW: Theories of adhesion. In Packham DE, editor: *Handbook of adhesion*, Essex, UK, 1992, Longman Scientific & Technical, pp 473–475.
4. Söderholm K-JM: Correlation of in vivo and in vitro performance of adhesive restorative materials: a report of the ASC MD156 Task Group on test methods for the adhesion of restorative materials. *Dent Mater* 7:74–83, 1991.
5. Rueggeberg FA: Substrate for adhesion testing to tooth structure: review of the literature. *Dent Mater* 7:2–10, 1991.
6. Buonocore MG: A simple method of increasing the adhesion of acrylic filling materials to enamel surfaces. *J Dent Res* 34:849–853, 1955.
7. Van Meerbeek B, Perdigão J, Lambrechts P, et al: The clinical performance of adhesives. *J Dent* 26:1–20, 1998.
8. Black GV: *A work on operative dentistry in two volumes*, ed 3, Chicago, 1917, Medico-Dental Publishing Company.
9. Peumans M, Van Meerbeek B, Lambrechts P, et al: The 5-year clinical performance of direct composite additions to correct tooth form and position. I. Esthetic qualities. *Clin Oral Investig* 1:12–18, 1997.
10. Coelho-de-Souza FH, Gonçalves DS, Sales MP, et al: Direct anterior composite veneers in vital and non-vital teeth: a retrospective clinical evaluation. *J Dent* 43:1330–1336, 2015.
11. Christensen GJ: Cementing porcelain-fused-to-metal crowns. *J Am Dent Assoc* 128:1165–1167, 1997.
12. Jokstad A: A split-mouth randomized clinical trial of single crowns retained with resin-modified glass-ionomer and zinc phosphate luting cements. *Int J Prosthodont* 17:411–416, 2004.
13. Layton DM, Clarke M, Walton TR: A systematic review and meta-analysis of the survival of feldspathic porcelain veneers over 5 and 10 years. *Int J Prosthodont* 25:590–603, 2012.
14. Heintze SD, Rousson V: Fracture rates of IPS Empress all-ceramic crowns–a systematic review. *Int J Prosthodont* 23:129–133, 2010.

15. Feigal RJ, Musherure P, Gillespie B, et al: Improved sealant retention with bonding agents: a clinical study of two-bottle and single-bottle systems. *J Dent Res* 79:1850–1856, 2000.
16. Fleming PS, Johal A, Pandis N: Self-etch primers and conventional acid-etch technique for orthodontic bonding: a systematic review and meta-analysis. *Am J Orthod Dentofacial Orthop* 142:83–94, 2012.
17. Diangelis AJ, Andreasen JO, Ebeleseder KA, et al: International Association of Dental Traumatology: International Association of Dental Traumatology guidelines for the management of traumatic dental injuries: 1. Fractures and luxations of permanent teeth. *Dent Traumatol* 28:2–12, 2012.
18. Kumbuloglu O, Saracoglu A, Ozcan M: Pilot study of unidirectional E-glass fibre-reinforced composite resin splints: up to 4.5-year clinical follow-up. *J Dent* 39:871–877, 2011.
19. Gupta A, Yelluri RK, Munshi AK: Fiber-reinforced composite resin bridge: a treatment option in children. *Int J Clin Pediatr Dent* 8:62–65, 2015.
20. Liu X, Zhang Y, Zhou Z, et al: Retrospective study of combined splinting restorations in the aesthetic zone of periodontal patients. *Br Dent J* 220:241–247, 2016.
21. Opdam NJ, Bronkhorst EM, Loomans BA, et al: Longevity of repaired restorations: a practice based study. *J Dent* 40:829–835, 2012.
22. Hickel R, Brüshaver K, Ilie N: Repair of restorations–criteria for decision making and clinical recommendations. *Dent Mater* 29:28–50, 2013.
23. Baur V, Ilie N: Repair of dental resin-based composites. *Clin Oral Investig* 17:601–608, 2013.
24. Latta MA, Barkmeier WW: Approaches for intraoral repair of ceramic restorations. *Compend Contin Educ Dent* 21:635–639, 642-644, 2000.
25. Combe EC, Shaglouf AM, Watts DC, et al: Mechanical properties of direct core build-up materials. *Dent Mater* 15:158–165, 1999.
26. Hagge MS, Lindemuth JS: Shear bond strength of an autopolymerizing core buildup composite bonded to dentin with 9 dentin adhesive systems. *J Prosthet Dent* 86:620–623, 2001.
27. Calamia JR, Styner DL, Rattet AH: Effect of Amalgambond on cervical sensitivity. *Am J Dent* 8:283–284, 1996.
28. Ferrari M, Cagidiaco MC, Kugel G, et al: Clinical evaluation of a one-bottle bonding system for desensitizing exposed roots. *Am J Dent* 12:243–249, 1999.
29. Veitz-Keenan A, Barna JA, Strober B, et al: Treatments for hypersensitive noncarious cervical lesions: a Practitioners Engaged in Applied Research and Learning (PEARL) Network randomized clinical effectiveness study. *J Am Dent Assoc* 144:495–506, 2013.
30. Gibson M, Sharif MO, Smith A, et al: A practice-based randomised controlled trial of the efficacy of three interventions to reduce dentinal hypersensitivity. *J Dent* 41:668–674, 2013.
31. Robinson C, Hallsworth AS, Weatherell JA, et al: Arrest and control of carious lesions: a study based on preliminary experiments with resorcinol-formaldehyde resin. *J Dent Res* 55:812–818, 1976.
32. Zhang N, Melo MA, Chen C, et al: Development of a multifunctional adhesive system for prevention of root caries and secondary caries. *Dent Mater* 31:1119–1131, 2015.
33. Macedo GV, Diaz PI, De O, et al: Reattachment of anterior teeth fragments: a conservative approach. *J Esthet Restor Dent* 20:5–18, 2008.
34. Ree M, Schwartz RS: The endo-restorative interface: current concepts. *Dent Clin North Am* 54:345–374, 2010.
35. Goncalves LA, Vansan LP, Paulino SM, et al: Fracture resistance of weakened roots restored with a transilluminating post and adhesive restorative materials. *J Prosthet Dent* 96:339–344, 2006.
36. Schmitter M, Huy C, Ohlmann B, et al: Fracture resistance of upper and lower incisors restored with glass fiber reinforced posts. *J Endod* 32:328–330, 2006.
37. Doyle MD, Loushine RJ, Agee KA, et al: Improving the performance of EndoRez root canal sealer with a dual-cured two-step self-etch adhesive. I. Adhesive strength to dentin. *J Endod* 32:766–770, 2006.
38. Teixeira FB, Teixeira EC, Thompson J, et al: Dentinal bonding reaches the root canal system. *J Esthet Restor Dent* 16:348–354, 2004.
39. Wu MK, Kontakiotis EG, Wesselink PR: Long-term seal provided by some root-end filling materials. *J Endod* 24:557–560, 1998.
40. Murray MJ, Loushine RJ, Weller RN, et al: Use of self-etching adhesives to seal resected apices. *J Endod* 30:538–540, 2004.
41. Buonocore MG, Matsui A, Gwinnett AJ: Penetration of resin dental materials into enamel surfaces with reference to bonding. *Arch Oral Biol* 13:61–70, 1968.
42. Gwinnett AJ: Histologic changes in human enamel following treatment with acidic adhesive conditioning agents. *Arch Oral Biol* 16:731–738, 1971.
43. Gottlieb EW, Retief DH, Jamison HC: An optimal concentration of phosphoric acid as an etching agent: part I. tensile bond strength studies. *J Prosthet Dent* 48:48–51, 1982.
44. Gwinnett AJ, Kanca JA, 3rd: Micromorphology of the bonded dentin interface and its relationship to bond strength. *Am J Dent* 5:73–77, 1992.
45. Soetopo , Beech DR, Hardwick JL: Mechanism of adhesion of polymers to acid-etched enamel: effect of acid concentration and washing on bond strength. *J Oral Rehabil* 5:69–80, 1978.
46. Barkmeier WW, Shaffer SE, Gwinnett AJ: Effects of 15 vs 60 second enamel acid conditioning on adhesion and morphology. *Oper Dent* 11:111–116, 1986.
47. Mardaga WJ, Shannon IL: Decreasing the depth of etch for direct bonding in orthodontics. *J Clin Orthod* 16:130–132, 1982.
48. Barkmeier WW, Gwinnett AJ, Shaffer SE: Effects of enamel etching time on bond strength and morphology. *J Clin Orthod* 19:36–38, 1985.
49. Nordenvall K-J, Brännström M, Malmgren O: Etching of deciduous teeth and young and old permanent teeth: a comparison between 15 and 60 seconds of etching. *Am J Orthod* 78:99–108, 1980.
50. Bastos PAM, Retief DH, Bradley EL, et al: Effect of etch duration on the shear bond strength of a microfill composite resin to enamel. *Am J Dent* 1:151–157, 1988.
51. Crim GA, Shay JS: Effect of etchant time on microleakage. *J Dent Child* 54:339–340, 1987.
52. Gilpatrick RO, Ross JA, Simonsen RJ: Resin-to-enamel bond strengths with various etching times. *Quintessence Int* 22:47–49, 1991.
53. Shaffer SE, Barkmeier WW, Kelsey WP, 3rd: Effects of reduced acid conditioning time on enamel microleakage. *Gen Dent* 35:278–280, 1987.
54. Barkmeier WW, Erickson RL, Kimmes NS, et al: Effect of enamel etching time on roughness and bond strength. *Oper Dent* 34:217–222, 2009.
55. De Munck J, Van Meerbeek B, Satoshi I, et al: Microtensile bond strengths of one- and two-step self-etch adhesives to bur-cut enamel and dentin. *Am J Dent* 16:414–420, 2003.
56. Swift EJ, Jr, Perdigão J, Heymann HO: Enamel bond strengths of "one-bottle" adhesives. *Pediatr Dent* 20:259–262, 1998.
57. Senawongse P, Sattabanasuk V, Shimada Y, et al: Bond strengths of current adhesive systems on intact and ground enamel. *J Esthet Restor Dent* 16:107–115, 2004.
58. Barkmeier WW, Erickson RL, Latta MA: Fatigue limits of enamel bonds with moist and dry techniques. *Dent Mater* 25:1527–1531, 2009.
59. Gwinnett AJ, Matsui A: A study of enamel adhesives: the physical relationship between enamel and adhesive. *Arch Oral Biol* 12:1615–1620, 1967.
60. Asmussen E, Hansen EK, Peutzfeldt A: Influence of the solubility parameter of intermediary resin on the effectiveness of the Gluma bonding system. *J Dent Res* 70:1290–1293, 1991.
61. Nakabayashi N, Kojima K, Masuhara E: The promotion of adhesion by the infiltration of monomers into tooth substrates. *J Biomed Mater Res* 16:265–273, 1982.

62. Yoshida Y, Van Meerbeek B, Nakayama Y, et al: Evidence of chemical bonding at biomaterial-hard tissue interfaces. *J Dent Res* 79:709–714, 2000.
63. Fukuda R, Yoshida Y, Nakayama Y, et al: Bonding efficacy of polyalkenoic acids to hydroxyapatite, enamel and dentin. *Biomaterials* 24:1861–1867, 2003.
64. Fu B, Sun X, Qian W, et al: Evidence of chemical bonding to hydroxyapatite by phosphoric acid esters. *Biomaterials* 26:5104–5110, 2005.
65. Yoshihara K, Yoshida Y, Hayakawa S, et al: Nano-layering of phosphoric-acid ester monomer at enamel and dentin. *Acta Biomater* 7:3187–3195, 2011.
66. Mitra SB, Lee CY, Bui HT, et al: Long-term adhesion and mechanism of bonding of a paste-liquid resin-modified glass-ionomer. *Dent Mater* 25:459–466, 2009.
67. Van Meerbeek B, Ionkoshi S, Braem M, et al: Morphological aspects of the resin-dentin interdiffusion zone with different dentin adhesive systems. *J Dent Res* 71:1530–1540, 1992.
68. Tay FR, Gwinnett AJ, Wei SH: Ultrastructure of the resin-dentin interface following reversible and irreversible rewetting. *Am J Dent* 10:77–82, 1997.
69. Veis A: Mineral-matrix interactions in bone and dentin. *J Bone Miner Res* 8(Suppl 2):S493–S497, 1993.
70. Brännström M, Lindén LA, Johnson G: Movement of dentinal and pulpal fluid caused by clinical procedures. *J Dent Res* 47:679–682, 1968.
71. Terkla LG, Brown AC, Hainisch AP, et al: Testing sealing properties of restorative materials against moist dentin. *J Dent Res* 66:1758–1764, 1987.
72. Van Hassel HJ: Physiology of the human dental pulp. *Oral Surg Oral Med Oral Pathol* 32:126–134, 1971.
73. Garberoglio R, Brännström M: Scanning electron microscopic investigation of human dentinal tubules. *Arch Oral Biol* 21:355–362, 1976.
74. Pashley DH: Dentin: a dynamic substrate—a review. *Scanning Microsc* 3:161–176, 1989.
75. Marchetti C, Piacentini C, Menghini P: Morphometric computerized analysis on the dentinal tubules and the collagen fibers in the dentine of human permanent teeth. *Bull Group Int Rech Sci Stomatol Odontol* 35:125–129, 1992.
76. Suzuki T, Finger WJ: Dentin adhesives: site of dentin vs. bonding of composite resins. *Dent Mater* 4:379–383, 1988.
77. Rosales-Leal JI, Osorio R, Holgado-Terriza JA, et al: Dentin wetting by four adhesive systems. *Dent Mater* 17:526–532, 2001.
78. Sattabanasuk V, Shimada Y, Tagami J: The bond of resin to different dentin surface characteristics. *Oper Dent* 29:333–341, 2004.
79. Adebayo OA, Burrow MF, Tyas MJ: Bonding of one-step and two-step self-etching primer adhesives to dentin with different tubule orientations. *Acta Odontol Scand* 66:159–168, 2008.
80. Zhang L, Wang DY, Fan J, et al: Stability of bonds made to superficial vs. deep dentin, before and after thermocycling. *Dent Mater* 30:1245–1251, 2014.
81. Bowen RL, Eick JD, Henderson DA, et al: Smear layer: removal and bonding considerations. *Oper Dent* 3(Suppl):30–34, 1984.
82. Ishioka S, Caputo AA: Interaction between the dentinal smear layer and composite bond strengths. *J Prosthet Dent* 61:180–185, 1989.
83. Pashley DH, Livingston MJ, Greenhill JD: Regional resistances to fluid flow in human dentine in vitro. *Arch Oral Biol* 23:807–810, 1978.
84. Eick JD, Cobb CM, Chappell RP, et al: The dentinal surface: its influence on dentinal adhesion: part I. *Quintessence Int* 22:967–977, 1991.
85. Pashley DH: The effects of acid etching on the pulpodentin complex. *Oper Dent* 17:229–242, 1992.
86. Torney D: The retentive ability of acid-etched dentin. *J Prosthet Dent* 39:169–172, 1978.
87. Opdam NJ, Feilzer AJ, Roeters JJ, et al: Class I occlusal composite resin restorations: in vivo post-operative sensitivity, wall adaptation, and microleakage. *Am J Dent* 11:229–234, 1998.
88. Bowen RL, Nemoto K, Rapson JE: Adhesive bonding of various materials to hard tooth tissue: forces developing in composite materials during hardening. *J Am Dent Assoc* 106:475–477, 1983.
89. Davidson CL, Feilzer AJ: Polymerization shrinkage and polymerization shrinkage stress in polymer-based restoratives. *J Dent* 25:435–440, 1997.
90. Davidson CL, de Gee AJ, Feilzer A: The competition between the composite-dentin bond strength and the polymerization contraction stress. *J Dent Res* 63:1396–1399, 1984.
91. Hegdahl T, Gjerdet NR: Contraction stresses of composite filling materials. *Acta Odontol Scand* 35:191–195, 1977.
92. Feilzer A, De Gee AJ, Davidson CL: Setting stress in composite resin in relation to configuration of the restoration. *J Dent Res* 66:1636–1639, 1987.
93. Davidson CL, de Gee AJ: Relaxation of polymerization contraction stresses by flow in dental composites. *J Dent Res* 63:146–148, 1984.
94. Perdigão J, Lambrechts P, Van Meerbeek B, et al: The interaction of adhesive systems with human dentin. *Am J Dent* 9:167–173, 1996.
95. Yoshikawa T, Sano H, Burrow MF, et al: Effects of dentin depth and cavity configuration on bond strength. *J Dent Res* 78:898–905, 1999.
96. Carvalho RM, Pereira JC, Yoshiyama M, et al: A review of polymerization contraction: the influence of stress development versus stress relief. *Oper Dent* 21:17–24, 1996.
97. Mantri SP, Mantri SS: Management of shrinkage stresses in direct restorative light-cured composites: a review. *J Esthet Restor Dent* 25:305–313, 2013.
98. Masouras K, Silikas N, Watts DC: Correlation of filler content and elastic properties of resin-composites. *Dent Mater* 24:932–939, 2008.
99. Gonçalves F, Kawano Y, Braga RR: Contraction stress related to composite inorganic content. *Dent Mater* 26:704–709, 2010.
100. Son SA, Park JK, Seo DG, et al: How light attenuation and filler content affect the microhardness and polymerization shrinkage and translucency of bulk-fill composites? *Clin Oral Investig* 21:559–565, 2017.
101. Munksgaard EC, Irie M, Asmussen E: Dentin-polymer bond promoted by Gluma and various resins. *J Dent Res* 64:1409–1411, 1985.
102. Soares CJ, Bicalho AA, Tantbirojn D, et al: Polymerization shrinkage stresses in a premolar restored with different composite resins and different incremental techniques. *J Adhes Dent* 15:341–350, 2013.
103. Park JW, Ferracane JL: Water aging reverses residual stresses in hydrophilic dental composites. *J Dent Res* 93:195–200, 2014.
104. Sideridou ID, Karabela MM, Vouvoudi ECh: Volumetric dimensional changes of dental light-cured dimethacrylate resins after sorption of water or ethanol. *Dent Mater* 24:1131–1136, 2008.
105. Wei YJ, Silikas N, Zhang ZT, et al: The relationship between cyclic hygroscopic dimensional changes and water sorption/desorption of self-adhering and new resin-matrix composites. *Dent Mater* 29:e218–e226, 2013.
106. Uno S, Asmussen E: Marginal adaptation of a restorative resin polymerized at reduced. *Scand J Dent Res* 99:440–444, 1991.
107. Kanca J, 3rd, Suh BI: Pulse activation: reducing resin-based composite contraction stresses at the enamel cavosurface margins. *Am J Dent* 12:107–112, 1999.
108. Park J, Chang J, Ferracane J, et al: How should composite be layered to reduce shrinkage stress: incremental or bulk filling? *Dent Mater* 24:1501–1505, 2008.
109. Alomari QD, Reinhardt JW, Boyer DB: Effect of liners on cusp deflection and gap formation in composite restorations. *Oper Dent* 26:406–411, 2001.

110. Ferracane JL, Hilton TJ: Polymerization stress - Is it clinically meaningful? *Dent Mater* 32:1–10, 2016.
111. Kemp-Scholte CM, Davidson CL: Marginal integrity related to bond strength and strain capacity of composite resin restorative systems. *J Prosthet Dent* 64:658–664, 1990.
112. Finger WJ: Dentin bonding agents: relevance of in vitro investigations. *Am J Dent* 1:184–188, 1988.
113. Kemp-Scholte CM, Davidson CL: Complete marginal seal of Class V resin composite restorations effected by increased flexibility. *J Dent Res* 69:1240–1243, 1990.
114. Lindberg A, van Dijken JW, Hörstedt P: In vivo interfacial adaptation of Class II resin composite restorations with and without a flowable resin composite liner. *Clin Oral Investig* 9:77–83, 2005.
115. Reis A, Loguercio AD: A 24-month follow-up of flowable resin composite as an intermediate layer in non-carious cervical lesions. *Oper Dent* 31:523–529, 2006.
116. Efes BG, Dörter C, Gömeç Y, et al: Two-year clinical evaluation of ormocer and nanofill composite with and without a flowable liner. *J Adhes Dent* 8:119–126, 2006.
117. Celik C, Ozgünaltay G, Attar N: Clinical evaluation of flowable resins in non-carious cervical lesions: two-year results. *Oper Dent* 32:313–321, 2007.
118. van Dijken JWV, Pallesen U: Clinical performance of a hybrid resin composite with and without an intermediate layer of flowable resin composite: a 7-year evaluation. *Dent Mater* 27:150–156, 2011.
119. van de Sande FH, Rodolpho PA, Basso GR, et al: 18-year survival of posterior composite resin restorations with and without glass ionomer cement as base. *Dent Mater* 31:669–675, 2015.
120. Condon JR, Ferracane JL: Reduced polymerization stress through non-bonded nanofiller particles. *Biomaterials* 23:3807–3815, 2002.
121. Palin WM, Fleming GJ, Nathwani H, et al: In vitro cuspal deflection and microleakage of maxillary premolars restored with novel low-shrink dental composites. *Dent Mater* 21:324–335, 2005.
122. Braga RR, Ferracane JL: Alternatives in polymerization contraction stress management. *Crit Rev Oral Biol Med* 15:176–184, 2004.
123. Papadogiannis D, Kakaboura A, Palaghias G, et al: Setting characteristics and cavity adaptation of low-shrinking resin composites. *Dent Mater* 25:1509–1516, 2009.
124. Xu HC, Liu WY, Wang T: Measurement of thermal expansion coefficient of human teeth. *Aust Dent J* 34:530–535, 1989.
125. Sideridou I, Achilias DS, Kyrikou E: Thermal expansion characteristics of light-cured dental resins and resin composites. *Biomaterials* 25:3087–3097, 2004.
126. Versluis A, Douglas WH, Sakaguchi RL: Thermal expansion coefficient of dental composites measured with strain gauges. *Dent Mater* 12:290–294, 1996.
127. Bullard RH, Leinfelder KF, Russell CM: Effect of coefficient of thermal expansion on microleakage. *J Am Dent Assoc* 116:871–874, 1988.
128. Asmussen E: Clinical relevance of physical, chemical, and bonding properties of composite resins. *Oper Dent* 10:61–73, 1985.
129. Borges AL, Borges AB, Xavier TA, et al: Impact of quantity of resin, C-factor, and geometry on resin composite polymerization shrinkage stress in Class V restorations. *Oper Dent* 39:144–151, 2014.
130. Hansen EK: Effect of Scotchbond dependent on cavity cleaning, cavity diameter and cavosurface angle. *Scand J Dent Res* 92:141–147, 1984.
131. Schroeder M, Reis A, Luque-Martinez I, et al: Effect of enamel bevel on retention of cervical composite resin restorations: a systematic review and meta-analysis. *J Dent* 43:777–788, 2015.
132. Mahn E, Rousson V, Heintze S: Meta-analysis of the influence of bonding parameters on the clinical outcome of tooth-colored cervical restorations. *J Adhes Dent* 17:391–403, 2015.
133. Jørgensen KD, Matono R, Shimokobe H: Deformation of cavities and resin fillings in loaded teeth. *Scand J Dent Res* 84:46–50, 1976.
134. Tonami K, Takahashi H: Effects of aging on tensile fatigue strength of bovine dentin. *Dent Mater J* 16:156–169, 1997.
135. Qvist V: The effect of mastication on marginal adaptation of composite restorations in vivo. *J Dent Res* 62:904–906, 1983.
136. Nikaido T, Kunzelmann KH, Chen H, et al: Evaluation of thermal cycling and mechanical loading on bond strength of a self-etching primer system to dentin. *Dent Mater* 18:269–275, 2002.
137. Toledano M, Osorio R, Albaladejo A, et al: Effect of cyclic loading on the microtensile bond strengths of total-etch and self-etch adhesives. *Oper Dent* 31:25–32, 2006.
138. Brudevold F, Buonocore M, Wileman W: A report on a resin composition capable of bonding to human dentin surfaces. *J Dent Res* 35:846–851, 1956.
139. McLean JW: Bonding to enamel and dentin [letter]. *Quintessence Int* 26:334, 1995.
140. McLean JW, Kramer IRH: A clinical and pathological evaluation of a sulphinic acid activated resin for use in restorative dentistry. *Br Dent J* 93:255, 1952.
141. Bowen RL: Adhesive bonding of various materials to hard tooth tissues: II. Bonding to dentin promoted by a surface-active comonomer. *J Dent Res* 44:895–902, 1965.
142. Barkmeier WW, Cooley RL: Laboratory evaluation of adhesive systems. *Oper Dent* 5(Suppl):50–61, 1992.
143. Albers HF: Dentin-resin bonding. *ADEPT Report* 1:33–42, 1990.
144. Alexieva C: Character of the hard tooth tissue-polymer bond: II. Study of the interaction of human tooth enamel and dentin with N-phenylglycine-glycidyl methacrylate adduct. *J Dent Res* 58:1884–1886, 1979.
145. Retief DH, Denys FR: Adhesion to enamel and dentin. *Am J Dent* 2:133–144, 1989.
146. Jendresen MD: Clinical performance of a new composite resin for Class V erosion (abstract 1057). *J Dent Res* 57:339, 1978.
147. Eick JD: Smear layer—materials surface. *Proc Finn Dent Soc* 88:225–242, 1992.
148. Asmussen E, Munksgaard EC: Bonding of restorative materials to dentine: status of dentine adhesives and impact on cavity design and filling techniques. *Int Dent J* 38:97–104, 1988.
149. Baier RE: Principles of adhesion. *Oper Dent* 5(Suppl):1–9, 1992.
150. Causton BE: Improved bonding of composite restorative to dentine: a study in vitro of the use of a commercial halogenated phosphate ester. *Br Dent J* 156:93–95, 1984.
151. Huang GT, Söderholm K-JM: In vitro investigation of shear bond strength of a phosphate based dentinal bonding agent. *Scand J Dent Res* 97:84–92, 1989.
152. Eliades GC: Dentine bonding systems. In Vanherle G, et al, editors: *State of the art on direct posterior filling materials and dentine bonding*, Leuven, 1993, Van der Poorten, pp 49–74.
153. Tyas MJ, Burns GA, Byrne PF, et al: Clinical evaluation of Scotchbond: three-year results. *Aust Dent J* 34:277–279, 1989.
154. Van Meerbeek B, Peumans M, Verschueren M, et al: Clinical status of ten dentin adhesive systems. *J Dent Res* 73:1690–1702, 1994.
155. Fusayama T, Nakamura M, Kurosaki N, et al: Non-pressure adhesion of a new adhesive restorative resin. *J Dent Res* 58:1364–1370, 1979.
156. van Dijken JWV, Horstedt P: In vivo adaptation of restorative materials to dentin. *J Prosthet Dent* 56:677–681, 1986.
157. Retief DH, Austin JC, Fatti LP: Pulpal response to phosphoric acid. *J Oral Pathol* 3:114–122, 1974.
158. Stanley HR, Going RE, Chauncey HH: Human pulp response to acid pretreatment of dentin and to composite restoration. *J Am Dent Assoc* 91:817–825, 1975.
159. American Dental Association Council on Scientific Affairs: *Revised American Dental Association acceptance program guidelines: Dentin and enamel adhesives* (pp. 1–9), Chicago, 2001, American Dental Association.
160. Mine A, De Munck J, Vivan Cardoso M, et al: Enamel-smear compromises bonding by mild self-etch adhesives. *J Dent Res* 89:1505–1509, 2010.
161. Suyama Y, Lührs AK, De Munck J, et al: Potential smear layer interference with bonding of self-etching adhesives to dentin. *J Adhes Dent* 15:317–324, 2013.

162. Eliades G: Clinical relevance of the formulation and testing of dentine bonding systems. *J Dent* 22:73–81, 1994.
163. Barkmeier WW, Suh B, Cooley RL: Shear bond strength to dentin and Ni-Cr-Be alloy with the All-Bond universal adhesive system. *J Esthet Dent* 3:148–153, 1991.
164. Kanca J, 3rd: Effect of resin primer solvents and surface wetness on resin composite bond strength to dentin. *Am J Dent* 5:213–221, 1992.
165. Kanca J, 3rd: Resin bonding to wet substrate: I. Bonding to dentin. *Quintessence Int* 23:39–41, 1992.
166. Swift EJ, Triolo PT: Bond strengths of Scotchbond Multi-Purpose to moist dentin and enamel. *Am J Dent* 5:318–320, 1992.
167. Perdigão J, Lambrechts P, Van Meerbeek B, et al: Morphological field emission SEM study of the effect of six phosphoric acid etching agents on human dentin. *Dent Mater* 12:262–271, 1996.
168. Ermis RB, Kam O, Celik EU, et al: Clinical evaluation of a two-step etch & rinse and a two-step self-etch adhesive system in Class II restorations: two-year results. *Oper Dent* 34:656–663, 2009.
169. Wilder AD, Swift EJ, Jr, Heymann HO, et al: A 12-year clinical evaluation of a three-step dentin adhesive in noncarious cervical lesions. *J Am Dent Assoc* 140:526–535, 2009.
170. Peumans M, De Munck J, Van Landuyt KL, et al: A 13-year clinical evaluation of two three-step etch-and-rinse adhesives in non-carious Class-V lesions. *Clin Oral Investig* 16:129–137, 2012.
171. Erickson RL: Surface interactions of dentin adhesive materials. *Oper Dent* 5(Suppl):81–94, 1992.
172. Panighi M, G'Sell C: Influence of calcium concentration on the dentin wettability by an adhesive. *J Biomed Mater Res* 26:1081–1089, 1992.
173. Armstrong SR, Vargas MA, Fang Q, et al: Microtensile bond strength of a total-etch 3-step, total-etch 2-step, self-etch 2-step, and a self-etch 1-step dentin bonding system through 15-month water storage. *J Adhes Dent* 5:47–56, 2003.
174. Peumans M, De Munck J, Mine A, et al: Clinical effectiveness of contemporary adhesives for the restoration of non-carious cervical lesions. A systematic review. *Dent Mater* 30:1089–1103, 2014.
175. De Munck J, Mine A, Poitevin A, et al: Meta-analytical review of parameters involved in dentin bonding. *J Dent Res* 91:351–357, 2012.
176. Tay FR, Pashley DH: Have dentin adhesives become too hydrophilic? *J Can Dent Assoc* 69:726–731, 2003.
177. Loguercio AD, Raffo J, Bassani F, et al: 24-month clinical evaluation in non-carious cervical lesions of a two-step etch-and-rinse adhesive applied using a rubbing motion. *Clin Oral Investig* 15:589–596, 2011.
178. Reis A, Pellizzaro A, Dal-Bianco K, et al: Impact of adhesive application to wet and dry dentin on long-term resin-dentin bond strengths. *Oper Dent* 32:380–387, 2007.
179. Perdigão J, Lopes L, Lambrechts P, et al: Effects of a self-etching primer on enamel shear bond strengths and SEM morphology. *Am J Dent* 10:141–146, 1997.
180. Watanabe I, Nakabayashi N, Pashley DH: Bonding to ground dentin by a Phenyl-P self-etching primer. *J Dent Res* 73:1212–1220, 1994.
181. Tay FR, Pashley DH, Suh BI, et al: Single-step adhesives are permeable membranes. *J Dent* 30:371–382, 2002.
182. Fukuoka A, Koshiro K, Inoue S, et al: Hydrolytic stability of one-step self-etching adhesives bonded to dentin. *J Adhes Dent* 13:243–248, 2011.
183. Clinician's preferences 2001, *CRA Newsletter* 25, 2001.
184. Perdigão J, Gomes G, Gondo R, et al: In vitro bonding performance of all-in-one adhesives. Part I – microtensile bond strengths. *J Adhes Dent* 8:367–373, 2006.
185. Perdigão J, Lopes MM, Gomes G: In vitro bonding performance of self-etch adhesives: II–ultramorphological evaluation. *Oper Dent* 33:534–549, 2008.
186. Ferrari M, Mannocci F, Vichi A, et al: Effect of two etching times on the sealing ability of Clearfil Liner Bond 2 in Class V restorations. *Am J Dent* 10:66–70, 1997.
187. Opdam NJ, Roeters FJ, Feilzer AJ, et al: Marginal integrity and postoperative sensitivity in Class 2 resin composite restorations in vivo. *J Dent* 26:555–562, 1998.
188. Miyazaki M, Sato M, Onose H: Durability of enamel bond strength of simplified bonding systems. *Oper Dent* 25:75–80, 2000.
189. Akimoto N, Takamizu M, Momoi Y: 10-year clinical evaluation of a self-etching adhesive system. *Oper Dent* 32:3–10, 2007.
190. Tay FR, Sano H, Carvalho R, et al: An ultrastructural study of the influence of acidity of self-etching primers and smear layer thickness on bonding to intact dentin. *J Adhes Dent* 2:83–98, 2000.
191. Pashley DH, Tay FR: Aggressiveness of contemporary self-etching adhesives: part II: etching effects on unground enamel. *Dent Mater* 17:430–444, 2001.
192. Peumans M, De Munck J, Van Landuyt K, et al: Thirteen-year randomized controlled clinical trial of a two-step self-etch adhesive in non-carious cervical lesions. *Dent Mater* 31:308–314, 2015.
193. Yoshihara K, Yoshida Y, Nagaoka N, et al: Nano-controlled molecular interaction at adhesive interfaces for hard tissue reconstruction. *Acta Biomater* 6:3573–3582, 2010.
194. Anchieta RB, Machado LS, Sundfeld RH, et al: Effect of partially demineralized dentin beneath the hybrid layer on dentin-adhesive interface micromechanics. *J Biomech* 48:701–707, 2015.
195. Wang Y, Spencer P: Physiochemical interactions at the interfaces between self-etch adhesive systems and dentine. *J Dent* 32:567–579, 2004.
196. Wang Y, Spencer P: Continuing etching of an all-in-one adhesive in wet dentin tubules. *J Dent Res* 84:350–354, 2005.
197. Cao Y, Mei ML, Xu J, et al: Biomimetic mineralisation of phosphorylated dentine by CPP-ACP. *J Dent* 41:818–825, 2013.
198. Tay FR, Pashley DH: Guided tissue remineralisation of partially demineralised human dentine. *Biomaterials* 29:1127–1137, 2008.
199. Zhang W, Luo XJ, Niu LN, et al: Biomimetic intrafibrillar mineralization of type I collagen with intermediate precursors-loaded mesoporous carriers. *Sci Rep* 5:11199, 2015.
200. Niu LN, Zhang W, Pashley DH, et al: Biomimetic remineralization of dentin. *Dent Mater* 30:77–96, 2014.
201. Akpata ES, Behbehani J: Effect of bonding systems on post-operative sensitivity from posterior composites. *Am J Dent* 19:151–154, 2006.
202. Browning WD, Blalock JS, Callan RS, et al: Postoperative sensitivity: a comparison of two bonding agents. *Oper Dent* 32:112–117, 2007.
203. Casselli DS, Martins LR: Postoperative sensitivity in Class I composite resin restorations in vivo. *J Adhes Dent* 8:53–58, 2006.
204. Wegehaupt F, Betke H, Solloch N, et al: Influence of cavity lining and remaining dentin thickness on the occurrence of postoperative hypersensitivity of composite restorations. *J Adhes Dent* 11:137–141, 2009.
205. Burrow MF, Banomyong D, Harnirattisai C, et al: Effect of glass-ionomer cement lining on postoperative sensitivity in occlusal cavities restored with resin composite—A randomized clinical trial. *Oper Dent* 34:648–655, 2009.
206. Reis A, Dourado Loguercio A, Schroeder M, et al: Does the adhesive strategy influence the post-operative sensitivity in adult patients with posterior resin composite restorations?: a systematic review and meta-analysis. *Dent Mater* 31:1052–1067, 2015.
207. Perdigão J: Dentin bonding as a function of dentin structure. *Dent Clin North Am* 46:277–301, 2002.
208. Tay FR, Pashley DH, Peters MC: Adhesive permeability affects composite coupling to dentin treated with a self-etch adhesive. *Oper Dent* 28:610–621, 2003.
209. Tay FR, Pashley DH, Yiu CK, et al: Factors contributing to the incompatibility between simplified-step adhesives and chemically-cured or dual-cured composites: part I. Single-step self-etching adhesive. *J Adhes Dent* 5:27–40, 2003.

210. Tay FR, Pashley DH: Aggressiveness of contemporary self-etching adhesives: part I. Depth of penetration beyond dentin smear layers. *Dent Mater* 17:296–308, 2001.
211. Ito S, Tay FR, Hashimoto M, et al: Effects of multiple coatings of two all-in-one adhesives on dentin bonding. *J Adhes Dent* 7:133–141, 2005.
212. Reis A, Leite TM, Matte K, et al: Improving clinical retention of one-step self-etching adhesive systems with an additional hydrophobic adhesive layer. *J Am Dent Assoc* 140:877–885, 2009.
213. Reis A, Albuquerque M, Pegoraro M, et al: Can the durability of one-step self-etch adhesives be improved by double application or by an extra layer of hydrophobic resin? *J Dent* 36:309–315, 2008.
214. Van Landuyt KL, Peumans M, De Munck J, et al: Extension of a one-step self-etch adhesive into a multi-step adhesive. *Dent Mater* 22:533–544, 2006.
215. Tay FR, Frankenberger R, Krejci I, et al: Single-bottle adhesives behave as permeable membranes after polymerization. I. In vivo evidence. *J Dent* 32:611–621, 2004.
216. Hanabusa M, Mine A, Kuboki T, et al: Bonding effectiveness of a new 'multi-mode' adhesive to enamel and dentine. *J Dent* 40:475–484, 2012.
217. Perdigão J, Sezinando A, Monteiro PC: Laboratory bonding ability of a multi-purpose dentin adhesive. *Am J Dent* 25:153–158, 2012.
218. Chen C, Niu LN, Xie H, et al: Bonding of universal adhesives to dentine–Old wine in new bottles? *J Dent* 43:525–536, 2015.
219. Sezinando A, Luque-Martinez I, Muñoz MA, et al: Influence of a hydrophobic resin coating on the immediate and 6-month dentin bonding of three universal adhesives. *Dent Mater* 31:e236–e246, 2015.
220. Tian F, Zhou L, Zhang Z, et al: Paucity of nanolayering in resin-dentin interfaces of MDP-based adhesives. *J Dent Res* 95:380–387, 2016.
221. Van Meerbeek B, Yoshihara K, Yoshida Y, et al: State of the art of self-etch adhesives. *Dent Mater* 27:17–28, 2011.
222. Yoshida Y, Yoshihara K, Hayakawa S, et al: HEMA inhibits interfacial nano-layering of the functional monomer MDP. *J Dent Res* 91:1060–1065, 2012.
223. Loguercio AD, Muñoz MA, Luque-Martinez I, et al: Does active application of universal adhesives to enamel in self-etch mode improve their performance? *J Dent* 43:1060–1070, 2015.
224. Marchesi G, Frassetto A, Mazzoni A, et al: Adhesive performance of a multi-mode adhesive system: 1-year in vitro study. *J Dent* 42:603–612, 2014.
225. Kanca J, 3rd: Wet bonding: effect of drying time and distance. *Am J Dent* 9:273–276, 1996.
226. Eick JD, Gwinnett AJ, Pashley DH, et al: Current concepts on adhesion to dentin. *Crit Rev Oral Biol Med* 8:306–335, 1997.
227. Maciel KT, Carvalho RM, Ringle RD, et al: The effect of acetone, ethanol, HEMA, and air on the stiffness of human decalcified dentin matrix. *J Dent Res* 75:1851–1858, 1996.
228. Perdigão J, Swift EJ, Cloe BC: Effects of etchants, surface moisture, and resin composite on dentin bond strengths. *Am J Dent* 6:61–64, 1993.
229. Goes MF, Pachane GC, García-Godoy F: Resin bond strength with different methods to remove excess water from the dentin. *Am J Dent* 10:298–301, 1997.
230. Magne P, Mahallati R, Bazos P, et al: Direct dentin bonding technique sensitivity when using air/suction drying steps. *J Esthet Restor Dent* 20:130–138, 2008.
231. Carvalho RM, Yoshiyama M, Pashley EL, et al: In vitro study on the dimensional changes of dentine after demineralization. *Arch Oral Biol* 41:369–377, 1996.
232. Perdigão J, Van Meerbeek B, Lopes MM, et al: The effect of a re-wetting agent on dentin bonding. *Dent Mater* 15:282–295, 1999.
233. Tay FR, Gwinnett JA, Wei SH: Micromorphological spectrum from overdrying to overwetting acid-conditioned dentin in water-free acetone-based, single-bottle primer/adhesives. *Dent Mater* 12:236–244, 1996.
234. Sasaki N, Odajima S: Stress-strain curve and Young's modulus of a collagen molecule as determined by the x-ray diffraction technique. *J Biomech* 29:655–658, 1996.
235. Sasaki N, Shiwa S, Yagihara S, et al: X-ray diffraction studies on the structure of hydrated collagen. *Biopolymers* 22:2539–2547, 1983.
236. Van der Graaf ER, ten Bosch JJ: Changes in dimension and weight of human dentine after different drying procedures and during subsequent rehydration. *Arch Oral Biol* 38:97–99, 1993.
237. Eanes ED, Lundy DR, Martin GN: X-ray diffraction study of the mineralization of turkey leg tendon. *Calcif Tissue Res* 6:239–248, 1970.
238. Perdigão J, Frankenberger R: Effect of solvent and re-wetting time on dentin adhesion. *Quintessence Int* 32:385–390, 2001.
239. Tay FR, Gwinnett AJ, Pang KM, et al: Resin permeation into acid-conditioned, moist, and dry dentin: a paradigm using water-free adhesive primers. *J Dent Res* 75:1034–1044, 1996.
240. Swift EJ, Perdigao J, Heymann HO: Bonding to enamel and dentin: a brief history and state of the art, 1995. *Quintessence Int* 26:95–110, 1995.
241. Luque-Martinez IV, Perdigão J, Muñoz MA, et al: Effects of solvent evaporation time on immediate adhesive properties of universal adhesives to dentin. *Dent Mater* 30:1126–1135, 2014.
242. Perdigão J, Reis A, Loguercio AD: Dentin adhesion and MMPs: a comprehensive review. *J Esthet Restor Dent* 25:219–241, 2013.
243. Tay FR, Pashley SH, Kapur RR, et al: Bonding BisGMA to dentin—a proof of concept for hydrophobic dentin bonding. *J Dent Res* 86:1034–1039, 2007.
244. Hosaka K, Nishitani Y, Tagami J, et al: Durability of resin-dentin bonds to water- vs. ethanol-saturated dentin. *J Dent Res* 88:146–151, 2009.
245. Salz U, Zimmermann J, Zeuner F, et al: Hydrolytic stability of self-etching adhesive systems. *J Adhes Dent* 7:107–116, 2005.
246. Hiraishi N, Nishiyama N, Ikemura K, et al: Water concentration in self-etching primers affects their aggressiveness and bonding efficacy to dentin. *J Dent Res* 84:653–658, 2005.
247. Hosaka K, Nakajima M, Takahashi M, et al: Relationship between mechanical properties of one-step self-etch adhesives and water sorption. *Dent Mater* 26:360–367, 2010.
248. Ito S, Hoshino T, Iijima M, et al: Water sorption/solubility of self-etching dentin bonding agents. *Dent Mater* 26:617–626, 2010.
249. Hashimoto M, Ohno H, Sano H, et al: In vitro degradation of resin-dentin bonds analyzed by microtensile bond test, scanning and transmission electron microscopy. *Biomaterials* 24:3795–3803, 2003.
250. Tay FR, Pashley DH, Suh BI, et al: Water treeing in simplified dentin adhesives–déjà vu? *Oper Dent* 30:561–579, 2005.
251. Kato G, Nakabayashi N: Effect of phosphoric acid concentration on wet-bonding to etched dentin. *Dent Mater* 12:250–255, 1996.
252. Sano H, Shono T, Takatsu T, et al: Microporous dentin zone beneath resin-impregnated layer. *Oper Dent* 19:59–64, 1994.
253. Tatsumi T, Inokoshi S, Yamada T, et al: Remineralization of etched dentin. *J Prosthet Dent* 67:617–620, 1992.
254. Okamoto Y, Heeley JD, Dogon IL, et al: Effects of phosphoric acid and tannic acid on dentine collagen. *J Oral Rehabil* 18:507–512, 1991.
255. El Feninat F, Ellis TH, Sacher E, et al: Moisture-dependent renaturation of collagen in phosphoric acid etched human dentin. *J Biomed Mater Res* 42:549–553, 1998.
256. Klont B, Ten Cate JM: Denaturation and degradation of dentin collagen during dentinal caries: possible mechanisms. In Thylstrup A, Leach SA, Qvist V, editors: *Dentin and dentin reactions in the oral cavity*, Oxford, 1987, IRL Press, pp155–164.
257. Ritter AV, Swift EJ, Jr, Yamauchi M: Effects of phosphoric acid and glutaraldehyde-HEMA on dentin collagen. *Eur J Oral Sci* 109:348–353, 2001.

258. Breschi L, Perdigão J, Gobbi P, et al: Immunolocalization of type I collagen in etched dentin. *J Biomed Mater Res A* 66:764–769, 2003.
259. Landis WJ, Hodgens KJ, Arena J, et al: Structural relations between collagen and mineral in bone as determined by high voltage electron microscopic tomography. *Microsc Res Tech* 33:192–202, 1996.
260. Landis WJ, Silver FH: Mineral deposition in the extracellular matrices of vertebrate tissues: identification of possible apatite nucleation sites on type I collagen. *Cells Tissues Organs* 189:20–24, 2009.
261. Wainwright SA, Biggs WD, Currey JD, et al: *Mechanical design in organisms*, Princeton, 1982, Princeton Univ. Press.
262. Rivera EM, Yamauchi M: Site comparisons of dentine collagen cross-links from extracted human teeth. *Arch Oral Biol* 38:541–546, 1993.
263. Miguez PA, Pereira PN, Atsawasuwan P, et al: Collagen cross-linking and ultimate tensile strength in dentin. *J Dent Res* 83:807–810, 2004.
264. Yang W, Sherman VR, Gludovatz B, et al: On the tear resistance of skin. *Nat Commun* 6:6649, 2015.
265. Prasad M, Butler WT, Qin C: Dentin sialophosphoprotein in biomineralization. *Connect Tissue Res* 51:404–417, 2010.
266. Eanes ED: Amorphous calcium phosphate. *Monogr Oral Sci* 18:130–147, 2001.
267. Weiner S: Biomineralization: a structural perspective. *J Struct Biol* 163:229–234, 2008.
268. Weiner S, Sagi I, Addadi L: Structural biology. Choosing the crystallization path less traveled. *Science* 309:1027–1028, 2005.
269. Dahl T, Sabsay B, Veis A: Type I collagen-phosphophoryn interactions: specificity of the monomer-monomer binding. *J Struct Biol* 123:162–168, 1998.
270. Tay FR, Pashley DH: Biomimetic remineralization of resin-bonded acid-etched dentin. *J Dent Res* 88:719–724, 2009.
271. Mai S, Kim YK, Toledano M, et al: Phosphoric acid esters cannot replace polyvinylphosphonic acid as phosphoprotein analogs in biomimetic remineralization of resin-bonded dentin. *Dent Mater* 25:1230–1239, 2009.
272. Mai S, Kim YK, Kim J, et al: In vitro remineralization of severely compromised bonded dentin. *J Dent Res* 89:405–410, 2010.
273. Mazzoni A, Pashley DH, Nishitani Y, et al: Reactivation of inactivated endogenous proteolytic activities in phosphoric acid-etched dentine by etch-and-rinse adhesives. *Biomaterials* 27:4470–4476, 2006.
274. Sulkala M, Larmas M, Sorsa T, et al: The localization of matrix metalloproteinase-20 (MMP-20, enamelysin) in mature human teeth. *J Dent Res* 81:603–638, 2002.
275. Martin-De Las Heras S, Valenzuela A, Overall CM: The matrix metalloproteinase gelatinase A in human dentine. *Arch Oral Biol* 45:757–765, 2000.
276. Gendron R, Grenier D, Sorsa T, et al: Inhibition of the activities of matrix metalloproteinases 2, 8, and 9 by chlorhexidine. *Clin Diagn Lab Immunol* 6:437–443, 1999.
277. Perdigao J, Denehy GE, Swift EJ, Jr: Effects of chlorhexidine on dentin surfaces and shear bond strengths. *Am J Dent* 7:81–84, 1994.
278. Pashley DH, Tay FR, Yiu C, et al: Collagen degradation by host-derived enzymes during aging. *J Dent Res* 83:216–221, 2004.
279. Ricci HA, Sanabe ME, de Souza Costa CA, et al: Chlorhexidine increases the longevity of in vivo resin-dentin bonds. *Eur J Oral Sci* 118:411–416, 2010.
280. Carrilho MR, Carvalho RM, de Goes MF, et al: Chlorhexidine preserves dentin bond in vitro. *J Dent Res* 86:90–94, 2007.
281. Hebling J, Pashley DH, Tjäderhane L, et al: Chlorhexidine arrests subclinical degradation of dentin hybrid layers in vivo. *J Dent Res* 84:741–746, 2005.
282. Loguercio AD, Hass V, Gutierrez MF, et al: Five-year effects of chlorhexidine on the in vitro durability of resin/dentin interfaces. *J Adhes Dent* 18:35–42, 2016.
283. Boushell LW, Kaku M, Mochida Y, et al: Immunohistochemical localization of matrixmetalloproteinase-2 in human coronal dentin. *Arch Oral Biol* 53:109–116, 2008.
284. De Munck J, Mine A, Van den Steen PE, et al: Enzymatic degradation of adhesive-dentin interfaces produced by mild self-etch adhesives. *Eur J Oral Sci* 118:494–501, 2010.
285. Sadek FT, Castellan CS, Braga RR, et al: One-year stability of resin-dentin bonds created with a hydrophobic ethanol-wet bonding technique. *Dent Mater* 26:380–386, 2010.
286. Bergenholtz G, Cox CF, Loesche WJ, et al: Bacterial leakage around dental restorations its effect on the dental pulp. *J Oral Pathol* 11:439–450, 1982.
287. Brännström M, Nyborg H: Pulpal reaction to polycarboxylate and zinc phosphate cements used with inlays in deep cavity preparations. *J Am Dent Assoc* 94:308–310, 1977.
288. Brännström M, Vojinovic O, Nordenvall KJ: Bacteria and pulpal reactions under silicate cement restorations. *J Prosthet Dent* 41:290–295, 1979.
289. Torstenson B, Nordenvall KJ, Brännström M: Pulpal reaction and microorganisms under Clearfil Composite Resin in deep cavities with acid etched dentin. *Swed Dent J* 6:167–176, 1982.
290. Reference deleted in pages.
291. Brännström M, Nordenvall J: Bacterial penetration, pulpal reaction and the inner surface of Concise Enamel Bond: composite fillings in etched and unetched cavities. *J Dent Res* 57:3–10, 1978.
292. Brännström M, Nyborg H: Cavity treatment with a microbicidal fluoride solution: growth of bacteria and effect on the pulp. *J Prosthet Dent* 30:303–310, 1973.
293. Mejàre B, Mejàre I, Edwardsson S: Acid etching and composite resin restorations: a culturing and histologic study on bacterial penetration. *Acta Odontol Scand* 3:1–5, 1987.
294. Mejàre I, Mejàre B, Edwardsson S: Effect of a tight seal on survival of bacteria in saliva-contaminated cavities filled with composite resin. *Endod Dent Traumatol* 3:6–9, 1987.
295. Turkistani A, Nakashima S, Shimada Y, et al: Microgaps and demineralization progress around composite restorations. *J Dent Res* 94:1070–1077, 2015.
296. Heintze SD: Systematic reviews: I. The correlation between laboratory tests on marginal quality and bond strength. II. The correlation between marginal quality and clinical outcome. *J Adhes Dent* 9 Suppl 1:77–106, 2007.
297. Perdigão J, Lopes M: Dentin bonding—questions for the new millennium. *J Adhes Dent* 1:191–209, 1999.
298. De Munck J, Van Meerbeek B, Yoshida Y, et al: Four-year water degradation of total-etch adhesives bonded to dentin. *J Dent Res* 82:136–140, 2003.
299. Hashimoto M, Ohno H, Sano H, et al: Degradation patterns of different adhesives and bonding procedures. *J Biomed Mater Res* 66B:324–330, 2003.
300. Hashimoto M, Ohno H, Kaga M, et al: In vivo degradation of resin-dentin bonds in humans over 1 to 3 years. *J Dent Res* 79:1385–1391, 2000.
301. Sano H, Yoshiyama M, Ebisu S, et al: Comparative SEM and TEM observations of nanoleakage within the hybrid layer. *Oper Dent* 20:160–167, 1995.
302. Li HP, Burrow MF, Tyas MJ: The effect of long-term storage on nanoleakage. *Oper Dent* 26:609–616, 2001.
303. Tay FR, Pashley DH, Yoshiyama M: Two modes of nanoleakage expression in single-step adhesives. *J Dent Res* 81:472–476, 2002.
304. Van Meerbeek B: The "myth" of nanoleakage. *J Adhes Dent* 9:491–492, 2007.
305. Pashley DH, Ciucchi B, Sano H, et al: Permeability of dentin to adhesive agents. *Quintessence Int* 24:618–631, 1993.
306. Lee HL, Orlowski JA, Scheidt GC, et al: Effects of acid etchants on dentin. *J Dent Res* 52:1228–1233, 1973.
307. Marshall GW, Inai N, Wu-Magidi IC, et al: Dentin demineralization: effects of dentin depth, pH and different acids. *Dent Mater* 13:338–343, 1997.

308. Selvig KA: Ultrastructural changes in human dentine exposed to a weak acid. *Arch Oral Biol* 13:719–734, 1968.
309. Wang J-D, Hume WR: Diffusion of hydrogen ion and hydroxyl ion from various sources through dentine. *Int Endod J* 21:17–26, 1988.
310. Uno S, Finger WJ: Effects of acidic conditioners on dentine demineralization and dimension of hybrid layers. *J Dent* 24:211–216, 1996.
311. Macko DJ, Rutberg M, Langeland K: Pulpal response to the application of phosphoric acid to dentin. *Oral Surg Oral Med Oral Pathol* 6:930–946, 1978.
312. Cox CF, Suzuki S: Re-evaluating pulp protection: calcium hydroxide liners vs. cohesive hybridization. *J Am Dent Assoc* 125:823–831, 1994.
313. Cox CF, Keall CL, Keall HJ, et al: Biocompatibility of surface-sealed dental materials against exposed pulps. *J Prosthet Dent* 57:1–8, 1987.
314. Fuks AB, Funnell B, Cleaton-Jones P: Pulp response to a composite resin inserted in deep cavities with and without a surface seal. *J Prosthet Dent* 63:129–134, 1990.
315. Inokoshi S, Iwaku M, Fusayama T: Pulpal response to a new adhesive restorative resin. *J Dent Res* 61:1014–1019, 1982.
316. Qvist V, Stoltze K, Qvist J: Human pulp reactions to resin restorations performed with different acid-etched restorative procedures. *Acta Odontol Scand* 47:253–263, 1989.
317. Snuggs HM, Cox CF, Powell CS, et al: Pulpal healing and dentinal bridge formation in an acidic environment. *Quintessence Int* 24:501–509, 1993.
318. Tsuneda Y, Hayakawa T, Yamamoto H, et al: A histopathological study of direct pulp capping with adhesive resins. *Oper Dent* 20:223–229, 1995.
319. White KS, Cox CF, Kanca J, 3rd, et al: Pulp response to adhesive resin systems applied to acid-etched vital dentin: damp versus dry primer application. *Quintessence Int* 25:259–268, 1994.
320. Kanca J, 3rd: Replacement of a fractured incisor fragment over pulpal exposure: a long-term case report. *Quintessence Int* 27:829–832, 1996.
321. Heitman T, Unterbrink G: Direct pulp capping with a dentinal adhesive resin system: a pilot study. *Quintessence Int* 11:765–770, 1995.
322. Cox CF, Hafez AA, Akimoto N, et al: Biocompatibility of primer, adhesive and resin composite systems on non-exposed and exposed pulps of non-human primate teeth. *Am J Dent* 11:S55–S63, 1998.
323. Gwinnett AJ, Tay FR: Early and intermediate time response of the dental pulp to an acid etch technique in vivo. *Am J Dent* 11:S35–S44, 1998.
324. Hebling J, Giro EM, Costa CA: Biocompatibility of an adhesive system applied to exposed human dental pulp. *J Endod* 25:676–682, 1999.
325. Pameijer CH, Stanley HR: The disastrous effects of the "total etch" technique in vital pulp capping in primates. *Am J Dent* 11:S45–S54, 1998.
326. Costa CA, Hebling J, Hanks CT: Current status of pulp capping with dentin adhesive systems: a review. *Dent Mater* 16:188–197, 2000.
327. Bjørndal L, Darvann T: A light microscopic study of odontoblastic and non-odontoblastic cells involved in tertiary dentinogenesis in well-defined cavitated carious lesions. *Caries Res* 33:50–60, 1999.
328. Bergenholtz G: Iatrogenic injury to the pulp in dental procedures: aspects of pathogenesis, management and preventive measures. *Int Dent J* 41:99–110, 1991.
329. Brännström M: The effect of dentin desiccation and aspirated odontoblasts on the pulp. *J Prosthet Dent* 20:165–171, 1968.
330. Stanley HR, Weisman MI, Michanowicz AE, et al: Ischemic infarction of the pulp: sequential degenerative changes of the pulp after traumatic injury. *J Endod* 4:325–355, 1978.
331. Ravn JJ: Follow-up study of permanent incisors with enamel fractures as a result of an acute trauma. *Scand J Dent Res* 89:213–217, 1981.
332. Murray PE, Smith AJ, Garcia-Godoy F, et al: Comparison of operative procedure variables on pulpal viability in an ex vivo model. *Int Endod J* 41:389–400, 2008.
333. Hannig M, Bott B: In-vitro pulp chamber temperature rise during composite resin polymerization with various light-curing sources. *Dent Mater* 15:275–281, 1999.
334. Baroudi K, Silikas N, Watts DC: In vitro pulp chamber temperature rise from irradiation and exotherm of flowable composites. *Int J Paediatr Dent* 19:48–54, 2009.
335. Bergenholtz G: Evidence for bacterial causation of adverse pulpal responses in resin-based dental restorations. *Crit Rev Oral Biol Med* 11:467–480, 2000.
336. Knezevic A, Zeljezic D, Kopjar N, et al: Cytotoxicity of composite materials polymerized with LED curing units. *Oper Dent* 33:23–30, 2008.
337. Eren D, Bektaş ÖÖ, Siso SH: Three different adhesive systems; three different bond strength test methods. *Acta Odontol Scand* 71:978–983, 2013.
338. Heintze SD, Thunpithayakul C, Armstrong SR, et al: Correlation between microtensile bond strength data and clinical outcome of Class V restorations. *Dent Mater* 27:114–125, 2011.
339. Paul SJ, Welter DA, Ghazi M, et al: Nanoleakage at the dentin adhesive interface vs microtensile bond strength. *Oper Dent* 24:181–188, 1999.
340. Eick JD, Robinson SJ, Chappell RP, et al: The dentinal surface: its influence on dentinal adhesion: part III. *Quintessence Int* 24:571–582, 1993.
341. Tam LE, Pilliar RM: Fracture surface characterization of dentin-bonded interfacial fracture toughness specimens. *J Dent Res* 73:607–619, 1994.
342. Fortin D, Swift EJ, Jr, Denehy GE, et al: Bond strength and microleakage of current adhesive systems. *Dent Mater* 10:253–258, 1994.
343. Sano H, Shono T, Sonoda H, et al: Relationship between surface area for adhesion and tensile bond strength: evaluation of a micro-tensile bond test. *Dent Mater* 10:236–240, 1994.
344. Nakajima M, Sano H, Burrow MF, et al: Tensile bond strength and SEM evaluation of caries-affected dentin using dentin adhesives. *J Dent Res* 74:1679–1688, 1995.
345. Phrukkanon S, Burrow MF, Tyas MJ: The influence of cross-sectional shape and surface area on the microtensile bond test. *Dent Mater* 14:212–221, 1998.
346. Shono Y, Ogawa T, Terashita M, et al: Regional measurement of resin-dentin bonding as an array. *J Dent Res* 78:699–705, 1998.
347. Reis A, Rocha de Oliveira Carrilho M, Schroeder M, et al: The influence of storage time and cutting speed on microtensile bond strength. *J Adhes Dent* 6:7–11, 2004.
348. Poitevin A, De Munck J, Van Landuyt K, et al: Critical analysis of the influence of different parameters on the microtensile bond strength of adhesives to dentin. *J Adhes Dent* 10:7–16, 2008.
349. Van Meerbeek B, Peumans M, Poitevin A, et al: Relationship between bond-strength tests and clinical outcomes. *Dent Mater* 26:e100–e121, 2010.
350. Heintze SD, Bluck U, Göhring TN, et al: Marginal adaptation in vitro and clinical outcome of Class V restorations. *Dent Mater* 25:605–620, 2009.
351. Armstrong SR, Boyer DB, Keller JC: Microtensile bond strength testing and failure analysis of two dentin adhesives. *Dent Mater* 14:44–50, 1998.
352. Pashley DH: Dentin bonding: overview of the substrate with respect to adhesive material. *J Esthet Dent* 3:46–50, 1991.
353. Pameijer CH, Louw NP: Significance of pulpal pressure during clinical bonding procedures. *Am J Dent* 10:214–218, 1997.
354. Yoshiyama M, Carvaho RM, Sano H, et al: Regional bond strengths of resins to human root dentine. *J Dent* 24:435–442, 1996.

355. Yoshiyama M, Sano H, Ebisu S, et al: Regional strengths of bonding agents to cervical sclerotic root dentin. *J Dent Res* 75:1404–1413, 1996.
356. Perinka L, Sano H, Hosoda H: Dentin thickness, hardness and Ca-concentration vs bond strength of dentin adhesives. *Dent Mater* 8:229–233, 1992.
357. Brännström M, Lindén LA, Aström A: The hydrodynamics of the dental tubule and of pulp fluid. A discussion of its significance in relation to dentinal sensitivity. *Caries Res* 1:310–317, 1967.
358. Harnirattisai C, Inokoshi S, Shimada Y, et al: Adhesive interface between resin and etched dentin of cervical erosion/abrasion lesions. *Oper Dent* 18:138–143, 1993.
359. Kwong SM, Cheung GS, Kei LH, et al: Micro-tensile bond strengths to sclerotic dentin using a self-etching and a total-etching technique. *Dent Mater* 18:359–369, 2002.
360. Heymann HO, Sturdevant JR, Bayne S, et al: Examining tooth flexure effects on cervical restorations: a two-year clinical study. *J Am Dent Assoc* 122:41–47, 1991.
361. Van Meerbeek B, Braem M, Lambrechts P, et al: Morphological characterization of the interface between resin and sclerotic dentine. *J Dent* 22:141–146, 1994.
362. van Dijken JWV: Clinical evaluation of three adhesive systems in Class V non-carious lesions. *Dent Mater* 16:285–291, 2000.
363. Braem M, Lambrechts P, Vanherle G, et al: Stiffness increase during the setting of dental composite resins. *J Dent Res* 66:1713–1716, 1987.
364. Heymann HO, Sturdevant JR, Brunson WD, et al: Twelve-month clinical study of dentinal adhesives in Class V cervical lesions. *J Am Dent Assoc* 116:179–183, 1988.
365. Heintze SD, Rousson V: Clinical effectiveness of direct class II restorations - a meta-analysis. *J Adhes Dent* 14:407–431, 2012.
366. Gladys S, Van Meerbeek B, Braem M, et al: Comparative physico-mechanical characterization of new hybrid restorative materials with conventional glass-ionomer and resin composite restorative materials. *J Dent Res* 76:883–894, 1997.
367. Miyazaki M, Hinoura K, Onose H, et al: Effect of filler content of light-cured composites on bond strength to bovine dentine. *J Dent* 19:301–303, 1991.
368. Browning WD, Brackett WW, Gilpatrick RO: Two-year clinical comparison of a microfilled and a hybrid resin-based composite in non-carious Class V lesions. *Oper Dent* 25:46–50, 2000.
369. Peumans M, De Munck J, Van Landuyt KL, et al: Restoring cervical lesions with flexible composites. *Dent Mater* 23:749–754, 2007.
370. Peumans M, Kanumilli P, De Munck J, et al: Clinical effectiveness of contemporary adhesives: a systematic review of current clinical trials. *Dent Mater* 21:864–881, 2005.
371. Heintze SD, Ruffieux C, Rousson V: Clinical performance of cervical restorations–a meta-analysis. *Dent Mater* 26:993–1000, 2010.
372. Loguercio AD, De Paula EA, Hass V, et al: A new universal simplified adhesive: 36-month randomized double-blind clinical trial. *J Dent* 43:1083–1092, 2015.
373. Lawson NC, Robles A, Fu CC, et al: Two-year clinical trial of a universal adhesive in total-etch and self-etch mode in non-carious cervical lesions. *J Dent* 43:1229–1234, 2015.
374. Sanares AME, Itthagarun A, King NM, et al: Adverse surface interactions between one-bottle light-cured adhesives and chemical-cured composites. *Dent Mater* 17:542–556, 2001.
375. Swift EJ, Jr, Perdigão J, Combe EC, et al: Effects of restorative and adhesive curing methods on dentin bond strengths. *Am J Dent* 14:137–140, 2001.
376. Bömicke W, Schürz A, Krisam J, et al: Durability of resin-zirconia bonds produced using methods available in dental practice. *J Adhes Dent* 18:17–27, 2016.
377. Seabra B, Arantes-Oliveira S, Portugal J: Influence of multimode universal adhesives and zirconia primer application techniques on zirconia repair. *J Prosthet Dent* 112:182–187, 2014.
378. Kalavacharla VK, Lawson NC, Ramp LC, et al: Influence of etching protocol and silane treatment with a universal adhesive on lithium disilicate bond strength. *Oper Dent* 40:372–378, 2015.
379. Perdigão J, Swift EJ, Jr: Universal adhesives. *J Esthet Restor Dent* 27:331–334, 2015.
380. Siqueira F, Cardenas AM, Gutierrez MF, et al: Laboratory performance of universal adhesive systems for luting CAD/CAM restorative materials. *J Adhes Dent* 18:331–340, 2016.
381. Pashley DH, Matthews WG, Zhang Y, et al: Fluid shifts across human dentine in vitro in response to hydrodynamic stimuli. *Arch Oral Biol* 41:1065–1072, 1996.
382. Markowitz K: Pretty painful: Why does tooth bleaching hurt? *Med Hypotheses* 74:835–840, 2010.
383. Gray A, Ferguson MM, Wall JG: Wine tasting and dental erosion: case report. *Aust Dent J* 43:32–34, 1988.
384. Grippo JO: Abfractions: a new classification of hard tissue lesions of teeth. *J Esthet Dent* 3:14–19, 1991.
385. Cuenin MF, Scheidt MJ, O'Neal RB, et al: An in vivo study of dentin sensitivity: the relation of dentin sensitivity and the patency of dentin tubules. *J Periodontol* 62:668–673, 1991.
386. Reinhardt JW, Stephens NH, Fortin D: Effect of Gluma desensitization on dentin bond strength. *Am J Dent* 8:170–172, 1995.
387. Kerns DG, Scheidt MJ, Pashley DH, et al: Dentinal tubule occlusion and root hypersensitivity. *J Periodontol* 62:421–428, 1991.
388. Nakabayashi N, Nakamura M, Yasuda N: Hybrid layer as a dentin-bonding mechanism. *J Esthet Dent* 3:133–138, 1991.
389. Scherman A, Jacobsen PL: Managing dentin hypersensitivity: What treatment to recommend to patients. *J Am Dent Assoc* 123:57–61, 1992.
390. Nikaido T, Burrow MF, Tagami J, et al: Effect of pulpal pressure on adhesion of resin composite to dentin: bovine serum versus saline. *Quintessence Int* 26:221–226, 1995.
391. Swift EJ, Hammel SA, Perdigao J, et al: Prevention of root surface caries using a dental adhesive. *J Am Dent Assoc* 125:571–576, 1994.
392. Felton DA, Bergenholtz G, Kanoy BE: Evaluation of the desensitizing effect of Gluma dentin bond on teeth prepared for complete coverage restorations. *Int J Prosthodont* 4:292–298, 1991.
393. Bowes JH, Cater CW: The reaction of glutaraldehyde with proteins and other biological materials. *J Royal Microsc Soc* 85:193–200, 1966.
394. Schüpbach P, Lutz F, Finger WJ: Closing of dentinal tubules by Gluma desensitizer. *Eur J Oral Sci* 105:414–421, 1997.
395. Duran I, Sengun A, Yildirim T, et al: In vitro dentine permeability evaluation of HEMA-based (desensitizing) products using split-chamber model following in vivo application in the dog. *J Oral Rehabil* 32:34–38, 2005.
396. Camps J, About I, Van Meerbeek B, et al: Efficiency and cytotoxicity of resin-based desensitizing agents. *Am J Dent* 15:300–304, 2002.
397. Finger WJ, Balkenhol M: Rewetting strategies for bonding to dry dentin with an acetone-based adhesive. *J Adhes Dent* 2:51–56, 2000.
398. Ritter AV, Heymann HO, Swift EJ, Jr, et al: Effects of different re-wetting techniques on dentin shear bond strengths. *J Esthet Dent* 12:85–96, 2000.
399. Sobral MA, Garone-Netto N, Luz MA, et al: Prevention of postoperative tooth sensitivity: a preliminary clinical trial. *J Oral Rehabil* 32:661–668, 2005.
400. Chermont AB, Carneiro KK, Lobato MF, et al: Clinical evaluation of postoperative sensitivity using self-etching adhesives containing glutaraldehyde. *Braz Oral Res* 24:349–354, 2010.
401. Cobb DS, Reinhardt JW, Vargas MA: Effect of HEMA-containing dentin desensitizers on shear bond strength of a resin cement. *Am J Dent* 10:62–65, 1997.
402. Swift EJ, Jr, Lloyd AH, Felton DA, et al: The effect of resin desensitizing agents on crown retention. *J Am Dent Assoc* 128:195–200, 1997.
403. Helfer AR, Melnick S, Schilder H: Determination of the moisture content of vital and pulpless teeth. *Oral Surg Oral Med Oral Pathol* 34:661–670, 1972.

404. Nalbandian J, Gonzales F, Sognnaes RF: Sclerotic age changes in root dentin of human teeth as observed by optical, electron, and X-ray microscopy. *J Dent Res* 39:598–607, 1960.
405. Kinney JH, Nalla RK, Pople JA, et al: Age-related transparent root dentin: mineral concentration, crystallite size, and mechanical properties. *Biomaterials* 26:3363–3376, 2005.
406. Mjör IA, Smith MR, Ferrari M, et al: The structure of dentine in the apical region of human teeth. *Int Endod J* 34:346–353, 2001.
407. Violich DR, Chandler NP: The smear layer in endodontics - a review. *Int Endod J* 43:2–15, 2010.
408. McComb D, Smith DC: A preliminary scanning electron microscopic study of root canals after endodontic procedures. *J Endod* 1:238–242, 1975.
409. Mader CL, Baumgartner JC, Peters DD: Scanning electron microscopic investigation of the smeared layer on root canal walls. *J Endod* 10:477–483, 1984.
410. Reis A, Loguercio AD, Bitter K, et al: Adhesion to root dentin: A challenging task. In Perdigão J, editor: *Restoration of root canal-treated teeth - an adhesive dentistry perspective*, Switzerland, 2016, Springer International Publishing, pp 137–151.
411. Bouillaguet S, Troesch S, Wataha JC, et al: Microtensile bond strength between adhesive cements and root canal dentin. *Dent Mater* 19:199–205, 2003.
412. Alster D, Feilzer AJ, de Gee AJ, et al: Polymerization contraction stress in thin resin composite layers as a function of layer thickness. *Dent Mater* 13:146–150, 1997.
413. Santos-Filho PC, Veríssimo C, Raposo LH, et al: Influence of ferrule, post system, and length on stress distribution of weakened root-filled teeth. *J Endod* 40:1874–1878, 2014.
414. Gomes GM, Gomes OM, Gomes JC, et al: Evaluation of different restorative techniques for filling flared root canals: fracture resistance and bond strength after mechanical fatigue. *J Adhes Dent* 16:267–276, 2014.
415. Farina AP, Chiela H, Carlini-Junior B, et al: Influence of cement type and relining procedure on push-out bond strength of fiber posts after cyclic loading. *J Prosthodont* 25:54–60, 2016.
416. Feilzer AJ, de Gee AJ, Davidson CL: Setting stresses in composites for two different curing modes. *Dent Mater* 9:2–5, 1993.
417. Bitter K, Perdigão J, Hartwig C, et al: Nanoleakage of luting agents for bonding fiber posts after thermo-mechanical fatigue. *J Adhes Dent* 13:61–69, 2011.
418. Bitter K, Perdigão J, Exner M, et al: Reliability of fiber post bonding to root canal dentin after simulated clinical function in vitro. *Oper Dent* 37:397–405, 2012.
419. Sarkis-Onofre R, Skupien JA, Cenci MS, et al: The role of resin cement on bond strength of glass-fiber posts luted into root canals: a systematic review and meta-analysis of in vitro studies. *Oper Dent* 39:E31–E44, 2014.
420. Skupien JA, Sarkis-Onofre R, Cenci MS, et al: A systematic review of factors associated with the retention of glass fiber posts. *Braz Oral Res* 29:pii: S1806-83242015000100401, 2015.
421. Mayer T, Pioch T, Duschner H, et al: Dentinal adhesion and histomorphology of two dentinal bonding agents under the influence of eugenol. *Quintessence Int* 28:57–62, 1997.
422. AlEisa K, Al-Dwairi ZN, Lynch E, et al: In vitro evaluation of the effect of different endodontic sealers on retentive strength of fiber posts. *Oper Dent* 38:539–544, 2013.
423. Aleisa K, Alghabban R, Alwazzan K, et al: Effect of three endodontic sealers on the bond strength of prefabricated fiber posts luted with three resin cements. *J Prosthet Dent* 107:322–326, 2012.
424. Demiryurek EO, Kulunk S, Yuksel G, et al: Effects of three canal sealers on bond strength of a fiber post. *J Endod* 36:497–501, 2010.
425. Hagge MS, Wong RD, Lindemuth JS: Effect of three root canal sealers on the retentive strength of endodontic posts luted with a resin cement. *Int Endod J* 35:372–378, 2002.
426. Kurtz JS, Perdigao J, Geraldeli S, et al: Bond strengths of tooth-colored posts, effect of sealer, dentin adhesive, and root region. *Am J Dent* 16 Spec No:31A–36A, 2003.
427. Boone KJ, Murchison DF, Schindler WG, et al: Post retention: the effect of sequence of post-space preparation, cementation time, and different sealers. *J Endod* 27:768–771, 2001.
428. Mesquita GC, Verissimo C, Raposo LH, et al: Can the cure time of endodontic sealers affect bond strength to root dentin? *Braz Dent J* 24:340–343, 2013.
429. Vano M, Cury AH, Goracci C, et al: The effect of immediate versus delayed cementation on the retention of different types of fiber post in canals obturated using a eugenol sealer. *J Endod* 32:882–885, 2006.
430. Tzanakakis EG, Tzoutzas IG, Koidis PT: Is there a potential for durable adhesion to zirconia restorations? A systematic review. *J Prosthet Dent* 115:9–19, 2016.
431. Morimoto S, Albanesi RB, Sesma N, et al: Main clinical outcomes of feldspathic porcelain and glass-ceramic laminate veneers: a systematic review and meta-analysis of survival and complication rates. *Int J Prosthodont* 29:38–49, 2016.
432. Sterzenbach G, Franke A, Naumann M: Rigid versus flexible dentine-like endodontic posts–clinical testing of a biomechanical concept: seven-year results of a randomized controlled clinical pilot trial on endodontically treated abutment teeth with severe hard tissue loss. *J Endod* 38:1557–1563, 2012.
433. Esquivel-Upshaw J, Rose W, Oliveira E, et al: Randomized, controlled clinical trial of bilayer ceramic and metal-ceramic crown performance. *J Prosthodont* 22:166–173, 2013.
434. Gracis S, Thompson VP, Ferencz JL, et al: A new classification system for all-ceramic and ceramic-like restorative materials. *Int J Prosthodont* 28:227–235, 2015.
435. Ozcan M, Vallittu PK: Effect of surface conditioning methods on the bond strength of luting cement to ceramics. *Dent Mater* 19:725–731, 2003.
436. Addison O, Marquis PM, Fleming GJ: The impact of hydrofluoric acid surface treatments on the performance of a porcelain laminate restorative material. *Dent Mater* 23:461–468, 2007.
437. Peumans M, Valjakova EB, De Munck J, et al: Bonding effectiveness of luting composites to different CAD/CAM materials. *J Adhes Dent* 18:289–302, 2016.
438. al Edris A, al Jabr A, Cooley RL, et al: SEM evaluation of etch patterns by three etchants on three porcelains. *J Prosthet Dent* 64:734–739, 1990.
439. Calamia JR: Etched porcelain facial veneers: a new treatment modality based on scientific and clinical evidence. *N Y J Dent* 53:255–259, 1983.
440. Bjorksten J, Yaeger LL: Vinyl silane size for glass fabric. *Mod Plast* 29:124, 128, 1952.
441. Bowen RL, Rodriguez MS: Tensile strength and modulus of elasticity of tooth structure and several restorative materials. *J Am Dent Assoc* 64:378–387, 1962.
442. Diaz-Arnold AM, Schneider RL, Aquilino SA: Porcelain repairs: an evaluation of the shear strength of three porcelain repair systems (abstract 806). *J Dent Res* 66:207, 1987.
443. Newburg R, Pameijer CH: Composite resin bonded to porcelain with a silane solution. *J Am Dent Assoc* 96:288–291, 1978.
444. Sheth J, Jensen M, Tolliver D: Effect of surface treatment on etched porcelain and bond strength to enamel. *Dent Mater* 4:328–337, 1998.
445. Chen L, Shen H, Suh BI: Effect of incorporating BisGMA resin on the bonding properties of silane and zirconia primers. *J Prosthet Dent* 110:402–407, 2013.
446. Kim RJ, Woo JS, Lee IB, et al: Performance of universal adhesives on bonding to leucite-reinforced ceramic. *Biomater Res* 19:11, 2015.
447. Chen C, Trindade FZ, de Jager N, et al: The fracture resistance of a CAD/CAM Resin NanoCeramic (RNC) and a CAD ceramic at different thicknesses. *Dent Mater* 30:954–962, 2014.
448. Fleming GJ, Jandu HS, Nolan L, et al: The influence of alumina abrasion and cement lute on the strength of a porcelain laminate veneering material. *J Dent* 32:67–74, 2004.

449. Garvie RC, Hannink RH, Pascoe RT: Ceramic steel? *Nature* 258:703–704, 1975.
450. Heuer AH: Transformation toughening in ZrO_2-containing ceramics. *J Am Ceram Soc* 70:689–698, 1987.
451. Hannink RHJ, Kelly PM, Muddle BC: Transformation toughening in zirconia-containing ceramics. *J Am Ceram Soc* 83:461–487, 2000.
452. Piconi C, Maccauro G: Zirconia as a ceramic biomaterial. *Biomaterials* 20:1–25, 1999.
453. Tanaka K, Tamura J, Kawanabe K, et al: Phase stability after aging and its influence on pin-on-disk wear properties of Ce-TZP/Al_2O_3 nanocomposite and conventional Y-TZP. *J Biomed Mater Res A* 67:200–207, 2003.
454. Benzaid R, Chevalier J, Saâdaoui M, et al: Fracture toughness, strength and slow crack growth in a ceria stabilized zirconia-alumina nanocomposite for medical applications. *Biomaterials* 29:3636–3641, 2008.
455. Kern M, Wegner SM: Bonding to zirconia ceramic: adhesion methods and their durability. *Dent Mater* 14:64–71, 1998.
456. Borges GA, Spohr AM, de Goes MF, et al: Effect of etching and airborne particle abrasion on the microstructure of different dental ceramics. *J Prosthet Dent* 89:479–488, 2003.
457. Kern M, Barloi A, Yang B: Surface conditioning influences zirconia ceramic bonding. *J Dent Res* 88:817–822, 2009.
458. Inokoshi M, De Munck J, Minakuchi S, et al: Meta-analysis of bonding effectiveness to zirconia ceramics. *J Dent Res* 93:329–334, 2014.
459. Özcan M, Bernasconi M: Adhesion to zirconia used for dental restorations: a systematic review and meta-analysis. *J Adhes Dent* 17:7–26, 2015.
460. Chen L, Suh BI, Brown D, et al: Bonding of primed zirconia ceramics: evidence of chemical bonding and improved bond strengths. *Am J Dent* 25:103–108, 2012.
461. Yoshida K, Tsuo T, Atsuta M: Bonding of dual-cured resin cement to zirconia ceramic using phosphate acid ester monomer and zirconate coupler. *J Biomed Mater Res B Appl Biomater* 77:28–33, 2006.
462. Maeda FA, Bello-Silva MS, de Paula Eduardo C, et al: Association of different primers and resin cements for adhesive bonding to zirconia ceramics. *J Adhes Dent* 16:261–265, 2014.
463. Amaral M, Belli R, Cesar PF, et al: The potential of novel primers and universal adhesives to bond to zirconia. *J Dent* 42:90–98, 2014.
464. Özcan M, Bock T: Protocol for removal of clinically relevant contaminants from zirconium dioxide fixed dental prostheses. *J Adhes Dent* 17:576–577, 2015.
465. Kobayashi K, Kuwajima H, Masaki T: Phase change and mechanical properties of ZrO_2-Y_2O_3 solid electrolyte after ageing. *Solid State Ion* 3/4:489–493, 1981.
466. Chevalier J, Cales B, Drouin JM: Low-temperature aging of Y-TZP ceramics. *J Am Ceram Soc* 82:2150–2154, 1999.
467. Chevalier J, Olagnon C, Fantozzi G: Subcritical crack propagation in 3Y-TZP ceramics: static and cyclic fatigue. *J Am Ceram Soc* 82:3129–3138, 1999.
468. Chevalier J, Gremillard L, Deville S: Low-temperature degradation of zirconia and implications for biomedical implants. *Annu Rev Mater Res* 37:1–32, 2007.
469. Studart AR, Filser F, Kocher P, et al: Fatigue of zirconia under cyclic loading in water and its implications for the design of dental bridges. *Dent Mater* 23:106–114, 2007.
470. Kohorst P, Dittmer MP, Borchers L, et al: Influence of cyclic fatigue in water on the loadbearing capacity of dental bridges made of zirconia. *Acta Biomater* 4:1440–1447, 2008.
471. Kohorst P, Borchers L, Strempel J, et al: Low-temperature degradation of different zirconia ceramics for dental applications. *Acta Biomater* 8:1213–1220, 2012.
472. Perdigão J, Pinto AM, Monteiro RC, et al: Degradation of dental ZrO_2-based materials after hydrothermal fatigue. Part I: XRD, XRF, and FESEM analyses. *Dent Mater J* 31:256–265, 2012.
473. Bergamo E, da Silva WJ, Cesar PF, et al: Fracture load and phase transformation of monolithic zirconia crowns submitted to different aging protocols. *Oper Dent* 41:E118–E130, 2016.
474. Ângela Maziero Volpato C, Francisco Cesar P, Antônio Bottino M: Influence of accelerated aging on the color stability of dental zirconia. *J Esthet Restor Dent* 28:304–312, 2016.
475. Frankenberger R, Hartmann VE, Krech M, et al: Adhesive luting of new CAD/CAM materials. *Int J Comput Dent* 18:9–20, 2015.
476. Kassotakis EM, Stavridakis M, Bortolotto T, et al: Evaluation of the effect of different surface treatments on luting CAD/CAM composite resin overlay workpieces. *J Adhes Dent* 17:521–528, 2015.
477. Coldea A, Swain MV, Thiel N: Mechanical properties of polymer-infiltrated-ceramic-network materials. *Dent Mater* 29:419–426, 2013.
478. Özcan M, Volpato CÂ: Surface conditioning and bonding protocol for polymer-infiltrated ceramic: How and why? *J Adhes Dent* 18:174–175, 2016.
479. Darr AH, Jacobson PH: Conversion of dual-cure luting cements. *J Oral Rehabil* 22:43–47, 1995.
480. Lührs AK, De Munck J, Geurtsen W, et al: Composite cements benefit from light-curing. *Dent Mater* 30:292–301, 2014.
481. D'Alpino PH, Silva MS, Vismara MV, et al: The effect of polymerization mode on monomer conversion, free radical entrapment, and interaction with hydroxyapatite of commercial self-adhesive cements. *J Mech Behav Biomed Mater* 46:83–92, 2015.
482. Gerth HU, Dammaschke T, Züchner H, et al: Chemical analysis and bonding reaction of RelyX Unicem and Bifix composites–a comparative study. *Dent Mater* 22:934–941, 2006.
483. Saskalauskaite E, Tam LE, McComb D: Flexural strength, elastic modulus, and pH profile of self-etch resin luting cements. *J Prosthodont* 17:262–268, 2008.
484. Vrochari AD, Eliades G, Hellwig E, et al: Curing efficiency of four self-etching, self-adhesive resin cements. *Dent Mater* 25:1104–1108, 2009.

Disclosure

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